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Technical Report No. 66

Application of Single-Use Systems in Pharmaceutical Manufacturing

2014

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Technical Report No. 66

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1.0 Introduction

Single-use technology, often described as single-use systems (SUSs) or single-use equipment, has the potential to transform pharmaceutical manufacturing by offering tremendous opportunities to reduce cost, improve flexibility or cycle time, and shorten the time needed to build a manufacturing process for new, lifesaving drugs. This success, however, is very much dependent on how effectively the industry approaches the development and implementation of single-use technology. Ultimately, a new drug can only be successful if it is effective, safe, and available. Traditionally, only a comprehensive understanding of the drug product and manufacturing process can achieve these goals. This remains true as SUS is introduced in place of traditional reusable equipment. Encouraging an open science- and risk-based dialogue during supplier audits and evaluation of SUS supply chains significantly improves an SUS implementation.

This document is intended to provide the reader with critical concepts or points to consider when implementing an SUS strategy in a pharmaceutical manufacturing process. These concepts are intended to be valid both for chemically synthesized small molecules and for bioprocesses that produce large-molecule biopharmaceutical products. However, to be truly effective, many of these critical concepts must start with the design, supply chain, manufacturing, and distribution of SUSs themselves, as many inherent quality attributes can impact either the product molecule or its production process. Pharmaceutical manufacturers and single-use technology suppliers have become partners whose success is dependent on the control strategies implemented.

This document discusses SUSs that are in either direct or indirect contact with the raw materials, intermediates, and pharmaceutical drug substances or drug products. The document does not intend to discuss disposable items related to laboratory activities, final delivery system to the patient, transfusion bags, packaging, or medical devices.

Successful SUS implementation needs a comprehensive approach balancing the product and process goals achieved by using single-use technology. Section 3, Manufacturing Strategy, of this technical report is intended to present an approach that ties together key considerations when evaluating single-use technology. A well-designed manufacturing strategy will address technical, quality, business, and implementation considerations. Each topic has a dedicated section in this technical report, providing a detailed discussion of the associated considerations.

Determining the optimal manufacturing strategy involves concepts from many disciplines. An effective evaluation will have a balanced viewpoint, with input from engineering, regulatory, quality, project management, and accounting. Balancing risks and rewards of an SUS over a multiple-use system (MUS) will help determine the most appropriate manufacturing strategy. Thus, a structured science- and risk-based approach is recommended and should be consistent with principles described in ICH Guidelines Q6, Q7, Q8, Q9, Q10, and Q11. Primary goals, when developing any manufacturing strategy, should focus on controlling impacts to patient safety, product availability, and product and process understanding ***(1–6)***.

Only a formal partnership with an SUS supplier can ensure that quality is as good as or better than what is achieved with traditional systems (e.g., a purchase order is not a partnership). SUS suppliers provide equipment that includes the outsourced, value-added activities that the end user no longer performs. These value-added activities are important for the success of both organizations, and a winning control strategy for SUS has elements in both the supplier's and the end user's quality systems.

The concepts and recommendations presented in this technical report were developed over several years of discussion within the task force, at PDA workshops, and at other industry meetings. The authors of this technical report recognize that the conversation regarding how best to implement SUSs is just beginning. Ultimately the success of these systems will be determined by the decisions suppliers and end users make during implementation, and the hope is that this report provides a foundation for the industry to build on.

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2.0 Glossary of Terms

Campaigning
Extending the period of time, or number of cycles a single-use system is operating in a closed process without breaking the sterile barrier processes. The end user is responsible to evaluate and determine if appropriate quality requirements are met for their application.

Closed System
An isolated system that has no interaction with its external environment, preventing contamination and release of the material contained.

Consumables
This refers to items (e.g., SUS, storage bags, tubing, filters, diaphragms, flasks, etc.) that form or are a part of process equipment and are used on a per batch basis.

Deployment
Activities involving the hands-on steps required to successfully assemble and make a system ready for use for a specific SUS application.

Extractables
Chemical substances that can be extracted from components of material process fluid contact surfaces by exertion of an exaggerated force (e.g., organic solvent, extreme elevated temperature, ionic strength, pH, contact time, etc.) Extractables may represent most but not all of the potential leachables that may be seen in process fluids.

Flexible 2D or 3D Bag
A flexible-wall container designed with 2 sides (two dimensional or "pillow" shape) or 6 sides (three dimensional cuboid shape) designed to hold process fluids or product.

In-Process Leachables
Chemicals substances that are leached, from product-contact or non-product-contact materials under typical process conditions and could be cleared or sufficiently diluted by downstream processes so as to be undetected as leachables in the final dosage. Alternate Terms: Transient Leachables, Migrant Leachable.

Packaged Raw Material for Single-Use
Procurement of a product such as liquid or powder format culture media or buffer and that has been supplied in single-use technology

Leachables
Chemicals substances that are leached, from product-contact or non-product-contact materials, under typical process conditions and detected in final dosage. Leachables may be a subset of Extractables, and can include their reaction or breakdown products.

Multi-Use System (MUS)
An engineered process equipment solution for process management and unit operations designed for repeated use.

Opportunity Costs
This is defined as the value of the next-best choice available when choosing between several mutually exclusive choices (e.g., the decision to expand a facility may result in losing the opportunity to invest the maintenance funds in the financial markets). Opportunity costs are often excluded from estimates of fixed operation costs because they can be difficult for comparative analyses in the overall decision process

Pressure Decay Test
A leak test in which a container or system is pressurized with air to a preset level. After the pressure has stabilized, the decay in pressure over a preset test time is measured and evaluated to determine if a leak (defect) is present.

Ready-To-Use
A marketing term often used to describe the benefits of single-use technology or SUS. This designation has no regulatory or scientific basis supporting suitability for use and the end user is responsible to evaluate and determine if appropriate quality requirements are met for their application.

Single-Use Component (SUC)
An individual part designed to perform a particular function when assembled into a single-use system.

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Single-Use System (SUS)

An engineered process equipment solution, most commonly assembled from components made using polymeric materials, which together create a system or unit operation designed for one time or campaign use.

SUS Interchangeability

Functionally equivalent substitution of an alternative SUS, for an existing SUS design providing a contingency or process improvement with equivalent process performance and product quality. The end user is responsible to evaluate and determine if appropriate quality requirements are met for their application.

Technical Diligence

The means by which the technical capabilities of the SUS, its supplier and its fit with the end user are verified. It is complimentary to a quality audit program and becomes a recurring theme over the implementation process and subsequent routine procurement.

Value-added Activities

Includes the management of materials, operations, qualification, verification, vender audits, materials testing, and shop-floor oversight that allows an SUS to be used in operations.

2.1 Acronyms

CIP Clean-in-Place

CMO Contract Manufacturing Organization

COGM Cost-Of-Goods-Manufactured

CPP Critical process parameters

CQA Critical Quality Attributes

E&L Extractables and Leachables

FMEA Failure Mode and Effects Analysis

HACCP Hazard Analysis and Critical Control Points

MU Multiple-use

MUS Multiple-use System

PEP Project Execution Plan

QRM Quality Risk Management

RAT Regulatory Assessment Table

SIP Sterilize-in-Place

SU Single-Use

SUS Single-Use System

UR User Requirements

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3.0 Points to Consider for Single-Use System Manufacturing Strategy

This section provides an introduction to and overview of the most important points to consider and the necessary steps the end user must accomplish in order to implement a successful single-use (SU) strategy, while a more detailed description of these points and a structured implementation strategy can be found in the subsequent sections of this document. Additionally, this section offers a guideline for a structured approach for an initial feasibility evaluation of a single-use system (SUS) manufacturing strategy.

The key to successful SU technology implementation is the development of an effective manufacturing strategy for the molecule or formulation to be produced. Many factors must be considered when an organization is making strategic decisions, and implementation of SU technology is no different. In this report we suggest developing a high-level manufacturing strategy after a science- and risk-based approach with consideration given to four main areas that influence the success of SUS implementation:

- Technologies and system integration
- Qualification of assembled products
- Business drivers
- Implementation

The resulting manufacturing strategy will then outline the questions for consideration needed for successful implementation of an SUS. This technical report has a dedicated section for each of these areas of consideration. An overview of each will be presented in this section.

As depicted in **Figure 3.0-1,** each area plays a crucial role in establishing successful use of SUSs. Questions within these areas of focus must be considered and balanced against an organization's needs and competencies when developing a manufacturing strategy (detailed in **Section 7.1**). Only a comprehensive approach ensures that the primary customer-based goals of patient safety, product availability, product understanding, and process understanding are met when implementing new technology.

Figure 3.0-1 Key Decision Areas for an SUS Manufacturing Strategy

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Each area has a situation-dependent role in the decision process and may be weighed differently according to the circumstances of the end user. For example, a contract manufacturer may value the ability to shorten changeover time between different products, whereas a large pharmaceutical company may weigh the risk of adopting new technology against utilization of existing capabilities. A small organization may not have the resources (manpower or capital) to invest in a fixed multiple-use system (MUS). Other important factors to consider are:

- The current stage of drug development
- The scale of post-launch manufacturing
- The likelihood of the facility being used to produce other products in the future
- The risk associated with depending on a sole SUS supplier

A detailed implementation process for SUS is described in **Section 7,** and this includes details of the workflow for the implementation process, starting with the preparation of the manufacturing strategy reviewed in this section.

A gated approach is used to facilitate the preparation of the manufacturing strategy and decision process. A "gate," as it is used here, is a decision point where the pros and cons of the strategy are addressed using a science- and risk-based approach. A proposal for the effective evaluation of important factors within the gated approach is described in **Figure 3.0-2.** Each stage is crucial to the successful implementation of a new technology. Negative results at any stage may halt the implementation of the SUS. An example of such a negative decision will be a case in which there will be an adverse effect on patient safety or where there is no feasible solution to a technical problem. The aim of the gated approach is to fail the process quickly when a major problem cannot be solved. The decision pathway is not necessarily linear—the end user may decide to enter the pathway at any given point. However, each stage must be addressed to have a balanced decision process prior to the implementation of an SUS.

A detailed implementation process for SUSs is described in Section 7, and the implementation workflow is developed based on the decisions made in the manufacturing strategy reviewed in this section. An overview of the different implementation stages is described in **Figure 7.0-1** and **Table 7.0-1**. Key documentary outputs for the whole implementation process are listed in Section 8, Appendix I: SUS Facility Overall User Requirement Specification.

The following sections provide an overview of a proposed decision-making process for implementing an SUS strategy.

Figure 3.0-3 offers an overview of how the sections in this document are related and support the structured approach for implementation of SUS.

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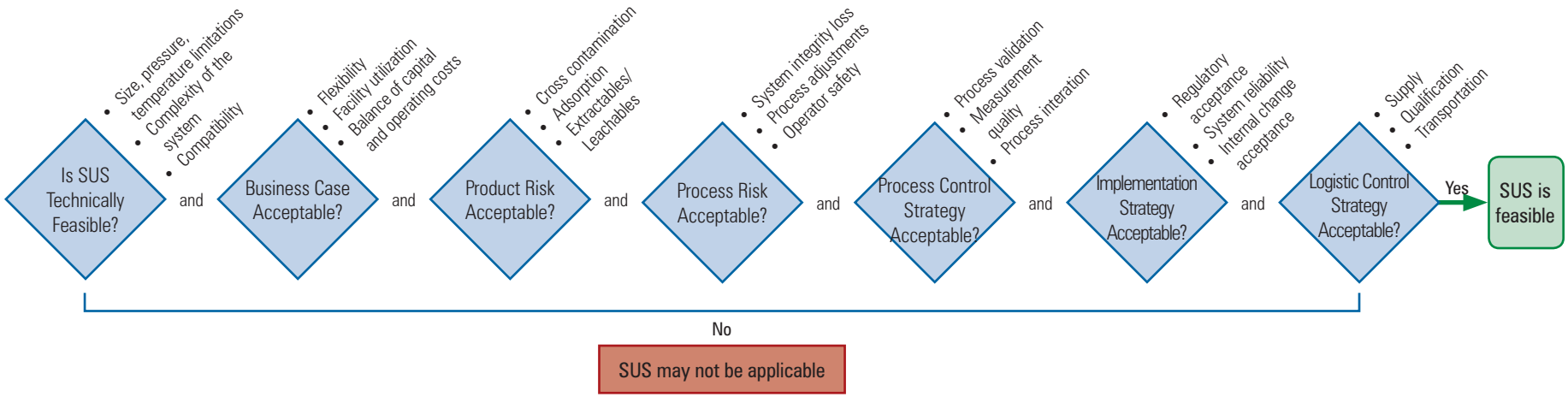

Figure 3.0-2 Proposed SUS Decision Pathway

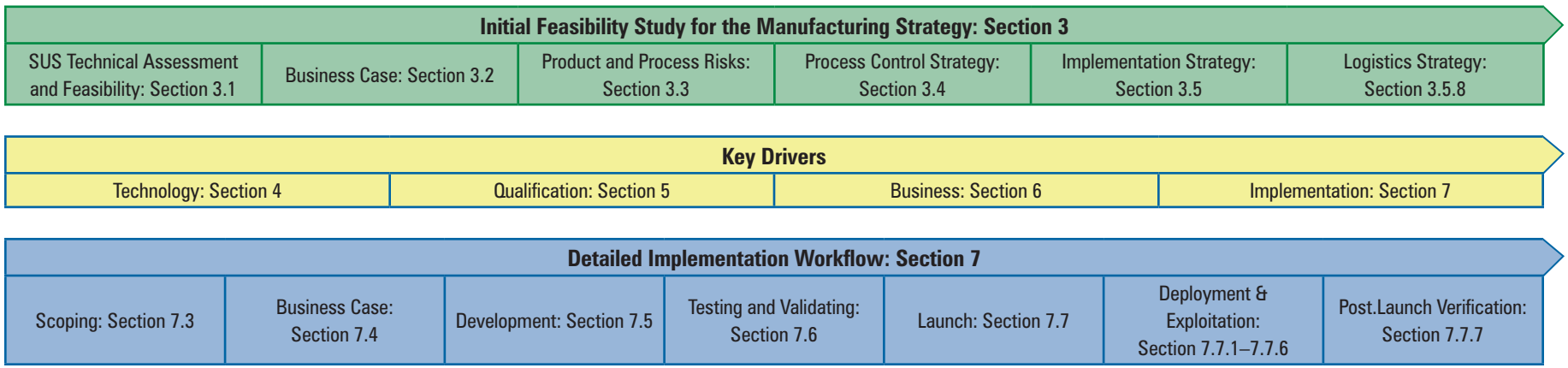

Figure 3.0-3 Technical Report Structure Overview

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3.1 Single-Use Technologies

SUSs differ in many ways from conventional MUSs. When examining whether a technically feasible single-use solution exists, the end user should evaluate process operating parameters and requirements in relation to traditional systems:

- Process and product are compatible with materials of construction
- Process volume, temperature, pH, pressure, and flow rate are within specifications
- Oxygen, moisture, or light barrier properties of the system are adequate for the process
- Specific process components can be sterilized
- Process monitoring and sampling strategy exists for the critical operating parameters
- Access ports and connection technology are available to perform the operation
- Mixing technology meets the uniformity needs of the process
- Weight, size, and physical strength of SUS meet the needs of the process and facility

After an organization determines that a technical solution exists for a single-use solution, other considerations to be included in the final decision to implement an SUS are as follows:

- Does the implementation of SUS impact existing process equipment that will need modification to optimize use of an SUS?
- What is the impact on cost and material management of customization, given that custom manufacturing of small quantities is less effective for suppliers?Although there are advantages to using a supplier's standard configurations, in some cases the SUS itself may require customization to accommodate an individual organization's needs.
- What is the potential impact on facility design and utility requirements?

The end user's approach to the determination of technical feasibility should be recorded in a risk-analysis document that is consistent with the methods outlined in ICH Q9, and it should include process-specific considerations such as *(4)*:

- The stage of the SUS application
- Process performance or capability
- Process scale
- Product quality attributes
- Batch production rates
- Factors affecting system design (e.g., process compatibility, component reliability, facility design, storage and materials handling, and/or transportation)

3.2 Business Drivers for the Adoption of Single-Use Systems

The decision between a single-use and a multiple-use process solution is often heavily influenced by a desire to reduce costs, increase the flexibility of a facility, and accelerate the drug development process. To that end, a structural comparative model that analyzes the cost of production, capital, quality, and process development, combined with the time constraints of the project, can be developed. The model's level of accuracy and detail should be suitable for the decision under consideration and should incorporate company-specific information wherever possible. A holistic model takes into account many factors that are useful for deciding if a greenfield facility is needed as opposed to a retrofit of existing applications. Often, a simplified model may be sufficient for an initial decision of whether SUS should be considered. The important point to consid-

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er is that an economic analysis needs to be done to support any important decision, and this is best done right from the start as part of developing a high-level manufacturing strategy. The following lists include examples of elements that may be considered when evaluating the business rationale for implementation of an SUS.

Capital Design

- Capital investment
- Equipment costs
- Utilities costs
- Facility floor space (cleanroom versus lower classification, production area versus warehouse)
- Validation—costs of validation clean-in-place and steam-in-place systems, facility environment
- Cost of retrofitting the facility for a new product or purpose
- Commissioning—costs associated with start-up and validation, testing, and documentation
- Cost of capital—borrowing costs

Operating Costs: Day-to-Day Variable Costs

- Labor
- Utilities and energy
- Waste management (solid and liquid, disposal versus recycling)
- Sterilization
- Cleaning
- Setup and changeover time and labor
- Amount lost to sterility failure or integrity losses
- Operation model make or buy SUS
- SUS run costs
- Inventory
- Media and buffer
- Preventive maintenance
- Value-added activities, quality audits of suppliers
- Multiple-product (MP) or single-product (SP) facility, changeover, testing, and regulatory submissions

Drug Development and Process Assessment

- Time-to-market and constraints
- Number of batches per year
- Number of products per year
- Product lifecycle—estimates of product demand over time
- Increased manufacturing time (may not fit existing manufacturing and quality processes)
- Process complexity
- Proximity to final drug presentation or container closures
- Workforce skill level
- Cost of active ingredients and intermediates
- Health, safety, and the environment (HSE) (e.g., biosafety, toxicity, occupational health, and industrial hygiene)
- Multiple-product or single-product facility

Time

- Process uptime
- Construction time
- Time constraints
- Regulatory milestones
- Expected market competition milestones
- Expected product launch date
- Product demand growth rate
- Inventory build time line
- Equipment and component assembly

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Green Manufacturing

- Environmental concerns for manufacturing process
- SUS waste generation
- Resources for waste management
- Design for sustainability

Value-Added Activities

- Cost of quality and cost of failure
- Quality, compliance, validation, calibration, operations
- Outsourced activities, make versus buy SUS
- Expedited transfer of manufacturing site to new location, reducing the time required to accommodate demand for additional product

Modeling will be highly dependent on specific circumstances and may result in different outcomes, dependent on model design, scale, timing, facility fit, product demand, and process frequency. For these reasons, it is not possible to make specific model design recommendations in this report. Every organization handles economic and process modeling differently. The concepts are presented in this report so the reader is aware that modeling plays an important role in supporting the decision process. Readers should reach out to others in their organization to decide how to best develop models for their organization. The following tools and resources may be useful to aid the decision process:

- ICH Q9 Risk Assessment Tools, FMEA, fishbone, fault tree, HACCP *(4)*
- Comparative matrix
- Internal cost databases
- Cost-of-goods modeling
- Comprehensive cost models

3.3 Qualification and Verification of Suppliers, Materials, Components, and Completed Assemblies

Any adverse effect on product quality and, hence, patient safety must be the main concern when the end user performs a risk assessment of an SUS. An assessment of critical operational parameters must be performed in order to determine the potential impact of an SUS on product quality. In this exercise, the end user must take into account the potential for interaction between process fluid and the materials of construction of the process equipment.

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3.3.1 Product Risk

The potential impact of the SUS on the product quality is a key area to be assessed through a risk-based approach considering its proximity to the final product, the processing stage, and product requirements. The following are examples of specific considerations for potential interactions between SUS and product fluids:

- An SUS is a defined system, which may minimize the risk of both microbial and chemical contamination.
- SUSs are susceptible to generation of particulates during the fabrication and operation of the system.
- SUS minimizes risk of cross-contamination when they are used once or in campaign manufacturing of a single drug substance.
- An SUS will contribute to the leachables present in a process. An assessment of impact can be made by 1) Identifying potential leachables through an extractables study and 2) Conducting further studies should an extractables study identify materials of concern.

Product impact considerations typically include:

- Sterility
- Degradation, or binding to the product molecule
- Adsorption of the product molecule or product pool
- Adsorption of the functional component of the media or buffer
- Interactions between leachables or their breakdown products with the product molecule or formulation or the process materials of construction
- Technical limitations of the SUS, which could lead to suboptimal processing, poor cell growth, low yields, or impact to product-critical quality attribute (e.g., limitation of temperature control may impact MAB glycan profile)
- Changes in the light, gas, moisture, or other barrier to the surroundings, which could cause oxidation or chemical reaction of the product molecule, product pool, or media components

In addition to product quality considerations, incorporating an SUS may affect key operating parameters at different process steps and potentially the control strategy for the entire process. An assessment for each process step, as well as the entire process, should include the following considerations:

- Impact on system integrity (may be positive or negative)
- Estimation of risk of equipment or system failure for both SUS and MUS, including downtime for correcting potential failures and backup equipment required (e.g., maintaining an ISO 5/Grade A-B area for connections if an SU connector fails or designing the connection for backup sterile tube welding if a connector fails)
- Size limitations of an SUS as compared to an MUS
- Productivity impacts when using SUS as compared to an MUS approach
- Verification activities, such as system integrity or leaks, compared to an MUS approach
- Supplier and vendor supply-chain transparency, change control, communication, product quality, and supply security
- Proximity of the SUS to the final drug product

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3.4 Process Control Strategy Considerations

The monitoring capability of single-use and multiple-use systems may differ. Some points to keep in mind include the following:

- When deciding between reusable and single-use probes, balance the cost per batch to generate the desired data, control of sensor calibration, cleaning validation requirements, and probe accuracy and reliability.
- Consider and assess impact from potential differences in automation level; SUSs typically include more manual controls dependent on operator training and execution. Some SUSs may incorporate proprietary support framework and control systems at comparable costs to traditional process skids; however, this significantly limits SUS interchangeability. To avoid this problem, the end user may wish to adopt cross-platform supervisory control and data application (SCADA) systems, which provide greater flexibility and portability.
- Consider that an SUS may affect how adjustments to the process are made (e.g., adjusting pH, adding solids, foam control); in some cases these adjustments may not be possible with SUS.
- Note that qualifying and calibrating probes and the measurement system before use may be more complicated or not possible if a single-use probe is built into a presterilized system. Batch documentation practices for recording sensor calibration may be different for SUS.
- Evaluate dependency on the supplier's and vendor's quality system and qualification activities.
- Account for the possibility of smaller batch sizes with increased frequency or sub-batch or batch pooling due to smaller container volumes.

3.5 Implementation of a Single-Use System

Implementation includes all of the activities required for the realization of the end user's manufacturing strategy. Since the manufacturing strategy is an assessment of feasibility informed by evaluation of the considerations from the sections on technology, business drivers, qualification, and implementation in this report, it provides a framework for implementation of the specific situation evaluated. However, each evaluation performed will be unique.

Furthermore, since time-to-clinic and time-to-market are becoming increasingly important factors for making implementation decisions, a comparative implementation assessment has become a crucial requirement. A well-thought-out and thorough implementation plan is important for successful implementation of an SUS. The implementation process is organized as a road map with a series of gateways between stages. The workflow should be organized around the following themes.

3.5.1 Stakeholder Management

Successful implementation of an SUS may require an active analysis of the effects of the implementation on the organization and stakeholder management. Broad implementation of SUS across an organization can impose certain organizational strains and stresses and problems may arise which have to be considered and dealt with.

3.5.2 Risk Management

The use of formal risk-assessment methods during SUS implementation provides a means for quantitative comparisons of risk related to MUS versus SUS to assist in decision making and stakeholder management *(7)*.

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3.5.3 Process Validation and Verification (PVV)

The PVV strategy for an SUS differs from that of an MUS. Therefore, the end user must be aware that traditional qualification activities may have limited value for an SUS implementation without the confidence that the supplier's production processes are of a high enough caliber and the end user's process is sufficiently robust to tolerate any variations of the SUS by the supplier *(8)*.

3.5.4 Scoping

This segment of the implementation plan contains a detailed description of the technical scope of the SUS project; it complements the manufacturing strategy by providing global guidance and policies to be applied for SU applications across the project.

3.5.5 Project Execution Plan (PEP)

A PEP provides a detailed description of the project, the execution strategy, management, and compliance in order to attain project goals in terms of budget, time, and quality. This includes a policy for the identification and selection of suppliers, which will encourage the creation of a transparent relationship and help prevent the selection of equipment and suppliers from becoming an ad hoc process. An example PEP is shown in Section 9, Appendix II: SUS Facility Project Execution Plan Example.

3.5.6 End-User Requirements

This section of the plan provides a description of the end user's requirements for the SUS, including E&L and standardization in preparation for the equipment procurement process.

3.5.7 Testing and Documentation

Following delivery and installation, the user will develop methods and documentation for equipment SOPs; maintenance; qualification and process validation; the HSE plan; and personnel training.

3.5.8 Supplier Management

The effective management of SUS suppliers must include periodic review meetings involving the analysis of the supplier's performance during the previous period for quality and punctuality. Business review meetings should also include collaborative planning and effective forecasting of future demands in order to take effective advantage of the supplier's resources, lower the costs of maintaining capacity, and eliminate spikes in demand that may affect lead times. Appropriate supply and quality agreements must be in place to clearly communicate expectations to critical suppliers.

Drug master files provided by either the polymer compound or film supplier can be referred to, however, they may not exist for all applications and the level of detailed of the information may vary by submission. The polymer compound formulations (type and levels of additives), for example, may only list the highest amount of the additives that can be included in the polymer rather than specifying the actual specification range of additives. In any case the information is not available to the end user so it is preferable to receive the information in the form of a regulatory support file.

3.5.9 Logistics Control Requirements

A logistics control strategy is a critical part of the implementation of an SUS because the end user relies on the supplier's production and quality systems to a higher degree than for the implementation of an MUS. The following factors are important for an effective logistics control strategy.

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3.5.9.1 Inventory and Supply Chain Management

A thorough analysis of the entire supply chain is necessary in order to safeguard the supply of important materials for undisrupted SUS production. Such an analysis must encompass the supply and inventory strategy of the vendors who supply components and materials to SUS manufacturers in addition to the supply and inventory strategies of the SUS suppliers themselves. SUS suppliers have a responsibility to protect the security and supply of their finished goods by appropriate risk analyses and risk-mitigation strategies for the materials and components they purchase. It is incumbent on the end user to confirm that these strategies are adequate to maintain the continuity of supply. If the supplier's control strategies fall short of expectations, the end user may mitigate the risk of supply interruption by qualifying secondary sources or putting specific inventory management and supply agreements in place.

3.5.9.2 Waste

Operation of an SUS leads to an increased amount of solid waste compared to an MUS. An SUS waste-management strategy must include:

- Collecting, storing, and inactivating waste in the facility
- Avoiding cross-contamination
- Monitoring the waste-management activities of subcontractors

3.5.9.3 Transportation

All aspects of transportation and holding must be understood for their role in maintaining integrity of the material throughout the supply chain. SU Technology requires additional care be taken during the entire supply chain; transportation, receiving, storage and delivery to the point of use. Some considerations are:

- Maintain integrity of material throughout the supply chain.
- An SUS must have a control strategy (such as supplier qualification, ID tests, or statistical analysis) of any product contacting materials to assure performance of critical components. Note that testing incoming components will in many cases be impractical beyond confirming the package and label, since SUS are typically not fully unpacked until the point of use.
- Ensure proper warehousing and inventory management.
- SU processes place additional demands on a warehouse. Consider shelf space and support functions for a warehouse and ways of improving delivery time, maintaining integrity, and documenting chain of custody from the vendor

3.5.9.4 Single Suppliers

In cases where an SUS component, film, or integrated assembly may be unique and proprietary, making it difficult to find and qualify a second supplier, the risk (including mitigation efforts and additional equipment needed to ensure uninterrupted supply) must be recorded in a risk-assessment document.

In some cases, sole source situations may exist with intellectual property (IP) protections, which need to be addressed with special contractual provisions in the event that the supplier is no longer able to provide the SUS in the future.

3.5.9.5 Change notifications

The end user must make sure that a change notification agreement is in place with suppliers (including third-party suppliers) and that such an agreement provides enough time to assess the effects of any changes on SUS functionally. Predictable, controlled transitions and ongoing operations of an SUS de-

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pend on strong communication between end users and suppliers. Identifying and qualifying alternative suppliers may help control costs, inventory, and time needed to plan and implement changes.

3.5.9.6 Technical Diligence

Technical diligence is the process by which the SUS end user obtains a thorough understanding of the SUS product, the supplier, and the compatibility of the SUS with the end user's processes and facility. These principles are based on established recommendations and procedures for critical raw materials.

3.6 Implementation Summary

Efforts required for implementation of a single-use process solution often differ from what is required for a reusable system. Thus, the critical path for the fastest implementation of a process solution might also differ. SUSs may allow for concurrent qualification and validation whereas MUSs require a more sequential approach. Successful implementation of an SUS hinges on a thorough comparative analysis.

Furthermore, the workflow performed in the implementation stages should be supported by best practices in process validation and verification, risk management, and stakeholder management.

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4.0 Single-Use Technologies and System Integration

4.1 Introduction

This section describes strategies for comparing single-use and traditional fixed multiple-use manufacturing methods with particular emphasis on the differences between SUS and the equivalent MUS. It also provides a framework for assessing the technical feasibility and risks associated with implementing an SUS. A description of the most common technical attributes of SUSs is also provided. Although SU manufacturing is undergoing rapid technological changes that may affect the specific factors used to compare and contrast SU with MU technologies, the fundamental principles used to assess the feasibility or development of risk assessment is expected to remain unchanged. The methodology and key points presented in this section will continue to be relevant as SUS suppliers continue to innovate.

4.2 Comparison of MUS and SUS

Most stainless steel systems are available in an SU format. Since SU suppliers use less costly polymeric materials, they have developed physical and mechanical operations to produce the functional equivalent unit operation in a format that is designed for an SU. A comparative description of the features of SU and MU technologies is presented in **Table 4.2-1.**

Table 4.2-1 Comparison of MU and SU Systems

Functional Requirement	MUS	SUS
Connections to sterile envelope	Stainless steel (SS), steam-in-place (SIP) valve arrays	Aseptic connectors, SIP connectors, and tubing welders. Alpha/beta ports
Disconnections from sterile envelope	SS SIP valve arrays	Aseptic disconnects, SIP connectors, and tubing sealers. Alpha/beta ports
Containment, storage, and transportation	SS tanks, fixed or portable	Polymeric film sealed to form 2D or 3D bags, unsupported, or as an insert to an external rigid support container (e.g., large bags, tank liners, and centrifuge liners)
Fluid transfer	SS piping, transfer panels	Tubing (flexible and rigid, polymeric)
Pumping	Multiple, fluid contact	Primarily peristaltic, nonfluid contact
Mixing (mass transfer)	SS impeller	Nonfluid contact (e.g., agitation and pumping) or SU impeller-based technology
Freezing	SS tanks or as integrated freeze-thaw systems	2D bags with or without shells, or as integrated SU freeze-thaw systems
Filtration	SS filter housings with filter elements (e.g., cartridges and plate/frame)	Capsule filters, hollow fiber cartridges, SU cassette technology
Chromatography	Multiple columns packed on-site with media (resin)	Prepacked columns and membrane adsorbers
Isolators	SS transfer canisters with docking port	2D bags with docking port and with or without tubing sets
Sampling	SS SIP sample valves	Sample bag or manifolds with or without SU sample valves. Aseptic connectors or tubing welders can be used to attach additional sample bags or manifolds if needed
Measurements	In situ reusable probe; can recalibrate during a process	In situ sensor may include a reusable probe or a single-use recalibrated sensor that cannot be adjusted or recalibrated during a process. Calibration is set during manufacturing and may be influenced by storage of the SUS

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4.3 SU Components and Assembly

The primary components for the assembly of an SUS include fittings, tubing, filters, connectors, bags (film), porting (bags), impellers, spargers, valves, sensors, filling needles, pump heads, docking port containers, and outer pouches (dust covers). The individual components do not require environmentally controlled storage rooms; they are stored in sealed bags and must be handled with gloves once opened to avoid contamination during assembly of the SUS. Several of the components can be assembled into a system and supplied presterilized, commonly via gamma irradiation. This is especially convenient for a completed fluid contact pathway of tubing preattached to a bag with all the tubing legs either heat-sealed or terminated with aseptic connectors. These systems are called bag assemblies, and they have tubing sets or tubing manifolds attached to them.

As shown in **Figure 4.3-1,** fully functional assemblies are often created by connecting components to a set of subassemblies, which in turn forms a functional SUS assembly that is capable of performing specific unit operations.

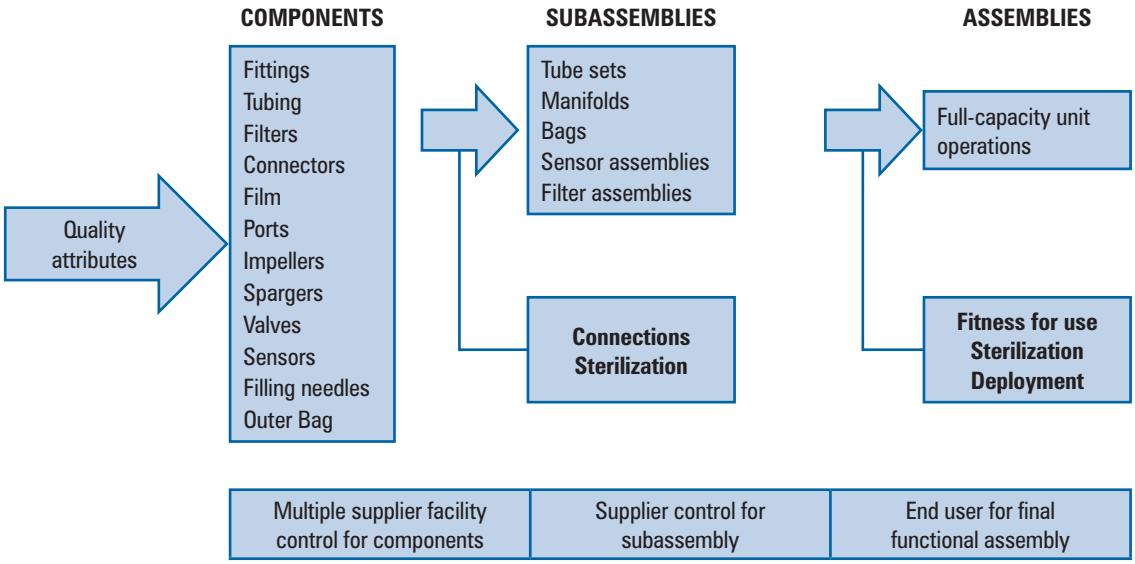

Figure 4.3-1 Anatomy of an SUS

MUSs are assembled in classified and environmentally monitored facilities, and the equipment interior is cleaned by a clean-in-place (CIP) technique. An SUS must be manufactured and assembled in a clean environment because there is no practical method to clean an SU assembly. Some tubing can be purged with ultrapure air or nitrogen for particle removal during fabrication; however, this is not practical for the assembled SUS. The area where the SUS will be assembled must be prepared in advance using standard clean-room practices. The outside of each component package must be wiped with alcohol before transport into the assembly area. Assembly rooms are typically ISO 7 or 8 (Grade B/C or Grade C/D, respectively), and it is normal to conduct a combination of nonviable particle and microbial sampling of the assembly room ***(9)*****.** The design of the room is similar to that of pharmaceutical manufacturing suites (i.e., HEPA-filtered air, low returns, smooth surfaces, and airlocks for entry and exit). Staff gowning must be appropriate for an ISO 7 or ISO 8 environment. Regular, consistent, and high-quality cleaning of the assembly room is required to maintain area classification and provide a consistent environment for manufacturing SU solutions. SUS components are manufactured and assembled in controlled environments in order to meet acceptable standards for particle, bioburden, and endotoxin levels. Since there are no established testing standards for an SUS, the end user must rely on the supplier's specifications

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and control strategies for particles, endotoxins, and bioburden. Suppliers should have quality systems in place to ensure their facilities and procedures meet end-user requirements. It is also critical that quality agreements are established between end users and suppliers to ensure that the suppliers' facilities are adequate to deliver an SUS that meets end-user requirements.

4.4 Technical Feasibility and Risk Assessment Framework

A technical feasibility assessment is needed to show that a proposed SUS can meet the functional and regulatory requirements expected for a particular application. This assessment is performed after the generation of user requirements (UR), which outline the particular technical requirements for an SUS application. The design of the SUS must be evaluated for each SUS application, taking into consideration process compatibility, facility fit, and the effect of SUS design on critical process parameters (CPPs) and critical quality attributes (CQAs). **Figure 4.4-1** delineates potential process and product risk factors and provides a framework for conducting a technical feasibility assessment. In using this figure, one should be aware that process risk may lead to product risk.

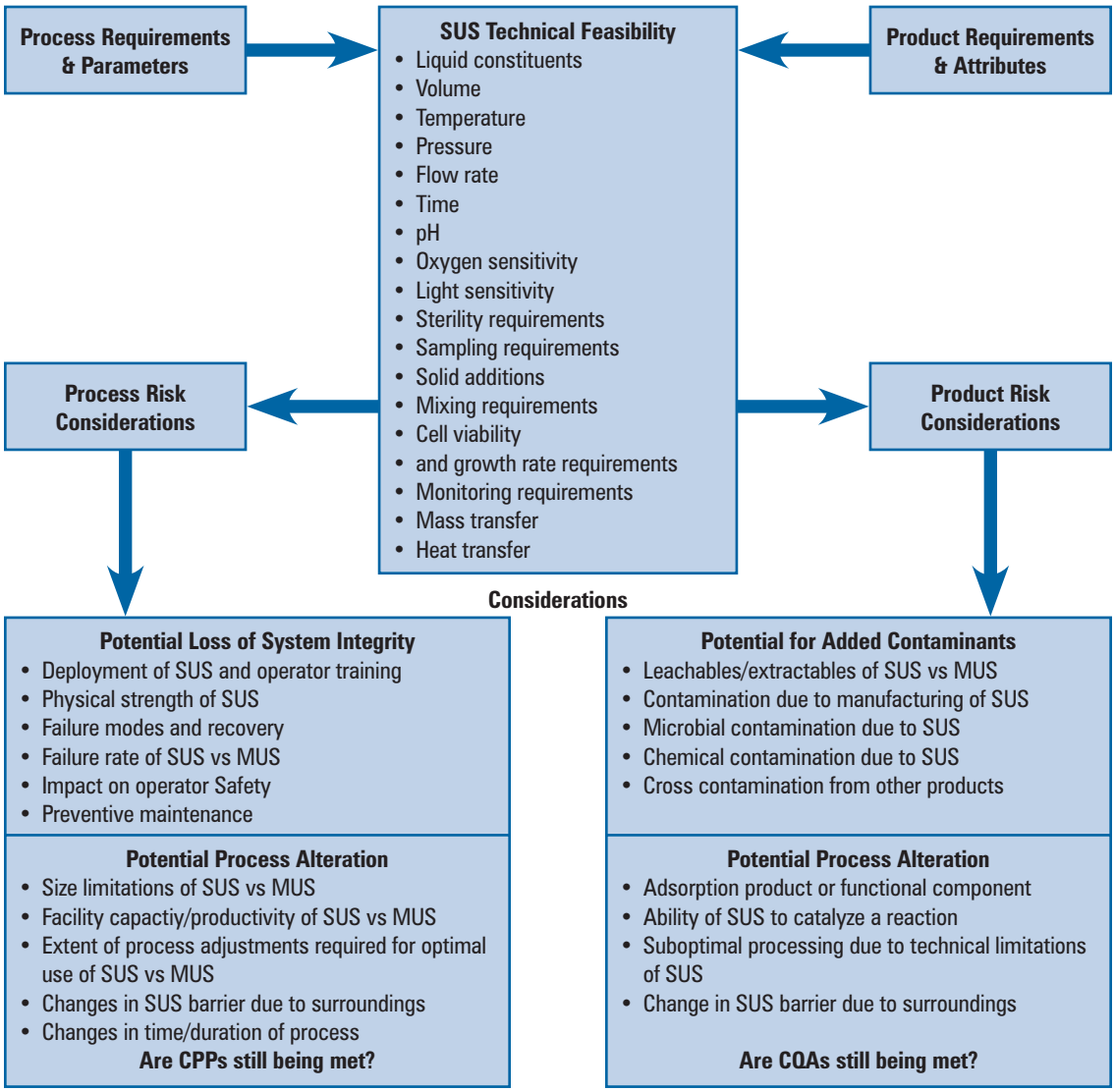

Figure 4.4-1 Implementation of an SUS and the Assessment of Drug Process and Product Risk

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4.5 Factors Affecting SUS Design

A technical feasibility study includes a thorough review of the design of a particular SUS in order to prevent mistakes stemming from component selection and system configuration. The scope of this design review will depend on both the functional objective and complexity of the SUS application in question. An inherent outcome of a technical feasibility study is the potential for identifying that a supplier's off-the-shelf SUS configuration is suitably designed with prequalified components (e.g., as proposed by a supplier). In some cases a customized, user-specified component or system will be identified during a technical feasibility study. Custom SUS configurations with specified components require additional efforts from an end user during a technical feasibility study if the supplier is not experienced with the user-specified components.

4.5.1 Process Compatibility

Another factor to consider during the technical feasibility study of an SUS is process compatibility. This involves an assessment of how each of the components of the SUS, plus the SUS as a complete integrated unit, may impact the process outcome. A description of a process compatibility assessment is provided in **Table 4.5.1-1.**

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Table 4.5.1-1 Assessment of Process Compatibility

Factor	Design Considerations
Construction materials	Chemical compatibility with the product and process fluids (liquid and solid), cleaning and sanitizing agents; biocompatibility; animal-derived components; extractable and leachable profiles; and nonspecific adsorption. Note that films may be permeable to some chemical agents (e.g., those used for wipe-down), which could have an impact on the process or product.
Physical properties	Attributes such as film or tube thickness, durometer, brittleness, surface smoothness, elongation, modulus, and impact and tear resistance will contribute to the ruggedness of the system and its resistance to breach of system integrity.
Microbial control methods	Ability to withstand gamma irradiation and/or other means of sanitization or sterilization
Time	Contact time (e.g., short passage in fluid conduit versus storage over an extended period of time), processing time (e.g., mixing time)
Volume	Size and scale of operation, holdup or residual volume (relative to drainage), dead volume
Temperature	Ability to withstand variations in temperature. Freezing applications require specialized systems to retain system integrity. The requirement for controlled addition of heat (e.g., to facilitate the flow of highly viscous fluids), and time for adjustment of temperature
Gas barrier	The loss of water vapor and moisture; oxygen, carbon dioxide, and nitrogen ingress and egress. (Barrier properties are temperature and relative humidity dependent.)
pH	Limitations in high or low pH; fixed versus variable pH over time or imposed profiles
Pressure	Limitations in absolute or differential pressure of the system and individual components (e.g., bags, which do not carry an absolute pressure rating; tubing burst pressure; and vacuum resistance ratings). Applications allowing for pressurized filling and draining of bags in enclosed rigid outer containers are possible following Good Engineering Practices)
Optical	Limitations in the visual process-monitoring capability (e.g., polymeric materials are often classified as transparent, translucent, or opaque)
Light	Sensitivity to light (including exposure time). The transparency of cell culture bags to light may or may not be a desirable feature.
Flow rate	Tubing ID, bag porting size, gravity drain versus pump capability
Particulates	Limitations due to normal particulate content or shedding as a result of abrasion and spallation
Mixing	Viscosity, inadequate mixing, and the generation of particles in cases where there is no terminal filtration. The ability to keep a solution homogeneous in applications involving suspensions. Make sure the design of the container allows for low-volume dispensing and mixing.
Sterilizing filtration	Location of the filter to ensure optimum product drainage; filtration inside sterile core or outside; venting or draining of filters during operation; pre-use flushing and integrity testing and post-use integrity testing; redundant sterilizing filters or need for prefiltration, filter sizing

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4.6 Facility Impact for SUS Setup and Deployment

Although SUSs can be set up and deployed in existing manufacturing facilities, a new or renovated facility may be more effective if a high level of SUS integration is required. To manage the risk of damage to the SUS, assessment of facility design and layout should include SUS material management, storage, and handling during operational setup. Including a defined setup and deployment stage, with standardized activities for each part, is a recognized success factor for SUS implementation and should be a routine part of the workflow and planning for each SUS application.

Setup includes those activities required to deliver the SUS to the point of use. Deployment involves the hands-on steps required to successfully assemble and ready the system for use in a specific SUS application. Examples of the facility setup and deployment assessments are described in **Table 4.6-1.**

Table 4.6-1 Assessment of SUS Facility Setup and Deployment

Factor	Design Considerations
Facility	Room classifications and the level of SUS integration between SUS and MUS and intra-facility transport of materials to progressively more stringent clean-room classification. Large SUS units may require hoisting or other mechanized deployment and transport. SUS implementation can increase demand on support areas (such as warehouses) and at use storage.
Staging, storage, and SUS deployment	SUS storage and space for staging SUS materials prior to operations; operator handling training; standardization of SUS deployment activities. Storage space for SUS is affected by run frequency and supplier lead times.
Floor space	Factors such as the lack of available space, plus the size and geometry of components (e.g., storage containers), may be the source of some obstacles to an SUS setup. Consider factors such as floor space and the geometry of the container when configuring an SUS, and optimize the use of stackable or collapsible reusable storage containers.
Changeover time	Product and campaign changeover time and individual unit processing setup times; sequence, bottlenecks, and process streamlining; and materials management. SUS can reduce facility changeover time, but SUS availability must be closely managed or the potential time savings will be lost.
Cleaning	Consider the effect of reusable storage containers, cleaning, and changeover time. Construction materials and surface finish of reusable storage containers (e.g., tolerance toward chemical cleaning agents, hygienic design and drainability).
Complexity	Ease of the SUS setup and deployment due to system complexity, ergonomic design, and error proofing should be assessed.
Labor	SUS and MUS may have differing labor demands, and how labor is impacted should be analyzed in detail.

4.6.1 Operational Requirements

SUS processes have inherently more manual connection and operational steps than MUS processes, which are often highly automated. Although the initial system design of an SUS generates an initial method of operation, it may be necessary to optimize the process operation for an end user's individual process requirements and facility. Examples of operational requirement assessments are reviewed in **Table 4.6.1-1.**

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Table 4.6.1-1 Assessment of Operational Requirements

Design Factor	Design Considerations
Campaigning	One use versus multiple fluid passages through an SUS
Deployment	Ability of bag and containment system to accommodate bag deployment during an SUS operation
Fluid integrity	Limitations in material strength, puncture resistance, and component and system integrity during normal operation
Process connectivity	Ability to make aseptic and sterile connections to SUS and MUS during the course of the process using aseptic connectors, including SIP connectors, and tube welders
Process disconnectivity	Ability to disconnect from the process stream aseptically, including SIP connectors (e.g., self-closing connectors, crimping tools, tube sealers)
Sampling	Ability to take representative and sterile samples from the process stream with considerations for point-of-use, volume, and frequency
Calibration	Practical considerations for the calibration of SU sensors
Space	Floor space for removing SUS after manufacturing

4.7 Applications and Technology

This section discusses several applications that are common in biopharmaceutical manufacturing, along with their cognate SU technologies. The structure to discussing these applications and technologies is as follows:

1. A description of the objective of the application
2. A description of current SU technologies available to meet the application needs
3. Specific considerations of the technology that one may use as part of a technical feasibility study

4.7.1 Process Connections

Connectors and tube welders are used primarily to connect:

- Components within an SU subassembly
- Multiple subassemblies to create a complete operational assembly
- An assembly to a neighboring unit operation

In addition, connectors are used with tube sealers or disconnectors to disconnect:

- Components (e.g., bags) within an SU subassembly
- A subassembly from a complete operational assembly

4.7.1.1 Technological Examples

A wide variety of connectors are commercially available for various user requirements. In general, two-piece connectors consist of two units with mating ends that may be joined to form a connection. Identical mating parts are denoted as "genderless," and nonidentical mating parts are usually denoted as "male" or "female." Although connectors are usually used to create a connection between parts, they may be installed internally within a pipe in order, for example, to open a flow path of a steaming process.

Single-piece insert connectors commonly are variants of standard straight, Y, or T configurations and employ hose barbs that can be inserted into tubing and secured with cable ties or clamps. A variant of a single-piece connector can be made by a specialized overmolding procedure that creates a connection with a uniformly smooth inner bore.

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Connectors with integrated valves and resterilizable SIP connectors have the additional functionality of enabling aseptic disconnections and reconnection.

As shown in **Table 4.7.1.1-1,** connectors may be grouped into three functional categories.

Table 4.7.1.1-1 Functional Categories of Connectors

Connector Category	Description
Adapter type	Hose barb and tri-clamp are the most common insert connection adapters. Hose barbs are for making tubing-to-tubing or tubing-to-device connections. Tri-clamp adapters are for making direct connections to stainless or filter with tri-clamp fittings. Overmolded connections can be converted into T, Y, 4-way, or tri-clamp configurations without insert connectors. Several connectors for aseptic requirements are commercially available.
Diameter of the flow path	Although there is a wide selection of commercially available connectors for tubing with standard diameters, in certain cases, it may be difficult to find an appropriate diameter of the flow path.
Aseptic connection or disconnection	Nonaseptic connectors are used in cases in which the connector is integrated into an SU assembly prior to sterilization via irradiation, whereas aseptic connectors are used to join two tubes that are part of a gamma-irradiated SU subassembly. Luer-lock connectors are examples of nonaseptic connectors that are used for small-bore (i.e., quarter inch or smaller) tubing connections during sampling applications. Steamable connectors are used to make aseptic connections between SU assemblies and traditional SIP stainless steel equipment; in this case, a resterilizable design system allows for aseptic disconnection. An aseptic connector with a built-in valve allows for aseptic disconnection from a process stream. A crimp-and-cut disconnector can also be used to achieve aseptic disconnection of an assembly. This is attained by installing a tightly fitting metal sleeve over the tubing. The sleeve is then crimped and cut with a specific tool.

A manifold can be constructed with pieces of tubing connected together with hose barb connectors and secured with cable ties, clamps, or other components so as to prevent leakage or bursting of the connection. Although organizations tend to adopt a particular type and style of connection technology to simplify logistics, it is important to complete a technology assessment for each new project to confirm that factors such as the specific connector technology, materials of construction, and connector size support end-user design requirements. **Table 4.7.1.1-2** contains specific considerations that apply to connector technology.

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Table 4.7.1.1-2 Specific Considerations for Connectors

Factor	Consideration
Ease of use	Determine the relative ease of use of the various connectors to ensure minimal operator error.
Technology	Evaluate the relative merits and feasibility of using tube-welding technology and aseptic connectors in a given application (such as tubing ID, space constraints, location of the needed weld, electricity/utilities requirements, and process requirements such as operating pressure). Examine risk factors of using tubing sets made with inserted hose barbs and overmolded connections with regard to the potential for leaks and fluid patch constrictions, which can affect flow rates and turbulence.
Sizing	Evaluate the effect of connector diameter on the flow rate requirement; ensure use of an appropriately sized connector.
Fitness for use	Check supplier qualification data for connector/disconnector, tube welding, and sealing technology to ensure the technology is suitable for the application. Make sure that connectors are qualified for use at pressures required in process. Consider how many connections are required for a particular application and location in the process design.
Connector and tubing materials	Evaluate the effect of materials on extractables and leachables and their potential impact on process and product.

The type of connection technology to be used should be considered when planning an SUS:

- **Tube welder:** A tube welder is an alternative type of connection technology that is designed to make a connection by cutting the two pieces of appropriate thermoplastic tubing to be connected and rapidly heating, aligning, and joining the two ends to form an aseptic, integral weld upon cooling. There is no need for a specific connector in the thermoplastic tubing, as the welded thermoplastic forms the seal between the two lines connected. Silicone tubing is not a suitable material for tube welding, as it is not a thermoplastic.

- **Tube sealer:** A tube sealer is a type of disconnection technology designed to seal a single thermoplastic tube using heat or energy to fuse together the lumen of one tubing end, segment, or fluid path. The sealed area is then separated from the rest of the tube by cutting it away. Silicone tubing is not appropriate for tube sealing, as it is not a thermoplastic.

4.8 Materials of Construction

Factors concerning the materials used for constructing the SUS may include inherent safety, traceability, physical properties, and compatibility with other components or product and process materials. The components used for the construction of SUSs are primarily comprised of plastic materials that may vary according to polymer class, molecular weight, and additives. This variability confers a wide range of performance characteristics on the plastic materials. **Table 4.8-1** contains a list of plastics commonly used for the production of SU components, along with their trade names, chemical abbreviations, properties, and applications.

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Table 4.8-1 Plastics Commonly Used in SUS

Family	Chemical Abbreviation	Chemical Name	Brand Name*	Applications	Reasons to Use	Comments
Fluoropolymers	PTFE	Poly(tetrafluoroethylene)	Teflon® Fluon®	Filters, tubing	Biologically inert Chemically resistant Extreme cold or hot temperature resistance Low extractables and leachables	Cannot be sterilized using gamma irradiation Limited mechanical properties
	PVDF	Poly(vinylidenefluoride) Poly(vinylidene fluoride-co-hexafluoropropylene)	Kynar® Kynar Flex®	Filters, fittings, tubing, bags	Biologically inert Greater strength and wear resistance than PTFE Broad temperature and chemical compatibility Low extractables and leachables Easily worked, molded, and sealed	Sterilizable by irradiation, steam, or chemical Some strong solvents can solubilize the material
Polycarbonate	PC	Polycarbonate	Lexan® Makrolon®	Fittings	Easily worked, molded, and thermoformed Excellent durability Temperature and impact resistant Autoclavable Clear, lightweight	Limited chemical compatibility range Some grades may leach bisphenol A (BPA), which has been implicated in a variety of adverse effects in some studies
Polyester	PET, PETE PBT	Poly(ethylene terephthalate) Poly(butylene terephthalate)	Rynite® Crastin®	Bottles, filter capsules	Semirigid to rigid Very lightweight Good gas barrier Naturally colorless with high transparency High strength Excellent stain resistance Good wear resistance Excellent dimensional stability	Radiation sterilizable Fair moisture barrier Naturally hygroscopic—must be dried before heat molding Copolymer variations include PETG (cyclohexane dimethanol co-polymer)

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Table 4.8-1 (Continued)

Family	Chemical Abbreviation	Chemical Name	Brand Name*	Applications	Reasons to Use	Comments
Polyolefin (polyalkene)	PP	Polypropylene	Moplen® Pro-fax®	Filters, housings, piping, fittings	Good chemical resistance Tough, impact resistant Good processing properties Higher temperature or chemical resistance over PE Rigidity, hardness Autoclavable Irradiatable (stabilized formulations only)	Nonstabilized formulations cannot be sterilized by gamma irradiation (reduces shelf life, elevates extractables) High thermal expansion Poor impact strength at low temperature
	PE	Polyethylene	Dowlex® Engage®	Bags, fittings	Good processing properties Impact resistant Good toughness High stress cracking resistance Lower extractables than EVA Cold temperature performance	Properties vary by density (high, low, ultralow, linear low) High thermal expansion Cannot be autoclaved
Copolymers	EVOH	Ethylene vinyl alcohols	Eval® Soranol®	Barrier layer in films	Superior gas barrier properties Good chemical resistance Very good clarity Can be coextruded with many different polymers	Hygroscopic, gas barrier properties diminished with moisture Moisture barrier properties lower than other polymers
	EVA	Ethylene vinyl acetate	Evatane® Elvax®	Bags, tubing	Soft and flexible Good clarity and gloss Good barrier properties Little or no odor Excellent cold temperature properties Better tear resistance than LDPE Excellent resistance to environmental stress cracking	Gamma-irradiated EVA can lower pH of contents Extractables may be undesirable Poor heat resistance Reduced barrier properties compared to LDPE Attacked by polar solvents, hydrocarbons, oxidants, and strong acids Reduced creep resistance compared to LDPE

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Table 4.8-1 (Continued)

Family	Chemical Abbreviation	Chemical Name	Brand Name*	Applications	Reasons to Use	Comments
Polyamides (nylons)	PA6,6	Polyamide 6,6 or nylon 6,6	Zytel® Ultramid®6,6	Films, filters	High melt point (autoclavable) Lower extractables than PA6 Moldable and extrudeable Good mechanical properties and wear resistance	Susceptible to degradation by strong acids, bases, and oxidizing agents
	PA6	Polyamide 6 or nylon 6	Zytel®	Films, tubes	Tough, higher impact resistance High-temperature resistant Easier to process Lower mold shrinkage Good fatigue resistance Greater elasticity and elastic recovery	Higher extractables Susceptible to degradation by strong acids, bases, and oxidizing agents Highest rate of water absorption and equilibrium water content
	PA11	Polyamide 11 or nylon 11	Rilsan® 11 Alamid® 11 Zytel®	Films, tubes	Improved chemical resistance versus other polyamides Some are bio-based Low water adsorption	More expensive than other PA Lower impact strength Minimal heat resistance (not autoclavable)
	PVC	Polyvinyl chloride Poly(chloroethanediyl)		Pipes, films, tubes	Tough, strong Inexpensive raw material and low processing temperature Good combination of stiffness and impact strength (rigid formulation), toughness, extensibility High ratio of strength to weight (flexible formulations)	Not autoclavable High levels of extractants Must be free of DEHP for fluid contact applications Low cost Disposal issues (generates dioxin on incineration)
Vinyl	CPVC	Chlorinated polyvinyl chloride		Pipes	Good resistance to acids and bases Inexpensive Improved chemical resistance over PVC More flexible than PVC	Not autoclavable Disposal issues (generates dioxin on incineration)

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Table 4.8-1 (Continued)

Family	Chemical Abbreviation	Chemical Name	Brand Name*	Applications	Reasons to Use	Comments
Silicone elastomers	Silicone elastomer	Poly(dimethylsiloxane) (May contain other silicone monomers)	Silastic®	Tubing, fitting, overmolding	Flexible, elastic Broad temperature resistance High tensile strength Elongation and tear resistant Low compression set at elevated and reduced temperatures compared to many organic rubbers	Not sealable or weldable Peroxide-cured types can have higher extractables than platinum-cured types Elevated silicone leachables can depress bubble point of downstream membranes
Thermoplastic elastomer	TPU	Thermoplastic polyurethane	Ellastolan® Irogran®	Tubing	Flexible—can be stretched to moderate elongations and return to close to original shape Processable as a melt at elevated temperature Absence of significant creep Autoclavable	TPUs have a large variety of chemical structures Critical to specify manufacturer and grade of TPU resins
Thermoplastic elastmomer blends	SEBS-PP	Styrene-ethylene-butyl-styrene + polypropylene	C-Flex® Kraton®	Tubing, fittings, overmolding	Flexible—can be stretched to moderate elongations and return to close to original shape Processable as a melt at elevated temperature Absence of significant creep Weldable	Autoclave conditions may cause tubing to deform—tubing should be evaluated on a case-by-case basis.

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4.8.1 Fluid Management

Fluid management describes the delivery of process fluids (e.g., buffers and culture media, process intermediates, and bulk drug product) to the point of use in the manufacturing stream.

4.8.1.1 Technological Examples

A fluid management system may include the following components:

- An SU bag to hold the process fluid
- A fluid-transfer manifold
- A storage container (usually reusable) to support the bag
- Packaging material (as required) for intra- or interfacility transportation (refer to **Section 4.17, Transportation).**

A fluid-transfer manifold can be integrated directly with a bag to establish a complete fluid delivery system. The specific considerations for fluid management are summarized in **Table 4.8.1.1-1.**

Table 4.8.1.1-1 Specific Considerations for Fluid Management

Factor	Consideration
Desired integration level	Simplify the design of the SUS by evaluating its functionality and complexity. Explore opportunities for implementing either a standard configuration or an application-specific configuration.
Packaging materials	Evaluate SUS or the reuse of storage containers and packaging material.
Microbial control	Investigate either the need to use presterilized components or a nonsterile system in a particular application. Compare the value of commercially available presterilized fluid pathway assemblies to that of traditional d fluid pathway assemblies and containers produced in-house.
Sterilization	Assess the value of presterilized fluid pathway assemblies supplied by the assembly supplier versus in-house cleaning and sterilization responsibilities.
Freedom from leaks	Evaluate supplier steps taken to assure a leak-free assembly. Inspect SUS for leaks through a standardized method prior to implementation.

4.8.2 Mixing

Mixing is the process used to achieve homogeneity of process liquids and solids. The process can involve liquid to liquid, solids dispersion, powder dissolution, powder to powder, or other combinations. Some mixing processes may require temperature regulation. The choice of mixing technology depends on the type of mixing required; the solubility, viscosity, or particle size of the constituents; and the scale of the operation.

4.8.2.1 Technological Examples

The particular method of mixing can be used to classify SU mixers into five main categories:

1. **Rocker-style bag mixer —** A rocker bag system is composed of a pillow-style bag attached to the top of a rocker platform. The bag is partially filled with liquid and inflated to create a pillow-shaped structure. The liquid is agitated in a back-and-forth or orbital movement to create a waveform on the liquid surface. The angle and frequency of rocking affects the size and speed of the resulting wave. Mixing is controlled by the intensity of the rocking action and the length of time the fluid is mixed. Addition of an electric blanket to the platform is used for temperature regulation.

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2. **Stirred-tank mixer —** A stirred-tank mixer is composed of a 3D bag inserted in an open-tank support structure made from rigid plastic or metal (e.g., aluminum or stainless steel). In this configuration, mixing is accomplished via an internal impeller driven by electromagnets. The purpose of the support structure is to support the shape of the bag and to counter the hydrostatic pressure in the bag. The bags are designed to be similar to the internal geometry of the support structure. An electric blanket or water jacket is used for temperature regulation. Although cylindrical systems are the most common, cube-shaped systems are available in various process sizes.
3. **Vibromixer —** A vibromixer is composed of a 3D bag with integrated polymeric disk impellers containing multiple conical perforations and a shaft, which is inserted into an outer support container or tank made of rigid plastic or stainless steel. The temperature regulation of a vibromixer is similar to that of stirred-tank mixers.
4. **Recirculation loop —** A recirculation loop provides a simple method for mixing using, for example, dip tubes and a peristaltic pump. These can be readily integrated into a bag.
5. **Unique configurations —** Novel mixing methods, including bellows, acoustic, and hybrid systems, continue to emerge. They rely on specific bag geometry and other unique design aspects for liquid movement (e.g., collapsible and expandable bellows sections designed into a container).

The specific considerations for selecting a mixing technique are summarized in **Table 4.8.2.1-1.**

Table 4.8.2.1-1 Specific Considerations for Mixing

Factor	Consideration
Process performance	Factors such as mixing time, agitator speed, and temperature may have to be adjusted to achieve desired process goals.
Mixing components	Because the components are mass produced, the ability to customize each component to suit a particular process may be limited.
Powder handling	Evaluate powder containment and handling options for ease of use, recovery, and potential interference with SUS materials.
Temperature control	Check the system for efficient heat transfer and proper alignment of mixing bags to container wall.
Mixing homogeneity	It is important for the operation of the SUS that a homogeneous solution can be produced from either emulsion or powder dispersion.
Heat transfer	To facilitate proper mixing, it may be necessary to transfer heat to the mixing bag. Consider impact of temperature control on component solubility during mixing.

4.8.3 Fermenters and Bioreactors

The fermenter or bioreactor is the system that maintains appropriate growth conditions to support the drug-product-producing strain.

4.8.3.1 Technological Examples

The following are the three main design categories of SU fermenters or bioreactors:

- **Rocker-style bag bioreactor —** A rocker-style bag bioreactor is a rocker mixer that can be used to monitor and control cell growth–related process parameters like pH and dissolved oxygen and also allows for product sampling. The rocking action is used to control mixing and dissolved oxygen. Headspace gas, sparge tubes, or a combination can also be used to regulate dissolved oxygen. Temperature regulation is achieved via heat transfer to the surface of the rocker.

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- **Stirred-tank bioreactor —** An SU stirred-tank bioreactor comprises a tube-shaped 3D bag inserted into an open-tank support structure made from plastic or stainless steel. The support structure retains the shape of the bag, counters hydrostatic pressure, and restrains the bag from moving around. Temperature regulation of the bioreactor is via an electric blanket or water jacket. A jacketed temperature regulation system allows for cooling of the culture during growth or prior to harvest and clarification. An agitator can be mounted either above or below the bioreactor. The impeller, shaft, and shaft cover (if used) and seal (if used) are integrated into the bag.
- **Other SU fermenters and bioreactors —** Hollow fiber systems remain in use for cell culture applications that are operated in perfusion mode. In addition, scaled-up versions of rigid, plastic cell-culture ware, novel bioreactor designs with unique mixing techniques (e.g., vibromixers, orbital shakers, paddle, and airlift) are being produced along with hybrid systems. These emerging systems often rely on bag geometry and other unique design aspects to achieve mass transfer. Novel SU fermenters and bioreactors are becoming commercially available. Technological advances in the method of agitation seem to be driving innovations in the new generation of SU fermenters and bioreactors. SU fermenters require unique strategies for heat and mass transfer in order to control temperature and the necessary oxygen transfer rates.

The specific considerations for fermenters or bioreactors are summarized in **Table 4.8.3.1-1.**

Table 4.8.3.1-1 Specific Considerations for Fermenters or Bioreactors

Factor	Consideration
Process control	SU sensors and the mating control systems may have different capabilities from a traditional MUS.
Process limitations	Compared to MUS, SU fermenters and bioreactors may be limited in the regulation of temperature, pressure, and oxygenation rates.
Polymer-specific effects	Make sure that the components neither adsorb nor produce reactive leachables, which could have adverse effects on cell viability, productivity, and the quality of the molecule of interest.
Design attributes	Investigate the quality of mixing and the magnitude of induced shear during the evaluation of specific, user-application requirements for the SUS bioreactor.
Light sensitivity	Determine the impact of exposure to ambient light on cells cultured in a transparent fermenter or bioreactor bag.
Mass transfer	Determine the capability of SUS to achieve desired power input, KLa*, and oxygen transfer rate.
Heat transfer	Determine the capability of SUS to transfer heat to and from the culture.
Cell viability and growth rate	Determine the capability of SU bioreactor to support cell viability and growth rate.
Sensors	Compare requirements for process sensors (e.g., temperature, pH, and dissolved oxygen) to the sensor technology used, effects on the sterile boundary, level of SUS integration, and sensor accuracy, robustness, and calibration.

* KLa = the volumetric mass-transfer coefficient that describes the efficiency with which oxygen can be delivered to a bioreactor for a given set of operating conditions

4.9 Storage

Appropriate storage is necessary in order to maintain the quality of process solutions (e.g., buffers and culture media), intermediates, and bulk drug substance until they are ready for further processing. Long-term storage of process intermediates often requires rapid, controlled freezing or cryogenic preservation. The storage bag must be qualified for the specific use, with a focus on maintaining bag integrity at the lower of the storage or freezing temperatures.

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4.9.1 Technological Examples

A storage bag with a polyethylene fluid contact layer is normally used to store aqueous-based liquid intermediates. Polyethylene polymers (particularly low-density grades) have broad chemical compatibility and very low reactivity to most intermediates and process fluids. Other materials like fluoropolymers are used to address specific process compatibility concerns.

Typical plastics used for freezing applications include poly(tetrafluoroethylene) (PTFE) and poly(ethylene vinyl acetate) (EVA). These materials are particularly good at retaining their mechanical strength at very low freezing temperatures. Specific bag systems have been integrated into commercially available controlled freeze-thaw systems. A gas barrier such as ethylene vinyl alcohol (EVOH) is typically used with EVA to create a multilayered composite film that offers mechanical strength and resistance to gas permeation.

For short-term storage, process intermediates in standard bags are normally kept at ambient temperature or in a refrigerator. Long-term storage of process intermediates often requires rapid, controlled freezing or cryogenic preservation (refer to **Section 4.10,** Freezing). Factors that affect the storage of process intermediates are summarized in **Table 4.9.1-1.**

Table 4.9.1-1 Specific Considerations for the Storage of Process Intermediates

Factor	Consideration
Potential for loss of quality (e.g., chemical, physical, or functional) of the process intermediate	In some cases, the process intermediates may have adverse interactions with the polymeric bag containers of the SUS. A special protective film may have to be applied to a bag containing oxygen-sensitive or photolabile process intermediates. It is vital to make sure that chemicals leaching from the polymer in the course of storage do not have adverse effects on the intermediates, buffer, or culture media that might get carried to unit operations.
Maintenance of system integrity	Ensure the integrity of the SUS by the careful choice and qualification of the bag, use of a supporting structure, and adherence to good laboratory practices.
Microbial quality	Ensure that sterility and bioburden control are maintained.

4.10 Freezing

The end user may use freezing techniques to separate process pools or intermediates and hold them for a short time at the process temperature and conditions or for a longer duration before another production step. For long durations, the end user may freeze the intermediate directly in an SUS that is designed for freezing. Uncontrolled freezing is performed in standard laboratory freezers, purpose-designed blast cells, or nitrogen freezing chambers. Controlled freeze-and-thaw equipment (e.g., a plate freezer) is used to regulate the rate of freezing and thawing of the contents of each SU bag. To maintain the integrity of the bag containing the frozen process intermediates or bulk drug substances, careful attention must be paid to the reusable storage container design and logistics of shipping and handling.

4.10.1 Technological Examples

- **Unprotected small-scale bag**: It is common practice to freeze an unprotected bag (less than 20 L) that is not specifically designed for the application. Compared to other technologies, the failure rates of frozen unprotected bags tend to be higher (**Table 4.10.1-1**).
- **Separate bag shell —** The purpose of a bag shell is to maintain the structural integrity of a frozen bag by protecting it from bumps and drops during handling at low temperatures. Bag shells are made of either rigid plastics or rigid foam construction.

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- **Integrated shell-protected bag —** The shell design includes a tubing and fitting compartment for security and storage at the most common point of failure. Off-the-shelf International Safe Transit Association (ISTA) and ASTM-qualified shipping containers can also be used ***(10)*.**
- **Controlled freeze-thaw system, bag, frame, and logistics equipment —** A controlled freeze-thaw system uses contact freezing technology to allow for the reproducible freezing of the contents of the bag. Contact freezing plates maintain freezing distances by maintaining the filled bag between two freezing contact plates. In addition, the rate of heat transfer from the bags is increased in a controlled freeze-thaw system, and this decreases the time required for freezing. Thus, a controlled freeze-thaw system is part of an integrated refrigeration system for maintaining the integrity of the bag and the quality of its contents during fill, transport, freezing, and thawing.

The specific considerations for freezing are summarized in **Table 4.10.1-1.**

Table 4.10.1-1 Specific Considerations for Freezing

Factor	Consideration
Storage temperature	At the typical storage temperature range of -40°C to -70°C, most polymer films, tubing, and fittings become brittle due to being below their glass transition temperature. Therefore, it is important to perform a structural integrity test on each freezing system.
Handling	Effective staff training is critical to the successful implementation of a freeze-thaw step in an SUS.
Shipping	Cold-chain logistics manufacturers are making robust, off-the-shelf solutions commercially available to the biopharmaceutical industry.

4.11 Filtration

Normal flow filtration (NFF) or direct flow filtration (DFF) devices are used in biopharmaceutical manufacturing to remove contaminants such as particulates, microbial cells, and virus particles. Cross flow filtration (CFF) or tangential flow filtration (TFF) devices are used to reduce the volume of a buffer and/or perform buffer exchange as part of the downstream process.

4.11.1 Technological Examples

CFF devices are commercially available in the following structural formats: flat-sheet cassettes, and hollow fiber cartridges. Currently, only a hollow-fiber cartridge can be classified as an SU filter device. Structurally, there is little difference between SU and MU CFF hollow fiber devices. Functionally, an SU CFF hollow fiber device provides a higher degree of cleanliness and sterility compared to that of a SU flat sheet device. Current flat sheet devices marketed as single-use contain a disposable flow path (e.g., ports, gaskets, and the filter device), which is then assembled into a stainless steel holder that provides the compression and support required for filtration NFF devices have been commercially available in an SU format for many years, and structurally, they contain a depth filter media or membrane for separation via size exclusion. The specific considerations for filtration are summarized in **Table 4.11.1-1.**

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Table 4.11.1-1 Specific Considerations for Filtration

Factor	Consideration
Device design	Versions of most NFF and hollow fiber CFF devices can be easily accommodated in a self-enclosed sterilized assembly (e.g., tubing, bags, and filters).
Microbial control	In general, flat-sheet devices are not supplied presterilized; however, they can be steam sterilized on-site. As an SU device, they allow a way to integrate a known separations technology into an SU unit operation.
Pre- and post-use integrity testing	From a technical perspective, it is less important to conduct a pre-use integrity test of a sterilizing filter that has been sterilized with gamma radiation. The pre-use test has a primary purpose of detecting a damaged filter, a purpose that has roots in steam sterilization, which has known mechanisms by which filters can be damaged. These mechanisms do not exist with gamma radiation sterilization.

4.12 Centrifugation

Centrifugation is often used for cell harvest as an alternative to filtration, and SU formats of this technology are emerging for low shear separations, including harvest clarification; cell concentration, wash, and harvest; and cell separation. These systems utilize a gamma-irradiated single-use module or polymeric liner for product contact and, as such, eliminate the need for CIP or manual cleaning and SIP or hot water for injection (WFI) for bioburden sterilization and control of traditional centrifuges. Flow rates vary depending on the manufacture and design of the equipment, ranging from <10 to 100 L per minute at <100X to 8000X G force. Continuous and batch centrifugation technologies are available.

4.13 Chromatography

Biomolecules are purified using chromatographic techniques that separate them according to differences in their specific physical or chemical properties. SU formats for various chromatography steps are available.

4.13.1 Technological Examples

Chromatographic media can be divided into two main categories: resin- or bead-based and membrane-based. Membrane-based chromatographic devices are also known as membrane adsorbers, and because they are membrane-based, they share similar physical and structural features with traditional filter devices. In order to produce membrane-based chromatographic devices with specific properties, the membrane is chemically derivatized with ligands.

Disposable bead-based chromatographic devices tend to contain the same resins and have the same general physical structure as traditional packed-bed columns that are made of glass and stainless steel. Bead-based chromatographic devices are commercially available as prepacked units made of plastic structural parts. They are useful for product campaigning because they can be used more than once. As with traditional packed-bed columns, sterilization methods such as gamma irradiation or moist heat cannot be used with disposable bead-based chromatographic devices; instead, rigorous chemical sanitization is used. As with multiple-use columns, the preservative solution in the SU device must be flushed before the unit is used. Conventional chromatography and SU chromatography both require large amounts of process fluids like buffers (refer to Section 4.8.1, Fluid Management, and **Section 4.17,** Transportation). The specific considerations for chromatography are summarized in **Table 4.13.1-1.**

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Table 4.13.1-1 Specific Considerations for Chromatography

Factor	Consideration
Scaling	Make sure that the method of choice can be scaled to a larger SUS or to an MUS.
Flow rate and capacity	Membrane adsorbers are used in polishing steps because they have a relatively high flow rate and low binding capacity.

4.14 Drug Product Final Filling

The impact of any failures in final fill operations is high, since further processing cannot reduce quality failures at this stage. Common strategies for reducing the risk of failure include performing filling operations in aseptic environments, use of physical barrier systems to separate operators from the process, and validating the sterility of process components (refer to **Section 4.15,** Isolators).

SU components are widely used in product final filling operations. For example, flexible plastic tubing is used between the filling pump head and needle. Another example is a peristaltic filling system, which employs disposable peristaltic tubing inside the pump head. Both types of tubing are normally replaced between batches.

4.14.1 Technological Examples

An SU dosing system comprises a product hold bag, supply tubing, and a filling needle. These components are assembled in a cleanroom and sterilized using gamma irradiation. The complete assembly is removed from its outer pouch and connected to the filling machine immediately before use. The filling machine may be as simple as a single pump or may contain multiple pumps to achieve commercial production speeds.

SU filling systems are controlled volumetrically or gravimetrically.

- **Volumetric dosing system —** This type of system involves a volumetric system that consists of a peristaltic pump and specialized tubing, within which flows the product to be dispensed. Peristaltic pumps work by mechanically compressing the tubing within the pump head to push a specified volume of fluid through the dosing path. Accurate dosing depends on factors such as the mechanical life of the pump tubing. A positive displacement pump-based system relies on the use of either a rotary-piston or rolling-diaphragm pump design. In both cases, fluid is forced out of the pump by the action of the piston. Accurate dosing depends on the precision of the piston stroke length.

- **Gravimetric dosing system —** A gravimetric system is normally used in low-throughput applications. It uses an optical sensor to dose a specified weight of the product based on the calculation of the interior volume of a given length of tubing or glass. The entire product path comprises SU components. The accuracy and precision of the system with aqueous solutions is comparable to that of other dosing systems. The dosing speed and accuracy are directly related to the fluid temperature and viscosity. Dosing time is the speed at which a liquid will flow through the tubing based on gravity; thus, viscosity influences dosing time. Relatively small temperature changes over the course of a filling event can affect product viscosity enough to have a significant effect on the dose volume.

The specific considerations for drug product final filling are summarized in **Table 4.14.1-1.**

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Table 4.14.1-1 Specific Considerations for Drug Product Final Filling

Factor	Consideration
Filling environment	Make sure that the filling system has appropriate connectors and disconnectors. Ascertain that the filling system is capable of being transferred into a restricted-access barrier system or an isolator. Check that the filling system is compatible with vapor-phase hydrogen peroxide (shadowing, material compatibility, and permeability). Alternatively, determine that the filling system can be aseptically transferred into a presanitized isolation system.
Sterility assurance	Maintain the integrity of packaging and assembly during transportation and installation.
Particulates	Ensure that filling system particulate contribution is low enough to meet USP <788> requirements ***(10).***
Endotoxins	Ensure that the filling system endotoxin contribution is low enough to meet USP <85>, EP 2.6.14 requirements, WHO 3.4, and JP XVI 4.01 ***(11–14).***
Physical properties	Check that the system is compatible with the required pressure and flow rate.
SU Part Lifetime	For SUS to be used in campaign manufacturing, qualify the lifetime of SU parts to be used to ensure they perform properly for the entire process
Needle design	Ensure that the needle design is compatible with the requirements for fill accuracy, flow rate, integration to filling machine, and vial opening. Make sure that the needle does not induce foaming or dripping.
Product handling and dispensing	Evaluate shear effects on product quality.
Accuracy and precision	Qualify the design and installation of the system to dispense the specified volume from the beginning to the end of the filling process (may need in-line weight check). Account for physical changes to polymeric parts (such as pump tubing) that can lead to a systematic drift in dosing volume.
Throughput	Peristaltic pump-based and most gravimetric dosing systems have relatively high fill times. The resultant throughput constraint may be reduced by using a more highly manifolded flow path compared to an MUS.

4.15 Isolators

If the canister of a traditional alpha-beta (A-B) port is replaced with an SU bag or connector, SU technology can be used for the aseptic transfer of liquids, powders, stoppers, plungers, tools, materials, and waste into and out of isolators or restricted-access barrier systems (RABS) ***(15).*** To avoid compromising the sterility of the isolator, SU bags are designed to prevent reuse. Categories of an SU product format include:

- **Gamma-sterilizable bag with mating A-B connector —** Manufacturers of stoppers use this as packaging for prewashed, siliconized, and sterile ready-to-use stoppers or plungers.
- **Gamma-sterilizable bag with mating A-B connector and tubing set —** This bag design includes one or more tubing sets for the transfer of liquids into the isolator or RABS. These are normally available as stand-alone components or as part of a presterilized SUS.
- **Autoclavable bag with mating A-B connector —** Manufacturers of stoppers use autoclavable bags as packaging for prewashed, siliconized, and ready-to-sterilize stoppers or plungers. In addition, autoclavable bags are used for in-house stopper processing, tool, and component transfer.
- **A-B connector —** A connector is used for liquid transfer. Connectors can be used with either an SUS or an MUS because they can be sterilized by gamma irradiation or autoclaving.

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The specific considerations for isolators are summarized in **Table 4.15-1.**

Table 4.15-1 Specific Considerations for Isolators

Factor	Consideration
Sterilization method	The SU transfer bag must withstand the chosen method of sterilization (e.g., gamma irradiation or autoclaving).
Port compatibility	Commercially available aseptic transfer ports have complementary bags that must be obtained from the same suppliers.
Film properties	Films with antistatic properties may be used to transfer small parts or powders.
Sleeve or pipe	Transfer bags may include a sleeve or pipe that extends past the port interface in order to avoid contact between the bag and the mechanical clearance in order to prevent possible contamination.
Port size	For complex SU assemblies containing filters, tubing sets, and needle canisters, make sure that the A-B port size is large enough for ease of use.

4.16 Sampling and Laboratory Analysis

Sampling is the process of removing a volume of process fluid from an SU assembly. This sample fluid is then usually submitted for laboratory analysis.

4.16.1 Technological Examples

Techniques for obtaining a sample fluid include:

- Use of a sampling port with or without the aid of a needle to collect the sample
- Use of a Luer-containing sampling bag or via the connection of a Luer-style syringe to a Luer adapter on the bag
- Use of a preassembled manifold of sampling bags that may be filled and detached from the assembly one at a time using crimping tools or a tube sealer
- Use of an SU sample valve integrated with one or more sample bags

The specific considerations for sampling and laboratory analysis are summarized in **Table 4.16.1-1.**

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Table 4.16.1-1 Specific Considerations for Sampling and Laboratory Analysis

Factor	Consideration
Sterility	Determine the degree to which maintenance of the sterility of the contents of the bioprocess bag or that of the sample is required on a case-by-case basis. Although using alcohol to sanitize a port before entry does not ensure sterility, it tends to reduce bioburden load. One should be cautious of the possibility of contaminating the sampled fluid with alcohol.
Number of samples	Use established standards to determine how many samples are required, and design the sampling aspects of the SUS accordingly.
Sample quality (uniformity and compatibility)	Assess the impact of the sample container and the sample storage conditions on the quality of the eventual analytical results. Evaluate the system for positive or negative interference in the analytical method and for sample degradation related to the sampling technology. Make sure that the sampling apparatus is designed to yield a representative sample. Recognize stagnant parts of the tubing in which fluid accumulation may occur, and flush the fluid out before sampling.
Sample handling	For critical process samples kept in long-term storage in bags or other SU sample containers, conduct a test and/or validation to ascertain sample stability and container integrity during handling and storage. The competence of the laboratory staff at sample handling is critical to the success of the process.

4.17 Transportation

Transportation is the movement of process solutions such as buffers, culture media, WFI, and process intermediates either within a facility or between facilities.

4.17.1 Technological Examples

The supply of process solutions in large-volumes bags (≤1000 L) from the point of manufacture to the point of use is a well-established practice involving the following components:

- A bag designed to fit a rigid-wall outer container
- A rigid-wall outer container such as a plastic drum or tote, or a stainless steel bin
- Secondary packaging materials (e.g., dunnage) and lids or other mechanical devices to suppress the fluid wave action in the bioprocess bag

Transportation of process solutions in bags in small volumes (nominally ≤20 L) is an equally well-established process that requires less complex packaging solutions. The exception is the transportation of frozen material, which requires special temperature-resistant materials and processes (e.g., bags, shells and secondary packaging materials, operating procedures, and cold chain logistics; refer to **Section 4.11,** Freezing).

Shipping systems must be qualified for their intended use through proper design and testing in consultation with a packaging engineer. The ISTA and ASTM D4169 are good references for testing standards ***(16,17)*.**

The specific considerations for transportation are summarized in **Table 4.17.1-1.**

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Table 4.17.1-1 Specific Considerations for Transportation

Factor	Consideration
Mode of transport	Compare the use of ground and air shipping over long distances versus wheeled dollies that are pushed from suite to suite.
Environmental controls	This involves temperature and relative humidity regulation and documentation. Check for material compatibility and protection from excessive cold or heat.
Functionality	Ensure forklift access plus the ability to stack outer containers. Make sure that process fluids can be transferred in and out of the bioprocess bag.
Room classification	This involves the deployment of the outer container and bioprocess bag in classified areas, and the transfer of fluids across the same grade level versus transfer across different grade levels (e.g., from a C to B grade).
Bioburden and particulate	Ensure that the bioprocess bag is suitable for room classification and protect it from contamination (e.g., by the progressive removal of various packaging layers, wipe and/or sanitize outer container and polybag, as appropriate) before placing it into a classified area.
Material management	This encompasses the complexity of logistics, freight efficiency, and availability and internal flows of supplies to the point of use while taking into account room classifications, in-suite staging, recycling of packaging materials, removal of SU components, and capacity of the staging area for cleaning of outer containers. Material management capability may vary significantly between an existing facility and a new facility when integrating SUS.

4.18 Sensors

An SU sensor enhances the functionality of an SUS by providing parametric data that allows for manual or automated process control or monitoring. Modern sensors can be directly integrated into SU systems, and traditional sensors equipped with a fluid-path barrier can be used to measure factors such as temperature and pressure. A hybrid sensor uses a combination of a sensor and a transducer. Although sensor and transducer technologies have been available in many industries for a considerable time, the application of SU sensor technology in bioprocessing is relatively recent.

4.18.1 Technological Examples

SU sensor technology is routinely used to measure factors such as temperature, pressure, and conductivity. Recently, advanced sensors for measuring pH, dissolved oxygen, and CO_2, as well as other sensors that employ chemical and/or optical technologies, have become available in an SU format. As the use of SU sensors becomes more widespread, challenges such as their compatibility with sterilization methods and the physical design of the SUS are beginning to emerge.

The dynamic range, accuracy, and robustness of a particular sensor must be taken into account during sensor evaluation for a particular SUS application. Other factors to consider are the need for a common sensing platform across several unit operations (i.e., from small- to large-scale operations); the impact of employing multiple sensing technologies, each with different quantification attributes (such as range, accuracy, and precision); and assessment of sensor calibration requirements and the ease of performing a post-use verification.

The specific considerations for sensors are summarized in **Table 4.18.1-1.**

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Table 4.18.1-1 Specific Considerations for Sensors

Factor	Consideration
Process compatibility	Assess the compatibility of polymeric materials used in the fluid contact of the sensor.
Connectivity	Sensors may come with standard hose barb or TC fittings for integration into a tube fluid path. Designs that are more specialized allow sensors to be integrated into the wall of an SU container (e.g., that of a bioprocess or bioreactor bag).
Sterilization method	Assess the suitability of an SU sensor for gamma irradiation or other means of sterilization.
Sensor performance	Reevaluate sensor parameters (e.g., dynamic range, accuracy, stability, reproducibility, and robustness) after sterilization. Some optical-chemical sensors are sensitive to moisture and drift after gamma irradiation.
Calibration	Determine the calibration and recalibration procedures of each sensor. For example, some chemical-optical sensors may require a four-point calibration while others may be precalibrated by the supplier.

4.19 System Integration

System integration encompasses all the activities and components related to creating a functional SUS or MUS. A schematic representation of a system integration process is shown in **Figure 4.19-1.**

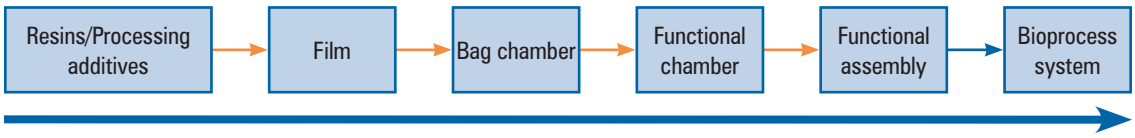

Figure 4.19-1 System Integration

4.20 Supply Chain Integration

The final SUS may involve a large number of supply-chain integration pathways that are currently in use. A company with an SU manufacturing strategy should have highly trained procurement and quality staff with a good understanding of how to manage the several supply chains related to the SUS products that will be used to produce their drug products.

4.20.1 Factors Which Affect the Quality of Supply Chain

The quality of the supply chain has a profound effect on the quality of the SUS. A SUS supplier will typically have a supply chain encompassing aspects of insourcing (i.e., the manufacture or assembly of a part in-house) and outsourcing (i.e., the procurement of a part from a subsupplier). Insourcing and outsourcing both have advantages and disadvantages. A high-quality supply chain ensures that the quality of raw materials and final products is consistent; that the supplier can maintain business viability and continuity; and that the supplier has access to technical experts for product design and support.

Having ready access to technical design and manufacturing experts ensures that both the design and the ongoing manufacture of parts will be executed correctly (refer to Section 6, Business Drivers for the Adoption of Single-Use Systems). If the end user chooses an insourcing model, technical experts would exist in-house, but they may be too few in number or may be sequestered due to business prioritization. With an outsourcing model, a well-chosen business partner would have core competency and experts, but the partner might be less accessible and require confidential disclosure agreements to be executed before substantive discussions can begin. What concerns the end user is the availability of technical experts to assist in troubleshooting any problems with the SUS.

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5.0 Qualification and Verification of Suppliers, Materials, Components, and Completed Assemblies

5.1 Introduction

The qualification and verification of suppliers, materials, components, and completed assemblies require attention to several factors. It is important to use a risk-based and science-based approach to qualify both materials and completed assemblies for SUSs in pharmaceutical development and manufacture ***(18)***. The assessment of risk must be based on the complexity of the system and its intended use (e.g., product contact versus non-product contact, upstream versus downstream use, short-term contact versus long-term storage). See PDA's Technical Report No. 54 on Quality Risk Management for more information ***(7)***. In addition, scientific principles must be applied to 1) identify and monitor extractable materials from films and other components; 2) investigate potential interactions with product CQAs or significant process parameters like cell culture media; and 3) assess the effects of assembly processes such as welding, fusing, mechanical stress, and sterilization (e.g., gamma irradiation) on the materials during manufacturing. In assessing the suitability of the SUS with respect to potential leachables in the final product, extractables may be considered as potential in-process leachables. Finally, the design and functional integrity of the completed SUS must be verified and maintained by appropriately trained end users via suitable processes for construction, packaging, shipping, and deployment.

The supplier of the SUS must provide sufficient data to qualify the materials and components used to assemble the final SUS. In this regard, the end user may use the documents from the supplier to meet most of the verification criteria required to make sure that the SUS is fit for the intended application and purpose. This requires a thorough qualification of the supplier by the end user to ascertain an acceptable supplier quality system, documentation, and an appropriate level of technical capability that extends all the way to the original sources of materials to the supplier. In addition, the suppliers must be able to provide documented evidence of performance claims for off-the-shelf component designs for the SUS. A supplier quality agreement must underpin any custom designs that the end user may need from the supplier.

The first step in a science- and risk-based approach is to identify and assess the risks associated with using an SUS. The end user must identify that the SUS supplier's approach to the design and selection of materials of construction includes components and materials such as:

- Filters
- Sensors
- Polymers
- Lubricants
- Slip agents

Supplier quality, manufacturing and analytical capabilities should also be evaluated. The end user must fully qualify and audit a number of potential suppliers before embarking on extensive studies of specific materials because it is preferable to have more than one supplier for the SUS.

It is important for the end user to have access to controlled documentation of the qualification studies conducted on the materials and components of the assembled SUS. Such documentation may include:

- Batch numbers traceable to the raw materials
- Product shelf life
- Certificate of quality
- Certificate of integrity
- Endotoxin certification (to required and specific endotoxin level)
- Animal-free certification
- Sterilization certification (if sterility is claimed)

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A robust change control communication and assessment process between the end user and the supplier (including sub-suppliers) must be established to monitor any changes in processing conditions, equipment, analytical methods, manufacturing site, resins, films, additives, and more. If the supplier does not have an appropriate assessment process in place, the end user may have to perform independent testing. After performing material and component qualification of the assembled products, a functional verification of the SUS should be performed according to the following:

- **Risk assessment —** To confirm that the assembled products are fit for purpose
- **Process verification and validation —** To confirm that the completed SUS functions within process constraints, including repetitive use or extended storage (if required)
- **Safety —** To make sure that the SUS conforms to safety requirements as a pressure vessel (especially for larger volumes) or for containment in the event of rupture or leak

If the SUS assembly includes SU sensors, calibration and additional verification documentation must be supplied.

5.2 Risks Associated with Using Single-Use Systems

The risks associated with the use of SUSs include that to the patient, the process, and operators. The risk to the process may occur as a result of process failure, but it rarely affects patients because of process discontinuation. Normal safety precautions for handling large volumes of liquids, which may also be hazardous, must be maintained. The main focus of this discussion is the risk to the patient. Begin risk assessment by considering the potential for the SUS to interact with product or process fluids. Some of the important parameters are point of use, duration of contact, process conditions (e.g., temperature, presence of solvents, and the ratio of surface area to volume). The greatest risk is associated with long-term (>30 days) storage containers of certain drug products ***(19).*** The degree of risk is different for upstream applications because of the ability to use further processing steps to remove modified products or leached compounds and the dilutive effect of multiple filtration and washing steps. Ishikawa (fishbone) diagrams can be used to map the process and then identify and characterize potential areas of concern (**Figure 5.2-1**). **Table 5.2-1** identifies the risk complexities of SUSs.

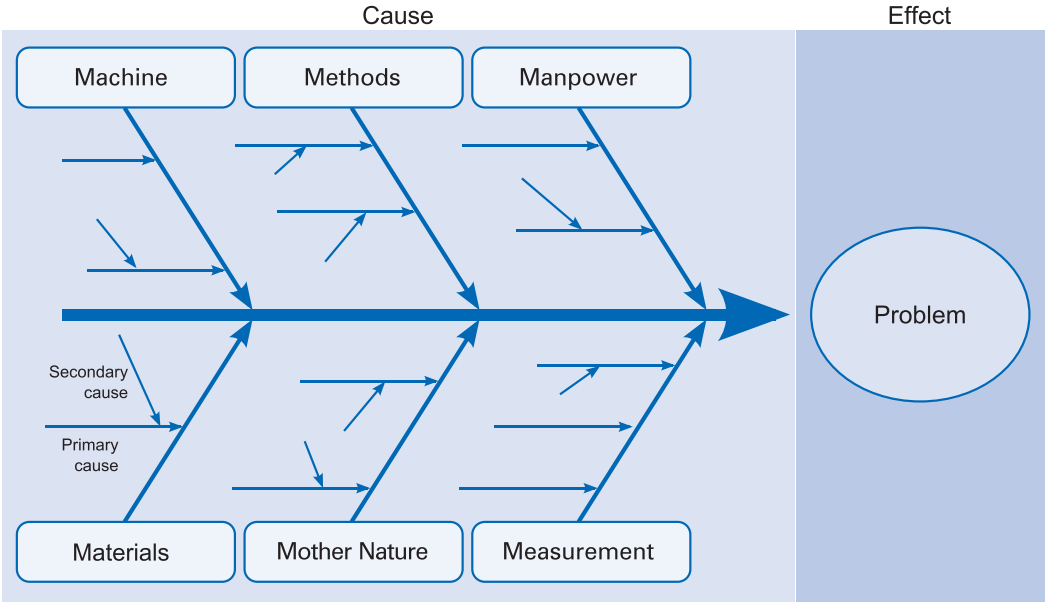

Figure 5.2-1 Example of an Ishikawa (Fishbone) Diagram for Determining Risk Sources

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Table 5.2-1 Risk Complexities of SUS Items and Applications

		System Complexity				
		Low	Moderate	High		
Impact to Process	Low	Buffer/Storage	UF*/DF†/ Concentration	Clarification/ Recovery	Low	Impact to Process
	Moderate	Transport/ Shipping	Connectors/Mixing/ Medium Storage	Cell Culture/ Fermentation	Moderate	
	High	Freeze/Thaw	Purification/ Product Storage	Fill and Finish	High	

*UF – ultrafiltration †DF – diafiltration

For assembly error or individual component failure, complex systems with multiple valves, manifolds, and sensors may present more risks than simple systems (horizontal axis). Applications that are more demanding or close to the final step may pose higher risks (vertical axis). Condensates may get back to the process if fluid pathways are not designed properly. In order of increasing risk are the following components:

1. Nonfluid contact systems

- Vent tubing and filter assemblies
- Samples and sampling systems for repetitive use

Sampling bags are considered a separate category, as the contained product is removed from the process; however, the potential for interactions with bags to interfere with the outcome of assays must be determined.

2. Fluid contact systems

- Short-term contact, simple components (e.g., bags for cleaning materials, buffer storage and concentrates, antifoam, additives, and media)

Mixing systems have additional requirements for establishing functional performance and are available in a range of designs (refer to **Section 4.8.3):**

- Bottom and top mixing
- Rocking mixing
- Cleaning, flushing, and draining
- Long-term storage (storage, holding, and delivery systems with impact on temperature, moisture loss, and gas adsorption)

3. Complex systems

- Bioreactors, including instrumentation (pH, temperature, pressure, conductivity, gases, nutrients with impact on mixing, cell viability, cell integrity, adhesion, and shear)
- Predispensed buffer or media additives (concentrates as well as commercial and proprietary blends of media)

4. Product contact and storage systems

- Tubing, filters, valves, connectors, and manifolds
- Process intermediate storage containers
- Bulk drug-substance storage containers
- Shipping containers
- Disposable filling needles

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5.3 Single-Use System Assembly

SUSs include fittings, tubing, filters, connectors, bag material (film), and outer packaging. SU components are normally received and stored in areas with minimal environmental control. It is critical for the components to remain in their sealed packaging, and they should be handled with gloves to avoid contamination. Transport into the assembly room must follow standard clean-room practices. All the external packaging of the SU components should be wiped with a suitable antimicrobial or cleansing agent before moving them into the assembly room. The assembly rooms are typically ISO 7 or 8 with a combination of particle and microbial sampling. It is important to design manufacturing suites with HEPA-filtered air, low returns, smooth surfaces, and airlocks for entry and exit. Avoid introducing any sharp edges or corners into the assembly room because they may perforate the SUS bags. Personnel must be trained and wear appropriate clothing for an ISO 7 operation and the assembly room must be cleaned regularly. Where SUSs are preassembled, the supplier takes these steps.

5.4 Supplier Qualification of Single-use Systems

Since a typical SUS installation requires the acquisition of several components from various external suppliers, it is important for the end user to qualify suppliers to ensure that they can deliver SU components that meet or exceed regulatory and customer expectations.

5.4.1 Supplier Audits and Technical Diligence

It is important to audit the supplier's quality systems as part of the supplier qualification process to ensure delivery of consistent SU components to the end user. An understanding of the supplier's capabilities may lead to a better understanding of end-user expectations and can help identify any gaps in the capabilities of the supplier. Therefore, the auditing step must encompass manufacturing and technical aspects, and both parties must agree on corrective actions as appropriate. Microbiological control measures for SUSs must be evaluated as part of a supplier audit.

The auditor should review the procedure and records for quarterly dose audits where the SUS is subjected to gamma-irradiation sterilization.

Standard auditing practices, extending to a review of quality systems and complaint handling, should be followed. Unique to SUSs for inclusion in audits would be the supplier's definition of change and how the change-control process functions, including customer notification. The specific particulate control and reduction schemes in place at the supplier should also be evaluated.

Many SUS suppliers may either be ISO 9001 or medical device manufacturers under FDA and ISO registration. ISO 9001 registration means a company operates in a quality management framework that has been audited by a registrar and confirmed as adhering to ISO 9001 guidelines ***(20).*** The manufacturing facilities of registered medical devices are subject to inspection and follow 21 CFR Part 820 (cGMPs for Medical Devices) and ISO 13485 ***(21,22).*** SUSs, including bulk drug containers, are considered process equipment rather than medical devices; therefore, the manufacturing facilities of SUS suppliers are not subject to regulatory registration or inspection. Although SUS manufacturers tend to design their quality systems along GMP guidelines, the auditor should not expect these systems to adhere strictly to cGMPs for drug or medical device manufacturing ***(21,23).*** ISO 9001 is considered a minimum requirement.

Pertinent factors for a technical diligence assessment and an interpretation of its implementation over the clinical development lifecycle are reviewed in greater detail in **Section 7.7.3.**

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5.4.2 Supplier Quality Agreements and Responsibilities

The purpose of a supplier quality agreement is to set quality expectations, identify and develop a mutual understanding of pertinent regulations and guidance documents, and provide advice on managing quality concerns such as audits, complaints, and change control. Change control can be a complex process for an SUS because of the diverse number of suppliers (e.g., compounders, film producers, and bag assemblers) and materials (e.g., resins and additives) involved in the supply chain. The supplier must provide full transparency on the full supply chain of the SUS and guarantee quality oversight of the supply chain. The supplier must have a robust supply chain, which may require improvements, which can be developed with the support of the customer in a partnership mind-set. The extent to which the primary supplier manages the overall supply chain should be assessed, and if necessary, audit and technical diligence activities should be extended to the original supplier of materials and components. It is important to establish a cordial working relationship between the end user and supplier because an SUS supplier will inevitably become an extended part of the end user's manufacturing and quality systems.

5.5 Qualification of Alternative Suppliers

The use of SUS permits considerable flexibility in the choice of materials, designs, and suppliers. This flexibility can be used to minimize risk by ensuring that multiple configurations are viable options for any given requirement. There are multiple combinations of films, bag designs, tubing, filters, connectors, and so on, that are capable of delivering or receiving a specified volume of process fluids to the required standard. Therefore, it is advisable for the end user to qualify an alternative supplier as part of the SUS risk management system *(7)*. Risk- and science-based assessment must be used to determine the level of testing and validation required. In a case involving minimal risk, it may be sufficient to treat an SUS from a secondary supplier as representative of the same class of system or material from the primary supplier. The data generated for the original SUS can be used and compared to that from the secondary supplier. With higher-risk cases (e.g., bioreactor bags or product storage), the secondary supplier's component must be subjected to the same comprehensive qualification process as that of the primary supplier. Note that different suppliers may adopt different approaches to tackle a particular issue and still remain within the bounds of an acceptable supplier quality system. For instance, supplier A may test systems for bioburden from every lot whereas supplier B may test representative samples from different lots as proof of an overall bioburden management system. Both approaches may be acceptable and should be reflected in the user specifications (**Table 5.5.2-1**). An industry effort to standardize SUS testing methods would improve end users' ability to compare alternative SUS suppliers and adequately assess testing differences between suppliers.

5.5.1 Qualification of Alternative Sources

End users typically have two options when it comes to alternative sources: 1) the same SUS supplier with alternative sources of raw materials, or 2) an alternative SUS supplier. Regardless of the choice, the qualification of an alternative source should be part of the drug substance manufacturer's risk-management system. The user should qualify the alternative supplier or raw material source in the same thorough way as the primary supplier or source.

With each option, the risk- and science-based assessment will continue to determine the level of testing and validation required.

The issues under consideration differ somewhat for each option. Suppliers may qualify alternate raw-material sources as part of their risk mitigation strategy at the level of the resins, additives, or the film itself (if sourced separately). For applications with low potential for process or product impact,

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it might be sufficient to review the data from the supplier's qualification. For applications with high potential for impact, the user may need to consider additional testing (e.g. cytotoxicity, stability). For films sourced separately, differences in formulations and manufacturing processes may occur, and the physiochemical properties and extractable profile of the alternate film can be quite different from the original. Such differences, however, should not be the deciding factor for the suitability of this alternative material for the SUS. The suitability of the alternative material should be determined by the end user depending on the intended use of the SUS.

For an alternative supplier, data generated for the original material can be used and compared to the data of the alternative supplier source. In high-risk cases (e.g., bioreactor bags or storage of bulk drug substance), the second supplier's component has to be treated as a specific instance, and full-scale qualification is required.

5.5.2 Interchangeability

It is important for end users to qualify alternative sources of films to safeguard against supply disruptions and possible shortages. In some situations this may be a minimal risk (e.g., sampling bag used for monitoring cell growth or nutrient conditions), so a comparison of extractables and leachables data from different suppliers may be acceptable; other situations may be more demanding (e.g., storage of bulk drug substance). SUS interchangeability can be considered if functionally equivalent substitution provides a contingency or process alternative while ensuring process performance and product quality. When considering SUS interchangeability, it is important to assess the component's intended function, anticipated processing conditions, and fitness for use (**Table 5.5.2-1**). In every case, the approach taken should be supported by a sound risk- and science-based assessment with appropriate evaluation and performance verification for the type of substitution desired.

For example, a bag or assembly used simply as a container for process fluids with no product contact has limited critical attributes—the bag should not leak and there should be an acceptable leachables profile. If the assembly has the additional function of acting as a mixer, the design would need to be assessed for functional performance, in this case minimum mixing conditions (time, speed, and so forth, depending on the mechanism) to achieve homogeneity. Once the design is qualified, this would not need to be established for every lot. For an in-process product container potential interaction with the product would need to be examined and deemed appropriate for the conditions (e.g., liquid, frozen, storage time). The presence of particulates may also need to be assessed, depending on where the alternative component is used in the process. For a complex assembly such as a bioreactor, a full-scale assessment should be conducted on the design for performance parameters and attributes (mixing, pH control, sparging, temperature control, cell growth, viability, titer/yield, etc.). The potential impact of differences in growth conditions on product quality should be included.

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Table 5.5.2-1 Component Interchangeability Evaluation

Change	Description	Risk Assessment Focus*	Performance Verification
Like-for-like	Identical components, dimensions, and materials of construction made by more than one supplier for the same processing conditions	1, 2	Design and user specification monitored through change control
Functionally equivalent	Combinations of materials with the same performance but different material suppliers (e.g., bag/filter/tubing) or processing conditions	1, 2, 3, 4, 6	Monitor performance attributes, leachables, and CQAs
Process or functional modification or improvement	Similar function or performance with different design and materials of construction and may require some process or equipment modification for operation (e.g., virus filter housing changed)	1, 2, 3, 4, 5, 6	Monitor performance attributes, leachables, and CQAs in a scaled-down development study
***Risk Assessment Focus Descriptions**			
1. Design, dimension, performance, and materials claims of the proposed substitute (form, fit, and function) 2. Materials of construction and process contact fluid compatibility, contact area, and time Availability and extent of supplier biosafety (USP <87> and/or <88>, as appropriate for the application) and extractables data *(24,25)*		4. Suitability of supplier extractables data to bracket process fluid and conditions 5. Application and potential for subsequent leachable removal or dilution 6. Product impact assessment	

When considering interchangeability, in most cases other than like-for-like, the end user will need to do some level of operational or extractable/leachable testing to verify fitness for use. Different dimensions or even top or bottom mixing can have product impact. As discussed in Section 5, Business Drivers for the Adoption of Single-use Systems, the cost of quality can be defined as "any cost that would not have been expended if quality were perfect." The additional end-user operational testing of SUS and related quality control activities to ensure process comparability for SU interchangeability could be included in the cost of quality and should then be considered in the overall SUS implementation costs. After all, if a supplier had perfect interchangeability, this end-user cost would be zero.

5.5.3 Using Supplier Quality Documentation

The end user may employ the supplier documentation as part of the SUS document verification process so long as the end user has audited the supplier and collected the requisite evidence of an acceptable supplier quality system, technical capability, and application of Good Engineering Practices (GEP) guidelines (**Table 5.5.3-1**) ***(18)***. If there is not strict adherence to GEP, then there should be assurance that documentation is accurate and suitable to meet the purpose of verification.

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Table 5.5.3-1 Example Supplier Testing and Reference Standards

	Vendor A	Vendor B	Vendor C
Compendial	USP <87>, <88>	USP <88>	USP <87>,<88>
Biocompatibility	ISO 10993-4	ISO 10993-4	ISO 10993-4
Testing	EP 5.2.8	EP 5.2.8	EP 5.2.8
Extractables	USP <661> 11 solvents	EP 3.1.5 USP <661>	Nothing unless the User Contracts
Testing	90 days at 40°C	"Pass" no values	"Pass" no values

5.6 Extractables and Leachables (E&L)

Extractables may be defined as chemical entities that can be extracted from the components of a material by exertion of an exaggerated force (e.g., organic solvent, extreme temperature, ionic strength, pH, contact time). *Leachables* may be defined as chemical entities that can migrate from product-contact or non-product-contact surfaces into a process stream. Leachables may also include chemical entities that can migrate from contact surfaces into the bulk drug substance or final product under specified conditions of production, storage, and use. While leachables are considered a subset of extractables, they can also be derived from product-contact or non-product-contact materials, under typical process conditions and detected in final dosage. The evaluation of extractables is an essential step in the accurate prediction of leachables and in the selection of a suitable SUS dedicated to the manufacture of a given product. When chemical substances are identified that could be cleared or sufficiently diluted by downstream processes so as to be undetected, these can be termed as in-process leachables for the final dosage and may need to be accounted for during risk assessment. Refer to **Table 4.8-1** for a summary of common materials of construction and comments regarding their respective E&L properties.

Drug manufacturers are encouraged to perform a risk-assessment analysis as part of an E&L evaluation within the overall context of enhancing process performance and product safety and efficacy. The accumulation of extractables and leachables in a process fluid is governed by thermodynamic (i.e., the extent to which the materials would migrate) and kinetic (i.e., the rate at which materials would migrate) factors, as well as the amount of time during which such migration would occur. Higher temperatures increase the migration rate of leachables from the bulk of the plastic to the surface in contact with the process stream or formulation. The user can determine the worst-case extent of the extractables by determining the highest temperature at which extractions cease and the entire additive has been removed from the plastic. E&L risk factors can be generally categorized as risk to the patient and risk to the process. In evaluating risks, an extractable or leachable and the level at which it appears will need to be evaluated for impact on process and product quality and for patient safety. The following risk factors may be considered:

- Proximity of the process steps undergoing a change to the final product
- Storage and processing conditions (e.g., duration of exposure, temperature, pressure, pH extremes, buffer extraction propensity)
- The type of process fluid (e.g., purification buffer versus formulated drug substance, presence of solubilizing agents)
- Construction materials (e.g., PVC containers, bags, and tubing are prone to leaching diethylhexyl phthalate, which has adverse effects on the liver, testis, mammary, nerve, immune system, blood, and fatty tissue) ***(26,27)***.
- Potential adverse effects, including synergistic and additive affects
- Drug dose, mode, and frequency of administration
- Therapeutic necessity of the drug (included in drug dose, mode, and frequency)

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The presence of leachables during any stage of the production process or storage may pose a safety risk due to their potential to cause toxicity, carcinogenicity, immunogenicity, and/or endocrine dysregulation ***(28,29)***. In addition, these substances may adversely impact the physicochemical characteristics (e.g., via aggregation, oxidation, degradation, and formation of particulates) of the final protein product. It has been demonstrated that leachables have adverse effects on cell growth and viability ***(30)***. Lastly, the potential for leachables to impact quality control tests should be considered. For these reasons, proper selection and handling of an SUS that supports the production and processing of safe and efficacious therapeutic biologic products is of paramount importance.

Potential sources of extractables and leachables include SU materials used in manufacturing, packaging, storage, filtration, and transfer systems. These include but are not limited to SU bioreactors, bags for intermediate and long-term storage, containers, filters, transfer tubing, valves, fittings, housings, and so on. Extractables and leachables are chemically diverse compounds that may include but are not limited to phthalates (e.g., plasticizers), organic acids, hydrocarbons, cyclic esters derived from urethane adhesives, antioxidants, residual solvents, antistatic and cleaning agents, as well as trace metals and degradation products from irradiation.

5.6.1 Material and Supplier Qualification

It is important for end users to qualify alternative sources of films to safeguard against supply disruptions and possible shortages. There will be situations where this may be a minimal risk (e.g., bags used for sampling), so a comparison of extractables data from different suppliers may be acceptable; other situations may be more demanding (e.g., storage of bulk drug substance). In every case, the approach taken should be supported by a sound risk- and science-based assessment.

Detection, characterization, identification, and quantitation of leachables may present analytical challenges, since these substances represent diverse chemical classes of organic and/or inorganic compounds that coexist in complex mixtures in trace amounts ***(31)***. Therefore, well-designed extractables studies are important to provide an analytical road map and to identify early warning signs regarding leachables released during upstream or downstream operations or in storage. Emerging analytical technologies with appropriate sensitivity and specificity may be considered in addition to the currently available methods during the design of such studies. If an extractable may have an adverse effect on the physicochemical or biological properties of a protein, performing characterization studies in which the extractable is spiked into the product may be of value in assessing the potential risk to product quality.

In order to isolate chemicals from plastic components, the study of extractables begins with the selection of a suitable exaggerated condition that is not stipulated for the normal manufacture, storage, or use of the SUS. It is recommended that the extraction studies be performed with material components using the exaggerated conditions with regards to pH, ionic strength, contact time, temperature, surface area–to–volume ratio, and, if applicable, organic solvents of varying polarity and surfactant properties as the extraction media. If the material component in question is not compatible with organic solvents, aqueous media may be used. If a prerinse step is used in routine manufacturing and use of material component, it should be included in the design of the leachables studies but not in the extractables studies because the latter are intended to represent a worst-case situation. Additional pretreatment of plastics such as gamma irradiation and steam treatment can also impact the E&L profile.

A representative sample of process fluid (e.g., cell culture media and buffer) can be used as the extraction media in addition to others listed above for accurate prediction of extractables. If a representative process fluid cannot be obtained (e.g., due to interference with the analytical method), a similar solution to the original may be an acceptable alternative. The extraction study may be performed using a soaking or a recycling method for a specific contact time, under accelerated temperature conditions and preselected surface area–to–volume ratio. USP chapters <661> and <381> contain recommenda-

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tions for extraction conditions with regards to surface area–to–volume ratio, temperature, and time of extraction ***(32,33).*** The USP <661> monograph also includes general analytical testing for the content of nonvolatile residue, residue on ignition, buffering capacity, and heavy metals. However, because such methods are not equipped to identify specific contaminants, their scope and applicability may not be sufficient in every instance. Several industry organizations, such as BioPhorum Operators Group (BPOG) and the Bio-Process Systems Alliance (BPSA), have developed standardized testing approaches. Please see the latest protocols at their websites as they are periodically updated ***(33-36).***

Testing of E&L substances includes the analysis of nonvolatile, semivolatile, and highly volatile organic compounds as well as analysis of trace light and heavy metals. Less specific tests such as pH, conductivity, chemical identity via Fourier transform infrared (FTIR) spectroscopy, and analysis of total organic carbon (TOC) may be performed. TOC analysis cannot be used with extract solvents that contain carbon. Highly selective analytical techniques for the detection, characterization, and quantitation of these chemical entities include high-performance liquid chromatography coupled with mass spectroscopy (HPLC-MS), gas chromatography coupled with mass spectroscopy (GC-MS), inductively coupled plasma with mass spectroscopy (ICP-MS), proton nuclear magnetic resonance (1H-NMR), and atomic spectroscopy (e.g., atomic absorption and atomic emission spectroscopy). For instance, nonvolatile compounds may be analyzed with HPLC-MS, whereas highly volatile and semivolatile organic compounds may be resolved with GC-MS. Other methods such as ICP-MS may be used for the detection of residual metals.

The extractables study must be complemented with an appropriately designed leachables study. In addition to monitoring the substances that are leaching under the recommended conditions of use and storage, leachable studies must be designed to identify any interactions and the resulting effect of such interactions on the in-process material and/or on the product. In many cases, drug substance and drug product stability studies may be used to support conclusions concerning the effects of these substances on product quality over time. This is particularly important where there is no downstream purification process to eliminate impurities (e.g., drug product formulation buffer storage container).

In addition to determining the chemical identity, quantity, and composition of E&L, and the impact on process fluid or product quality, E&L must be assessed for cytotoxicity and toxicity in animals in order to preserve patient safety and clinical efficacy ***(24,25).*** This information may be obtained from publicly available literature sources such as ICH Q3C (R3) or Product Quality Research Institute Leachables and Extractables Working Group, which stipulates the safety threshold levels for such impurities ***(37,38).*** In the absence of specific toxicity data, the threshold of toxicological concern (TTC) approach may be applied in order to judge the risk associated with a particular leachable.

When an SUS is used to grow cell culture, the end user must determine the effect of cell growth on both individual components and the complete SUS. This is important because a recent study showed that the breakdown products of gamma-irradiated antioxidants can have adverse effects on the viability of the cell culture. Spiking studies showed that the phenomenon can occur at levels well below parts per million in a cell line-specific manner ***(39).***

Cell-based assays can be used to identify the particular combinations of cell lines and plastic extractables that may reduce cell culture viability and growth profiles. End users must evaluate the compatibility of the supplied SUS with their specific application, because certain applications (e.g., media storage, bioreactor bags) have been reported to be sensitive to certain combinations of materials and cell lines ***(30).***

The mechanism of interaction of the SUS and contact solution can include leaching from the plastic into the contact solution or the adsorption of the contained solution or its constituents onto the SUS. These interactions can be modeled through science-based assessments of the physicochemical interactions of the solution and SUS. The use of model solvents for the extractables and leachables is an industry standard that limits the

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cost and time required for their assessment ***(40)***. It is important for the model solvent to be representative of the contact solutions in the end user's process or formulation. Common model solvents include:

- WFI aqueous extraction (neutral pH)
- Aqueous extraction (pH <2)
- Aqueous extraction (pH >9)
- Solvent extraction—ethanol/water mixtures
- Phosphate-buffered saline
- Surfactant systems—anion (neutral pH)

The key aspect of the model solvent selection is to closely replicate the contact solution and minimize interference with qualified analytical methods for detecting the extractable or leachable.

Tables 5.6.1-2 through **5.6.1-4** show examples of testing methods for extractables from filters and containers from different suppliers ***(40)***. The amounts detected would depend on the methods used and materials tested.

Table 5.6.1-2 Example—Quantitation of Extractables from SU Components after 50 kGy Irradiation

Test	Compounds Identified	Concentration
FTIR spectroscopy on ethanol-extracted nonvolatile residue (NVR)	Acrylates (derived from membrane surface modification)	NVR <0.5 mg (ppm)
GC-MS on ethanol extract	2-ethylhexanoic acid	0.56
	1,3-di-tert-butyl-benzene	0.52
	2,4-di-tert-butyl-phenol	0.12
	Lauryl acetate	0.13
	Lauryl acrylate	0.64
GC-MS on derivatized ethanol extract	Ethanedioic acid, dibutyl ester (oxalic)	0.16
	Propanedioic acid, dibutyl ester (malonic)	0.10
	Dodecanoic acid, butyl ester (lauric)	0.23
	Butanedioic acid, butyl ester (succinic)	0.17
	Tetradecanoic acid, butyl ester (myristic)	0.09
	Hexadecanoic acid, butyl ester (palmitic)	0.61
	Octadecanoic acid, butyl ester (stearic)	1.23
	1,4-benzenedicarboxylic acid, bis(2-methylpropyl) ester	2.24
LC-MS on ethanol extract	2-ethylhexanoic acid	144.21
	Lauric acid	200.32
	Myristic acid	228.37
	Palmitic acid	256.42
	Stearic acid	284.48

Note: Identity and quantities of compounds derived via ethanol extracts from a polypropylene filter capsule with modified polyvinylide fluoride membrane.

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Table 5.6.1-3 Identified Extractables from Membrane Filter Cartridges from Several Manufacturers

	Water Extract	Ethanol Extract
Volatiles	2-methoxy-2-propanol	2,3,4-trimethylpentane
Semivolatiles	No peaks	Low MW aliphatic hydrocarbons 1,3-ditertbutylbenzene (1,3-DTBB) 2,4-ditertbutylphenol (2,4-DTBP) 1-tridecanol Lauryl acrylate
Nonvolatiles and heat-sensitive compounds	No peaks	Irgafos®* antioxidant and its degradants
Organic acids	Oxalic acid	Myristic, palmitic, and stearic acid
Inorganic elements	Na, Fe, Zn, and K <10 ppb; Ca 53 ppb	Na, Al, K, and Ca no higher than for the negative control

* Irgafos is a proprietary stabilizer and trademark of Ciba Holding AG in Basel, Switzerland.

Table 5.6.1-4 Identification of Extractables from Polyethylene Biocontainers with Ethyl Vinyl Alcohol Interlayer

	Water Extract	Ethanol Extract
Volatiles	2-methyl-2-propanol Butanal Hexanal 2-octanone	2-methylpentane Hexane Trimethylpentane 3-methylheptane 1-octene; octane
Semivolatiles	No peaks	Low MW oligomers of polyethylene; 1,3-DTBB 2,4-DTBP 2-octanone 1-heptadecanol 1-octadecanol
Organic fatty acids	Oxalic acid 0.07 ppm	Succinic, palmitic, and stearic acid all below 0.06 ppm
Inorganic elements	Na, B, Mg, K, Ca, and Ba all at ppb levels	B 8.3 ppb

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5.6.2 Toxicity of E&L

The toxicity of leachables must be evaluated with regards to the effect on the patient or process. Containers and storage bags are a potential source of leachables that may have adverse effects on the patient. Although some chemicals may be considered unacceptable at almost any concentration in pharmaceutical products (e.g., ICH Q3C class 1 solvents), determination of the toxicity of extractables or leachables must be viewed in the broader context of the following criteria ***(37):***

- Actual concentration of leachables in the final drug product
- Dose size and regimen
- Mode of delivery
- Patient population
- Risk-benefit assessment

Therefore, toxicity cannot be derived from extractable identification and concentration alone, or from the amount of leachables in process fluids or drug product intermediates.

The data for the identification and quantitation of extractables is used to determine the concentration of potential leachables in the final drug product after considering further processing—including downstream purification, diafiltration, dilution, and formulation. If an identified leachable in the final drug product is not covered by existing guidelines and there is limited or no toxicity data, the threshold of toxicological concern (TTC) may be applied ***(41).*** The purpose of the TTC is to define a common acceptable exposure level for unstudied chemicals. Therefore, it is applicable to the risk assessment of leachables derived from SUSs. The concept of TTC may also be applied to extractables that may be potential leachables in the final drug product.

Leachables may affect the function of various components of the SUS. For instance, it has been reported that residual silicone from silicone tubing significantly suppressed the post-use bubble points of sterilizing-grade filters, resulting in false integrity test failures ***(42).*** Therefore, it is important to make sure that SUS test samples removed for sampling do not contain leachables that may interfere with test measurements or present unknown peaks in routine assays.

Leachables may have an adverse effect on the viability of cell cultures without posing a risk to the patient. Recent presentations from different users have reported that certain cell lines may be more sensitive to some materials ***(39,43-46).***

In one case, cell growth media that was warmed overnight in disposable containers produced lower than expected growth in glass bioreactors. In another case, the effect of prefilters on growth performance was studied, and it was noted that extensive flushing was required for optimal performance. Differences were observed in the surfaces and texture of films, and certain cell lines showed different sensitivities to different films ***(39).***

5.6.3 Using Supplier Documentation for Extractables

The end user may rely on extractables data from the supplier as a starting point for general information on E&L components. This kind of data is typically generated using model extraction solvents under exaggerated conditions. Given the specificity of each product and diversity of extractables and leachables, as well as production and final use conditions, it would be prudent for the end user to perform an independent assessment to ascertain that the data are representative of the actual SUS process. Supplier documentation may be sufficient for certain low-risk applications (e.g., short-term exposure, no contact with the product, or position in the process stream); but more detailed studies may be required for high-risk applications (e.g., where long-term contact and extreme conditions exist). This is critical for qualifying alternative sources of materials.

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The end user must review the underlying methods in order to conduct an effective comparison of supplier extractables data for components constructed of like materials. In addition, it is critical for the end user to use the same components and processes for an extractables study as that used in the original supplier's manufacturing process. If the potential for adverse results is high (e.g., a high severity score in a risk assessment), data from extended storage of product and a leachables study may need to be generated and supported by actual functional testing to ensure that there would be no adverse effect on either the product or the process (e.g., stability test of the final product container, cell growth, and the effect of the bioreactor bag on cell viability).

5.6.4 Extractable Testing Standardization

To assist in defining common extractable study requirements for biopharma drug substance requirements, including test conditions, analytical techniques, and format of the results provided to users, some level of standardization is needed in the industry. Such standardization would facilitate SUS implementation ***(36).***

5.7 Bovine Spongiform Encephalopathy (BSE)/Transmissible Spongiform Encephalopathy (TSE) Concerns

SUSs have low-risk BSE/TSE concerns and they do not contain high-risk animal-derived components. Certain mold release and slip agents and plastics resins (e.g., polypropylene or polycarbonate) may contain trace levels of additives (e.g., calcium stearate), which are manufactured from bovine tallow and have low BSE/TSE risk. Tallow is a lubricant sometimes used in pelletizing operations for bulk resin or as a mold release agent for the manufacture of SU components. Tallow is produced by treating beef fat with steam to create a mixture of fats. The fat (e.g., glyceryl tristearate) is then boiled in a sodium hydroxide solution, resulting in a mixture of sodium stearate and glycerin.

Tallow-derived additives are not considered as specified BSE risk materials according to the current revision of the U.S. Code of Federal Regulations, Title 21 Part 189.5 ***(47).*** Furthermore, the CPMP's "Note for Minimizing the Risk of Transmitting Animal Spongiform Encephalopathies via Human and Veterinary Medicinal Products" (EMA 410/01 rev 3), gives specific consideration to tallow derivatives ***(48).*** It states that they are unlikely to be infectious due to the rigorous processing steps (e.g., transesterification, or hydrolysis, at not less than 200°C under pressure for not less than twenty minutes) used during their manufacture. A BSE/TSE risk assessment should be completed and periodically updated to reflect whether an SUS supplier is using components in the fluid pathway that were fabricated from plastic resins containing bovine-derived additives at trace levels.

5.8 The Quality Systems of the End User and Supplier

It is important for the end user to have instituted a robust quality system to receive materials, quarantine unreleased and rejected materials, inspect those materials, and release them to production. Since the testing and release of an SUS can vary widely from end user to end user, a complete and careful documentation in the form of a purchase specification and inspection plan should accompany any off-the-shelf or customized SUS requisition. A supplier quality agreement is another useful way to manage the requirements and expectations of both supplier and end user.

5.9 Quality Management for Single-Use System Implementation

5.9.1 Determination of Expiration Date and Shelf Life

The SUS supplier is expected to provide estimates for expiration dates and the shelf life of the SUS according to validation of maintenance of sterility tests and key functional properties. The expiration

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date is normally set post-sterilization. Any extension of the shelf life of the SUS must be supported by pertinent data. Real-time aging studies provide the best data to ensure that both the materials and the functional integrity of a sterile barrier system do not degrade with time. Accelerated studies may be a suitable substitute if real-time aging studies cannot be performed. The ASTM F1980–07(2007) Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices is a widely used test for accelerated aging of the plastic components of SUSs ***(49).***

The term Ready-To-Use is a marketing term often used to describe the benefits of single-use technology or SUS. This designation has no regulatory or scientific basis supporting suitability for use, and end users are responsible for evaluating and determining whether appropriate quality requirements have been met for their application.

5.9.2 Dealing with Particulates

The level of particulates found in the SUS depends on the application (fluid path contents) of the system. Particles embedded in the bulk plastic as well as the presence of loose particles or fibers in an SUS are undesirable and their limits should be clearly specified in purchase agreements. If the presence of particulates in the fluid path would not have adverse effects on the product or process, then components used upstream of final filters may not require stipulated particulate levels. However, SUS components employed downstream of final filters or without particulate filtration may require stringent limits for particulates to ensure that the final product meets pharmacopeia standards. An acceptable solution is to qualify fluid path rinse volumes to meet USP <788> and USP <790> or equivalent pharmacopeia standards for large-volume injectables ***(10,50).*** In addition, the frequency of sampling and analysis may be extended from lot samples to periodic audits of model SUSs—this would be akin to the bioburden audits that are conducted for the validation of sterilization by irradiation. Applicable SUS model system design and particulate audit practices should be established between the supplier and the end user.

5.9.3 Sanitation and Sterilization

The use of an SUS in biopharmaceutical manufacturing can vary so widely that in one instance there may be no need for a specific bioburden control (e.g., as in the mixing of nonsterile buffer formulations or other raw materials) whereas in another, stringent restrictions on the incidence of a bioburden are called for (e.g., validated sterility for final formulation and filling of sterile drug product). It is critical for the end user to review processes, fluids, contact, and storage conditions for microbial quality to determine the appropriate measures to be taken.

5.9.4 Bioburden Control

The molding and extrusion of SUS components at high temperatures plus their assembly into a functional SUS in a clean and controlled environment confers advantages such as intrinsically low incidence of bioburden on the fluid paths of the SUS. Although this inherently low degree of bioburden may be adequate for nonsterile processes, many SU components or systems may be autoclaved or subjected to gamma irradiation for additional control of bioburden. Doses of irradiation as low as 8 to 10 kGy are enough to reduce intrinsic bioburden to zero; however, in practice, doses >25 kGy are used to achieve a sterility assurance level (SAL) of 10^{-6}. The increased irradiation dosage has the advantage of attaining a bioburden level comparable to that of validated sterility without conducting the necessary validation protocols or periodic sterilization audits that would otherwise be required for a statistical claim of sterility of 10^{-6} ***(51).*** This may be suitable for bacterial fermentation applications and aseptic processes such as the preparation and storage of downstream chromatography and ultrafiltration buffers and intermediates.

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5.10 Sterilization

5.10.1 Irradiation Sterilization

SUSs are especially suitable to sterilization by irradiation because of the relative transparency of plastics to ionizing radiation, which allows complex systems to be sterilized without the risk of spreading the sterilizing agent throughout the fluid path. Irradiation may be used to sterilize plastics or SUS assemblies that may be unsuitable for steam sterilization. The practice of sterilizing via irradiation is well established in the medical device industry with transferable standards that can be applied to SU process equipment ***(51).*** Although no specific industry standard has been developed for the sterilization of SUSs, the industry and regulatory communities have adopted the ANSI/AAMI/ISO 11137 standard, *Sterilization of Health Care Products—Radiation*, and the closely related AAMI Technical Information Report TIR 33:2005, *Substantiation of a Selected Sterilization Dose—Method VD max* ***(51,52).*** Adaptation and implementation of these standards to the implementation of SUS have been discussed in the literature ***(53).*** The most common alternative methods mentioned in the standards are the dose substantiation methods (i.e., the verification dose max, or "VD max," methods).

In practice, a typical audit involves the selection of a single-use product (SUP) as representative of the SUS. A supplier quality audit must consider how the product family and the representative product design were derived. In cases where testing the representative product would be impractical (e.g., due to size and complexity), a portion of the sample item may be used with mathematical scaling between the actual product and the sample item. A supplier quality audit must include an understanding of the mathematical scaling technique. Since the standards require that a supplier must perform quarterly dose audits for each product family, the end-user quality audit of the SU supplier must ascertain that there is a procedure in place for the supplier to conduct quarterly dose auditing. Validation normally involves the determination of an average bioburden level in a representative master system, assessment of radiation resistance by sterility testing of the fluid path of the master system after a low "verification" dose (about 8 to 10 kGy), and calculation of a suitable dose to provide a sterility assurance level or a probability of nonsterility of 10^{-6} for the model product and its variants. The supplier and the end user should review the applicability of validation via master system sterilization to the end user's SUS configuration.

5.10.2 Sterilization with Moist Heat

Several methods of sterilization are acceptable for use with SUS. Each is appropriate for different applications and should be evaluated for suitability.

- **Autoclaving —** This technique is often used to sterilize an SUS or SU components assembled by end users ***(54).*** After that, most end users may find it more convenient to source preassembled SUSs that are sterilized by gamma irradiation.

- **Sterilize-in-place (SIP) —** Steam-to-connector configuration is one example of SIP. The physical limitations of plastics limit the ability of an SUS to be sterilized by steam pressure in situ. Some SU connectors are designed to be physically attached to a stainless multiple-use port so that the tank connection can be sterilized with steam. Steam-to or steam-through connectors tend to be preconnected to tubing with an SUS at one end and are presterilized by autoclaving or irradiation. A valve or barrier mechanism is used to isolate the sterile system. The steam-to or steam-through connector is then connected to a tank, and the interface is sterilized with steam ***(55).***

- **Gas sterilization —** Although it is not a common practice, an SUS may be sterilized using a gas such as ethylene oxide. Gas sterilization is not as popular as other methods of sterilization for two reasons: 1) ethylene chlorohydrin and ethylene glycol reaction by-products can form; and 2) some ethylene oxide may be retained in the filter matrix, when a filter is present. Some of the factors that

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affect the use of gas sterilization are preconditioning, concentration, relative humidity, temperature, and time. The monitoring and validation of gas sterilization is conducted by destroying a suitable biological indicator such as the spores of *Bacillus subtilis*. To achieve successful sterilization using ethylene oxide, the SUS should be dry and wrapped in a way that allows for the influx of water vapor and gas into the fluid path. The appropriate concentration of gas and other parameters must be established during development and validation processes. It is important to rid the SUS of residual gas and other undesirable by-products after gas sterilization.

- **Resterilization —** SU components are not amenable to multiple autoclave cycles. Gamma irradiation is a cumulative process and most SU components cannot be re-irradiated without causing significant degradation of physical integrity, creation of extractables, and reduction of shelf life. In addition, irradiation may reduce the resistance of the SUS to subsequent steam exposure, which may increase the formation of extractables and leachables. In general, an SUS is not designed to be resterilized.

5.10.3 Sensor Technology

The pace of technological evolution of SU sensors is so rapid that the end user may decide to use several strategies at the same time. The end user may:

- Choose a novel method
- Adapt an existing technology for SU
- Apply traditional multiple-use sensors in an SUS

The end user must evaluate the technology for suitability according to the accuracy, repeatability, range, and calibration requirements. Other questions to consider are:

- How compatible is the SU sensor with the sterilization technique?
- What is the shelf life of the sensor?
- What is the in-service life of the sensor?
- How appropriate is the sensor for the SUS in question?

Early examples of SU sensors for pH and dissolved oxygen in an SU bioreactor (SUB) were based on optical differences observed on a chemical patch immersed in the process fluid. Optical sensors are calibrated with a batch-specific calibration standard instead of a zero-and-span calibration technique. The end user must be aware that optical sensors may be sensitive to photo bleaching and drift. In general, steam sterilization cycles may be more damaging to a reusable sensor than a single gamma sterilization cycle. Some SU sensors may be calibrated using a traditional two-point calibration technique prior to use without any adverse effects on sterility. Thus, SU sensors may be inherently more reliable than reusable sensors. The end user must pay attention to the advantages and limitations of a chosen SU sensor, including the specific requirements and circumstances of application.

5.10.4 Qualification and Validation

The end user is responsible for process validation, and the method for validating an SUS must be determined using a risk-based approach, which considers the effective integration of the SUS design into the end user's quality systems ***(8)***. Although purchasing a preassembled, sterilized SUS allows the end user to take advantage of the qualification data generated by the SUS supplier, it is incumbent on the end user to understand the supplier's qualification approach and how it applies to the end user's specific application. The scope of process validation at the manufacturing site must encompass the materials for construction, relevant testing procedures (e.g., for heavy metals, particulates, pyrogenic-

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ity, cytotoxicity, leachables, and extractables), and information on any sterilization method used. Any deficiencies discovered during the validation process may be mitigated by applying additional targeted checks or other controls. Additional information may include pressure limitations, correct procedures for using aseptic connectors and tubing fusers, and how many times pump tubing can be reused. The validation study should ensure and document process-specific fitness for use of the SUS.

Pertinent validation concerns for an SUS include the following:

- SU products, and qualified alternative designs, used in bulk drug substance and sterile filling should be incorporated into the media fill program.
- SU filtration systems may need to be designed with alternative receiving containers to accommodate pre-use wetting and integrity testing.
- Since it may not be feasible to test the SUS for leaks, the manufacturing, shipping, handling, and operation procedures should be assessed for their assurance of avoiding issues (punctures, tears, stresses) that may lead to leaks.
- While extractables data may be suitable for predicting worst-case in-process leachables, actual testing of leachables may be necessary for final drug formulation in bulk storage containers.
- When proprietary sterile connections are incorporated into the SUS, the end user must review supplier validation data to determine whether any additional process-specific validation may be required.
- Proper handling and manipulation of the SUS by trained operators should be qualified.

5.11 Integrity

5.11.1 Testing an SUS for Leaks

The requirements for testing an SUS will depend on its intended use. The retention of structural integrity (i.e., containers do not leak) is the main concern and is a key concern in process-contact applications. For low-risk applications (e.g., sample and buffer bags), an understanding of supplier testing, performance, and training in appropriate handling procedures may be sufficient.

SUSs are increasingly used for the production of vaccines, and the egress of a live virus could have serious consequences;, therefore, specific testing of individual systems is recommended.

When an SUS is used for holding media, the potential for microbial ingress that may compromise sterility becomes a significant concern. Attempts have been made to assess the maximum size of a defect that will compromise the effectiveness of a flexible bag ***(56)***. This is a complex undertaking because it depends on factors such as the surface tension of the liquid in the bag, time, temperature, and the method of measurement. In addition, the permeability of individual components (e.g., tubing and films) may differ according to the test agent (e.g., gas or dye) used.

The structural integrity of the SUS should be carefully evaluated in a collaborative manner by the end user and the supplier. Post-shipment/transport testing to evaluate the method of packaging the SUS will need to be incorporated. To test for the integrity of the fluid path during qualification—including tubing manifolds and connectors—a suggested method is to internally pressurize tubing with air and hold it under water. Provided that testing will not compromise the sterility or cleanliness of the equipment, manifolds can be inspected by manipulation to check for air bubbles at both tubing connector engagements and along the entire length of tubing. For upstream applications, the user should confirm the supplier's claims during verification and accept materials based on their documentation

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thereafter. A special case may need to be made for customized SUS. Demands are generally stricter for product containers; therefore, end users may wish to ask the suppliers what testing is feasible for larger containers at their sites versus the end user's site (after installation).

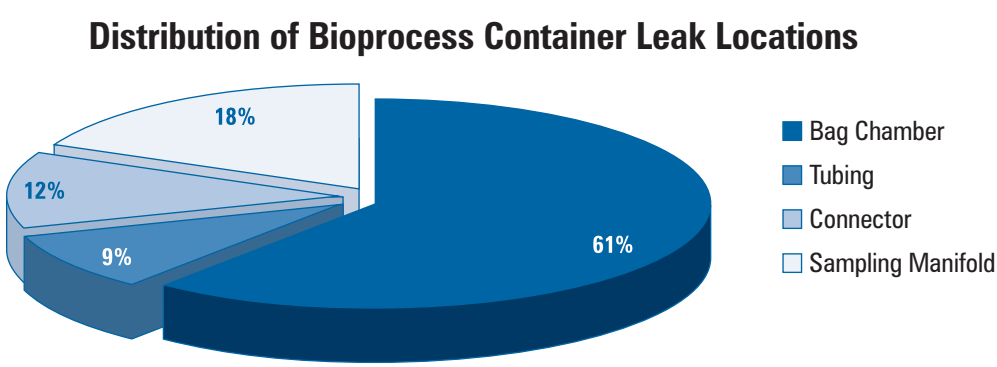
To test a bag container or a SUS for leakage, a suggested method (depending on the size of the SUS) is to fill the plastic bag or SUS with water inside the bin, drum, or tank and check for leaks at maximum volume. If the location of the leak is not apparent, fill the plastic container with water outside of the bin, drum, or tank on a flat, absorbent surface such as cardboard or paper. Locate the leak by identifying wet spots on the absorbent surface corresponding to a position on the container. Using the mapped location of the leak on the bag, examine the bin, drum, or tank for sharp points that may have caused the leak. Examine the leak under magnification to determine whether the leakage was caused by puncture, tear, creasing, pinhole, or another means. Track leak locations to help with root cause analysis of failure modes (e.g. handling, structural/design considerations, and material failure).

If the end user determines that a leak was caused by a manufacturing defect, the failed sample should be returned to the supplier for quality investigation and corrective action. The returned sample should include full information regarding the conditions of use and location of the leak. Suppliers should maintain records of system failures and their root causes for further analysis.

5.11.2 Leak Prevention in an SUS

Leaks in fluid management systems can lead to contamination, personnel safety risk, and product loss. A thorough investigation must be conducted to determine the cause, identify the source of the leak, and monitor corrective and preventive actions. The investigation strategy should include a leak-investigation procedure for generating standard reports. Such a report must include the location of the leak on the assembly, the surrounding temperature conditions, composition of the fluids, any adverse mechanical events, and an analysis of the failure mode. This information is particularly useful for trend analysis. Consider the example depicted in **Figure 5.11.2-1** in which a bioprocess container (BC) was investigated for leaks. Individual reports for each leak were analyzed and the location of each leak was mapped and identified according to type and cause ***(57)*.**

Figure 5.11.2-1 Investigation of a Bioprocess Container for Leaks



The leaks were mapped onto a deconstructed view of the BC (**Figure 5.11.2-2**). The map indicated the leaks were more likely to occur at the bottom and top of the bag (i.e., where the BC was handled by operators). Subsequently, the operators were trained to avoid damage due to mishandling, and this led to a significant decline in leak rates ***(57)*.**

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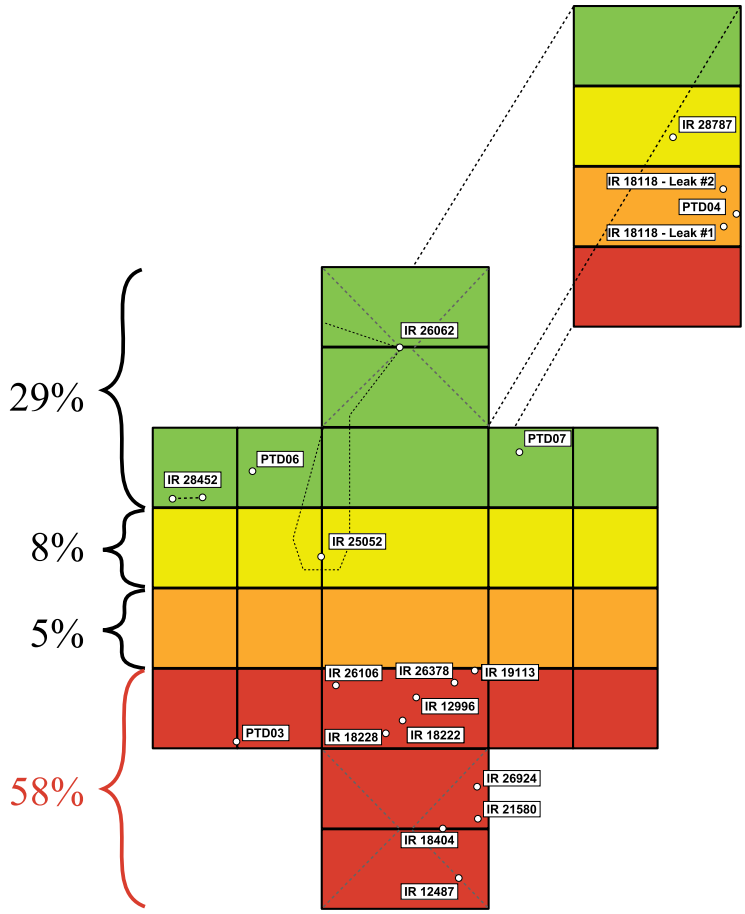

Figure 5.11.2-2 Identifying the Location of Leaks on a Bioprocessing Container

This, in turn, led to the following recommendations.

For end users

- Train operators to be aware of the root causes of leaks.
- Remove sharp objects from the vicinity of a BC.
- Improve lifting equipment for handling the BC.
- Make sure that the BC fits properly into the container in order to minimize handling.

Partnering with suppliers

- Share leak investigation reports when returning defective products.
- Use supplier-developed methods to improve reliability.
- Identify and implement a more robust BC.

The summation of activities involving the hands-on steps required to successfully assemble and make a system ready for use for a specific SUS application is called deployment. This includes a defined setup and deployment stage, with standardized activities for each part, and is a recognized success factor for SUS implementation. Deployment should be a routine part of the workflow and planning for each SUS application.

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5.12 Campaigning

Although they are disposable systems designed to be used once and discarded, there are a few exceptional circumstances in which an SUS may be reused. Examples would be the use of a container as a mixing vessel or reservoir for buffer concentrates, in which there are no broken connections and the contents of the container are replenished. Extending the period of time or number of cycles an SUS is operating in a closed process without breaking the sterile barrier processes is called campaigning. The end user is responsible for evaluating and determining whether appropriate quality requirements are met for their application.

The end user must also consider whether the reuse of an SUS would compromise the definition of a batch. Conversely, strict adherence to single usage of a prepacked, presanitized chromatography column may be prohibitively expensive; therefore, it would be economical for the end user to run the column for multiple cycles (e.g., for preparing clinical material). The column may then be held in storage conditions and tested before being reused. The effectiveness of the storage conditions and tests should be verified as for a brand-new column.

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6.0 Business Drivers for the Adoption of Single-Use Systems

This section provides a detailed description of the key business drivers that inform the evaluation and implementation of single-use versus traditional MUSs. The business drivers for the introduction of an SUS are many, varied, and unique to each company's individual situation and business model. Several factors must be considered for a successful analysis. These factors include:

- Supplier-demonstrated technical and quality capabilities
- Product sustainability
- Guarantee of business continuity
- The business model of the company considering an SUS
- Process design
- The product and its potential market volume
- Available facilities
- Anticipated location

Each company will approach SUS implementation with a different set of objectives developed from business drivers that are unique to a specific business model.

SUS implementation confers advantages such as reduced capital investment, flexible operation, faster changeover, and the application of lean manufacturing principles. Since a company's individual situation and business model will affect the role of pertinent business drivers, it is critical to start the evaluation with an understanding of the business in question. For example, the main concern of a contract manufacturing company (CMO) may be the elimination of cross-contamination in a multiple-client, multiple-product facility. A stand-alone manufacturer with a facility dedicated to the production of a single product may consider a hybrid approach of an SUS and MUS to reduce SU variable costs yet allow for the expansion of production capacity at peak periods. An effective implementation of an SUS would help the company to meet important business goals in addition to cost savings.

6.1 Evaluation of Business Drivers

The decision to implement an SUS tends to be driven by economic incentives such as 1) potential elimination of cleaning and sterilization processes; 2) reduction of dedicated equipment or changeover time; and 3) increased flexibility and capacity of the manufacturing facility. The end user must consider both tangible cost savings and less quantitative benefits such as low risks of cross contamination. A cost-of-goods-manufactured (COGM) analysis can be used to support claims of economic savings. Other tools, such as risk assessment, value-stream mapping, CO_2 footprint, green manufacturing, and value-added analysis, can provide invaluable data to inform a business decision on whether to implement SUS or not.

6.2 Lifecycle Approach

The importance of specific business drivers will depend on the needs of each individual company. The scope of the project may be as narrow as using an SUS to boost the capacity of a single process step or to eliminate cleaning steps. Such a tactical comparison depends on technological, process compatibility, and economic factors. Conversely, the project scope could be a corporate-wide initiative to evaluate the feasibility of SUS implementation for an entire production process or part of an overall lifecycle evaluation for a product or facility. A lifecycle approach involves an evaluation of the costs incurred throughout the lifetime of the project in order to provide a complete economic basis for making decisions. Costs for evaluation may include the initial capital investments, production and quality overheads, and decommissioning. The lifecycle approach incorporates time and risk factors as relevant strategies are applied, and this leads to a more comprehensive economic evaluation.

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For instance, the forecast for a new product may include an initial launch phase with a relatively uncertain production demand followed by a full-scale production phase with options for capacity expansion or additional products. In such a situation, operational flexibility becomes important to the business case because of the uncertainty in product demand. Such flexibility comes with an increase in capital investment for traditional MUSs, whereas the implementation of an SUS tends to reduce the cost of operational flexibility because it is less reliant on investment for utilities and equipment installation. An SUS is less costly to modify or expand to a new facility when a change in product lifecycle occurs.

6.3 Opportunity Cost

Opportunity cost is defined as the value of the next-best choice available when choosing between several mutually exclusive options. These costs are often excluded from estimates of fixed operation costs because they can be difficult for comparative analyses in the overall decision process. For example, deciding to expand a facility may result in forgoing the opportunity to invest in other projects, contract batch production at a CMO, or maintain cash reserves. Opportunities can be difficult to compare; hence, opportunity cost calculations between competing projects may help identify the best path forward.

6.4 Cost of Quality

The cost of quality can be defined as any cost that would not have been expended if quality were perfect. The inherent advantages of an SUS may lead to reductions in the cost of quality for the end user. In order for an SUS to be used in a manufacturing process, the end user must be able to demonstrate that it is fit for purpose and meets applicable quality and regulatory requirements. SUS suppliers may take advantage of automation, economies of scale, and lean manufacturing to improve consistency and reduce the cost of manufacturing. This arrangement makes the end user dependent on the supplier's quality system. Therefore, it is critical for the quality system of the supplier to be robust enough to manage subsuppliers (e.g., resin and component suppliers). In addition, the quality system of the supplier must be able to monitor and detect the unique failure modes associated with SUS (e.g., bag breakage, variations in the profile of extractables and leachables, and effects on cell growth). It is the responsibility of the end user to ensure that the SUS meets applicable quality standards. Any tests conducted by the end user to ascertain process compatibility due to changes made by the supplier may be included in cost-of-quality estimates.

6.5 Quantitative Evaluation of Business Drivers

A quantitative evaluation of business drivers for an SUS project may be based on determining the average cost of a batch in order to examine the feasibility of the SUS project. An overall economic model for a given project may be obtained by combining parameters from a series of submodels. A list of submodels and their interactions for setting up the overall economic model is shown in **Figure 6.5-1.** The actual split between submodels and the amount of modeling work necessary depends on the project scope and level of accuracy necessary for the study. A more comprehensive comparative study may include all submodels, but in some cases, depending on the level of accuracy needed, an approach including only the Process, Capital Cost, and Variable Cost models may be sufficient for giving an estimate of the feasibility of the project.

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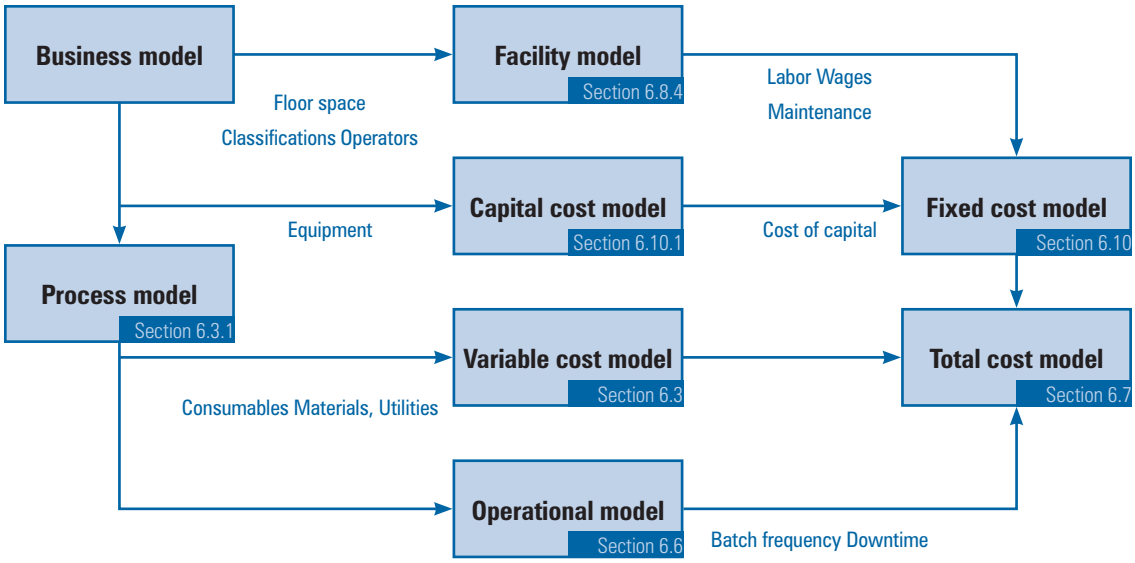

Figure 6.5-1 Cost Comparison Studies Reference Model

As shown in **Figure 6.5-1,** a process model is created first, followed by the generation of submodels according to the project scope. The resulting submodels for facility, investment, variable, and fixed costs may then be combined to form the total cost model. The factors to consider for the process model are summarized in **Table 6.5-1.**

Table 6.5-1 Factors That Affect the Process Model

Process Model Factors	Considerations for SUS Implementation
Unit operation	Processes involving large volumes may be done with only one reusable vessel, whereas several vessels may be required for an SUS. The effect on the function, layout, and classification of the facility should be taken into account.
Storage of solution	The storage and distribution of solutions may be performed with mobile tanks or bags or in fixed installations. The concentrations of the solutions may be at use concentration or highly concentrated for subsequent dilution prior to use.
Support system requirements	This includes utility demand, washing and sterilization requirements, and the need for water and steam.
Batch duration and frequency	The time required for performing the process steps in question needs to be determined.

6.5.1 Process Assessment

The process model is an assessment of the process with respect to an SUS. The process model forms the basis for any cost comparison exercise by defining process setup and deployment for either an SUS or an MUS. Data from the process model feeds into the facility, capital cost, and variable cost model to provide the basis for the fixed-cost model.

6.5.2 Batch Frequency

With a higher batch frequency, variable costs arising from using more consumables over time may become prohibitive for an SUS design. If the batch frequency is low, varies, or becomes difficult to predict, then an SUS may improve the business case by keeping fixed costs low. For batch sizes exceeding feasible SUS volumes, there may be adverse economic effects from frequent use of the SUS.

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6.6 Operational Model

Operational data about the facility is captured by the operation model and then added to the data from the process model in order to estimate utilization and batch output from the facility. When this information is combined with other cost models, the average batch cost for the various scenarios can then be calculated. The main factors that affect the selection of operation model are summarized in **Table 6.6-1.**

Table 6.6-1 Factors That Affect the Operational Model

Operational Model Factors	Considerations
Production (weeks/year)	The number of weeks scheduled for production minus holidays and scheduled maintenance.
Downtime duration	The time required for cleaning, dismantling, sterilizing, calibrating, testing, setting up, and deploying the next batch to be processed.
Changeover duration	The time required for switching between campaigns or products. This may include the dismantling and cleaning of the production train (e.g., reusable storage containers, equipment, or instrumentation screens), and clearance of items used in the last batch of production.
Batches per year	An SUS may produce more batches per year compared to an MUS because of the reduction in changeover time. The average cost of a batch does not include variable costs only, but it is influenced by fixed costs and the number of batches processed per year.
Scrap ratio	This refers to the number of batches lost to contamination, loss of integrity, and other problems. It also refers to the cost of materials and labor stemming from the loss of sterility or the integrity of the system.

6.6.1 Single- or Multiple-Product Facility

The business case may be improved with an SUS because it may reduce downtime and changeover time. In a single-product facility, capacity may be increased via a reduction in the time required for equipment cleaning. In a multiple-product facility, improved capacity may be achieved through a decrease in both changeover and equipment-cleaning time. It is important to factor in differences in facility throughput when technologies are compared.

6.7 Total Cost Model

A project may involve either the construction of a new facility or the retrofitting of an existing one. The main points of consideration for both cases are summarized in **Table 6.7-1** *(58)*.

Table 6.7-1 Comparative Evaluation of a New Versus Retrofitted Facility During SUS Implementation

New Facility	Retrofitting of an Existing Facility
Emphasis on facility model for capital cost estimate	Ability to limit facility model by constraints already given
Ability to eliminate or reduce functions or utilities by using SUS	Possibility to overcome capacity limitations by using an SUS
Design for SUS technology	Interfacing and connecting SUS technology to existing areas and systems
Reduction in total project time as a result of SUS, time to market	Possible reduction of interruptions in productivity

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6.7.1 Shortcut Cost Model Based on Cost of Goods

In a preliminary evaluation in which the speed of comparison is more important than accuracy, a yes-or-no type of feasibility study incorporating a minimum of factors would be sufficient. This approach may not be rigorous enough for a final binding decision where funds would be committed. In such a case, elements of the more comprehensive approach must be applied.

The minimum factors to consider in the shortcut model are:

- Process equipment
- Utility equipment
- SU consumables
- Cost of capital assumptions

6.7.2 Example of a Shortcut Cost Model

Use the following lists and the preceding paragraphs to estimate the following for each scenario:

Process and Operational Basis

- Process equipment involved
- Number of batches produced annually

Investment Costs

- Equipment costs
- Area needed
- Area costs
- Total investment costs

Cost of Capital

- Annual depreciation on investment
- Annual interest rate paid on capital investment
- Total cost of capital

Operating Costs

- Consumables costs
- Materials costs
- Labor costs

Evaluate the Following According to the Business Model:

- Investment costs
- Operating costs
- Number of batches produced
- COGM = (total operating costs + total cost of capital)/number of batches produced
- Payback time = difference in investment cost/difference in COGM

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6.7.3 Comprehensive Cost-of-Goods Model

A more comprehensive cost model may include factors selected from the preceding sections (**Tables 6.5-1, 6.6-1, 6.7-1** and **Section 6.7.2-1**). The structure of the model must be flexible enough so that it can be improved for accuracy with additional data. A sensitivity analysis is the most effective way to use an economic model because it incorporates a range of input factors for arriving at a good understanding of the potential risks involved.

6.8 Investment Costs

6.8.1 Process Equipment

The first step in evaluating an SU project is normally an equipment-to-equipment comparison of SUS versus MUS direct investment cost. Reduced demand for cleaning and sterilization allow for simplifications in piping and instrumentation, which leads to a reduction in the investment cost for SUSs. While this will give an estimate of whether the technology is economically feasible for the project, **Table 6.8.1-1** shows additional factors for evaluating investment costs that should be considered for an SUS project ***(59)***.

Table 6.8.1-1 Contributory Factors to Investment Costs

Investment Costs	Considerations
Process equipment	This refers to the process equipment (e.g., bioreactors, centrifuges, tanks, filters, columns) and support equipment (e.g., welders, sealers, process controller, pumps, and bag supports for SUS) required to run the process.
Utility equipment	This refers to the secondary equipment or systems (e.g., clean steam, CIP, and COP) required to run the process.
Installation	Installations for CIP/SIP and traditional process tanks should be considered.
Engineering design	A SUS may either be customized or purchased with the off-the-shelf configuration.
Commissioning	Implementation of an SUS may eliminate I/O points associated with CIP and SIP.
Validation quality shop floor	A SUS may require investigations into potential leachables and other studies, and reusable equipment would require CIP and SIP validation. Consider the cost of doing facility acceptance test (FAT) and site acceptance test (SAT) as well as verification activities.
Facility floor space	This refers to the effect on both footprint and cleanliness classification for operation and SUS storage.
Operators	The number of operators affects training considerations as well as the number of workstations and lockers required.

6.8.2 Utility Equipment

The effect of utility systems on the investment costs should be explored because it may be possible to remove entire system groups from the scope (e.g., clean steam, CIP stations) or achieve enough reduction in capacity to obtain cost reductions (WFI for rinsing, kill systems). The need to inactivate solid waste may increase the investment costs of an SUS, while the absence of a CIP in an SUS design may lead to a reduction in the generation of wastewater.

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6.8.3 Indirect Equipment Costs

The estimates for engineering and installation costs (e.g., design, commissioning, and qualification and validation) are normally calculated as a percentage of the equipment costs. One of the advantages of an SUS is that reductions in piping and instrumentation may decrease the number of both installed hardware and communication I/O points in the automation system. However, an SUS may incur a higher engineering percentage cost compared to similar MUS designs because of lower initial investment costs and the cost of validating leachables and extractables for the SUS. The installation percentage is lower for an SUS than an MUS. A similar approach can be used to determine the percentage costs for construction and project management ***(60)***.

6.8.4 Facility

An SUS uses stackable rectangular containers, which results in a smaller footprint compared to MUS. The absence of a CIP and washing/sterilization function further reduces the space requirements of SUS. However, inefficient movement of containers, installation, and dismantling of bags can easily waste the space saved by an SUS. In addition, proper storage and handling of an SUS is very important, since any damage might result in a lost batch or contamination. The transport route of an SUS may increase compared to that of a hard-piped system, and an SUS requires space to store process waste prior to final disposal. Storage space for SUS is typically greater than that for traditional MUS because of the need to manage SUS inventory and associated reusable storage containers. The main contributions to variable operating costs are summarized in **Table 6.8.4-1** ***(61)***.

Table 6.8.4-1 Main Variable Operating Costs

Variable Costs	Description
Raw materials	Input materials such as media, buffer, and various process aids like antifoam and chromatography gels
Consumables	Items such as storage bags, tubing, filters, and diaphragms that form or are a part of process equipment and are used on a per batch basis
Utilities	Steam, water, power, cooling, and chemical consumption for CIP/SIP
Waste	Energy consumption for inactivation systems and costs for disposing of solid waste
QC/QA and cost of quality	Analytical, documentation, batch documentation, release activities, investigations, supplier audits

6.8.5 Materials

Often, input materials (e.g., SUS materials prefilled with hydrated or powdered media and buffer components and chromatography gels) may be independent of SUS or MUS technology ***(62)***.

6.8.6 Consumables

The best way to determine the component cost of consumables is from the specific SUS process design under consideration because there can be a significant discrepancy between the costs of a supplier's standard configuration and that of the specific assembly required for a particular process step. Whenever possible, using the supplier's standard configurations would be the most cost-effective way to determine the cost of consumables. Designing the SUS with backup systems such as redundant tubing, filters, and connectors will increase the costs per batch of that SUS. However, if these backup systems are necessary, they must be reflected in the final cost analysis of the project.

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It is possible to minimize SUS overdesign via a thorough understanding of the real risks determined in a process risk assessment such as a failure mode and effects analysis (FMEA) ***(7)***. It is also important to get a feel for the actual cost of the SUS once specific connectors, filters, and tubing sets are added to the design. Since it may be possible to obtain discounts with large-volume orders from prospective suppliers, it is necessary to understand SUS configurations and how they can be customized to meet individual SUS process requirements. It would be advisable to incorporate a sensitivity analysis for the purchase cost of an SUS because final configurations and prices can be difficult to predict. In general, the sensitivity analysis should be extended to any variable with uncertain cost estimates ***(63)***.

6.8.7 Utilities

The reductions in cleaning and sterilization associated with an SUS leads to a decrease in the cost of utilities, especially for clean water generation, CIP, and SIP systems. It is important to use carefully verified cost data since large volumes may be involved in these processes. The cost of water for injection is significantly lower for large on-site systems than for commercially supplied WFI. It may be calculated from system depreciation, labor, and energy costs.

6.8.8 Waste

The cost of inactivating solid waste can be significant depending on the degree of containment required. The solid waste may be 1) inactivated on-site, which will lead to increased load on kill autoclaves; or 2) transported off-site for incineration or disposal at a landfill. Both approaches would incur transport and disposal fees. A third approach involves chemical inactivation. The disposal strategy for each SUS must comply with the laws and requirements of the municipal authority in charge ***(64–66)***.

6.9 QC/QA and Cost of Quality

Since some variable costs (e.g., analyses, preparing batch documentation, and batch release) are rarely affected by SUS implementation, the baseline costs of MU and SU systems tend to be similar. Batch review and investigation efforts may decrease and result in additional savings because SUS implementation results in reduced cleaning, SIP, and changeover activities. However, certain quality oversight activities (e.g., quality agreement, supplier audits, SUS documentation review, subsupplier audits, and QC-receiving inspection) may increase with an SUS implementation. The operation of several quality activities within the supplier's quality system rather than that of the end user may make process investigations more complex, and this additional effort should be taken into account.

6.10 Fixed Operating Costs

Fixed operating costs represent the expenditure on the facility regardless of the state of production. The main factors involved in evaluating fixed operating costs are summarized in **Table 6.10-1.**

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Table 6.10-1 Main Fixed Operating Costs

Fixed Costs	Description
Cost of capital	This is the cost of borrowing capital, and it depends on the internal interest rate of the project. Asset capitalization strategies may differ for a CMO, an established producer, or a startup.
Depreciation	This is the value loss over time; process equipment depreciates faster (e.g., 7 to 15 years) than buildings.
Staff	This refers to the cost of staff including wages and salaries. Additional functions of quality personnel may include monitoring incoming supplier documents, site auditing, and verification of control strategy.
Value-added activities	These include the management of materials, operations, qualification, verification, vendor audits, materials testing, and shop-floor oversight for allowing the SUS to be used in operations.
Maintenance	Cost of maintenance of production systems, scheduled renewal of valve diaphragms, calibration, and systems inspection. Although costs will decrease for SUSs, not all maintenance costs can be eliminated.
Opportunity costs	This is defined as the value of the next-best choice available when choosing between several mutually exclusive choices (e.g., the decision to expand a facility may result in losing the opportunity to invest the maintenance funds in the financial markets). Opportunity costs are often excluded from the estimates of fixed operation costs because they can be difficult to compare.
Facility operation	This is the cost of maintaining and keeping the facility running (e.g., CIP/WFI/HVAC, gowning, and revalidation), and it may vary according to the scope of the project. There can be significant differences in facility operation costs between SU and MU systems.

6.10.1 Cost of Capital

The cost of capital represents a fixed cost that is independent of the number of batches produced. The capital required for the investment may be borrowed at a certain interest rate or it could be at hand. Because fixed costs are added to variable operating costs to obtain the average batch cost, capital costs would affect batch costs through depreciation and amortization.

6.10.2 Depreciation

The rate of depreciation may be faster for an SUS compared to an MUS because new models and technology appear on the market regularly. In addition, some equipment (e.g., reusable storage containers, SUS holders) undergoes inevitable wear and tear with time. It is important to distinguish between building and equipment depreciation. SU facilities tend to have more storage and warehouse space but less utility and process space. The type of space required may affect the overall cost of the facility and its depreciation.

6.10.3 Staff

The cost of staff depends on the number of batches produced through direct processing operations and indirectly through activities like sample analyses, preparation of batch documentation, and the release of batches. Since over the short term the payment of wages, hiring, and training would occur irrespective of the number of batches produced, it is a good simplification to keep staff costs separate. However, if a production slot is allocated at a CMO, the cost of staff needs to be included as a variable operating cost. This is because both process suite occupancy and the cost of staff can be viewed as direct variable costs.

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The cost of training may increase for an SUS because additional effort may be required to ensure that the facility is established and run in a reproducible manner. Staff costs can be broken into two groups: direct-production (production staff) and indirect-production related (e.g., administration, management, automation and maintenance). A time study or production modeling may be used to determine the comparative cost of staff between SUSs and MUSs.

6.11 Project Duration, Time, and Productivity

6.11.1 Overall Project Duration

Overall project duration may be an overarching decision driver that makes all other economic analysis secondary. The comparison of the project duration for a facility that depends heavily on SUS to one that utilizes primarily MUSs may be compelling enough to make further analysis unnecessary. In general, SUS-based facilities have shown significantly shorter project durations when compared to MUS-based facilities. This reduced project duration can be used to delay capital investment or to speed time-to-clinic, time-to-market, or a combination thereof. The benefit of delay of capital investment may be measured by both the reduced cost of capital and the significantly reduced risk of a bad investment due to more clinical data being available at the time the decision must be made. For novel drugs under patent protection, the benefit of reduced time-to-clinic or time-to-market may be calculated based on the assumed revenue and profits to be realized during the period of exclusivity. Reduced time-to-clinic and time-to-market effectively extends the period of exclusivity by the difference in the project duration.

6.11.2 Process Operation Time

A SUS may increase the capacity of the facility by reducing process downtime for:

- Cleaning and sterilization
- Equipment setup
- Product changeover
- Calibration
- Maintenance

With an SUS, there is an opportunity to streamline the process by combining process steps because of better turndown ratios. However, the end user must consider the entire process because limitations in mass and heat transfer capacities may delay important setup or preparative operations for SUS, such as hydration, process fluid transfers, temperature adjustments, pH adjustments, and charges and discharges.

6.11.3 Time Constraints

An SUS may remove time and capacity constraints and improve scheduling because of the reduced need for WFI, clean steam, and cleaning. Increases in solid waste may be tackled without any adverse effects on the main process schedule. In this case, an SUS still requires time to bring components to the facility for setup and deployment. The planned batch time should account for time spent in set up, deployment, process adjustments, temperature and pH control, and dismantling the systems ***(67)***.

6.11.4 Speed of Process Development

The factors mentioned above, as well as the inherent flexibility of an SUS, lead to an increase in the speed of process development without a major change in the design of the facility.

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6.11.5 Equipment and Process Validation

The implementation of an SUS reduces the need to install support systems, and this leads to a reduction in the time required for validation activities. Similarly, the time required for cleaning and sterilization validation as part of the process validation activity is also reduced. These time savings are somewhat countered by the need to evaluate E&L during material and equipment qualification (refer back to **Section 5).** This additional activity for SUS should be based on a risk assessment that is performed prior to process validation.

6.11.6 Construction Time

For a retrofitting facility project, the effect of an SUS on construction time may be evaluated by comparing lead times for SU and MU systems. A new facility project may benefit from SU technology because processes and buildings are decoupled and many systems then can be installed and validated in parallel. The end user must include the following factors in the evaluation study:

- Design and specification
- Qualification and validation
- Space considerations
- Piping tie-ins
- Impact on utility capacity
- Training
- Project risk
- Number of welders and testers in the facility
- The ability to perform commissioning and qualification activities in parallel

SUS equipment skids may reduce custom engineering design and computer-aided design piping and instrumentation drawing (CAD PID) because they can be built from standardized parts with shorter delivery times.

6.11.7 Self-Assembly

It is possible to create a custom SUS by assembling and sterilizing the various components (e.g., tubing, strips, barbs, filters, and bags). If the quality is similar to that of a commercially available ready-to-use SUS, the customized SUS should be evaluated against a combination of time considerations and fixed and variable costs incurred from the following factors:

- Operator time
- Training for making assemblies
- Validation of assembly and sterilization procedures
- Documentation of assembly and sterilization
- Parts storage
- Sterilization equipment capacity

The quality system of the end user must be able to source SUS components as well as their assembly, testing, and release in-house. This is a good exercise for a quality unit within a company to perform as part of developing control strategies for a commercially supplied SUS.

6.12 Logistics

A significant difference between SU and MU systems is that the essential components of an SUS must be sourced from outside on a continuous basis. An MUS is also dependent on outside supply, but interchangeability between different suppliers is much better. The need for continual sourcing imposes another cost, which is addressed in the following paragraphs.

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6.12.1 Supply

Maintaining a steady supply of an SUS requires additional process activities that comprise a single step for an MUS. These include:

- Procurement
- Establishing service and confidentiality agreements
- Auditing suppliers
- QC testing of leachables and extractables

Apart from the testing of extractables and leachables, the preceding activities may be construed as extensions of existing functions with some additional complexities. In addition, the quality system will need to interact with the supplier's quality systems and their supply chain.

6.12.2 Storage

The number of pallet spaces required for storage must be included in the evaluation of the SUS. In addition, storage space depends on the lead times and inventory levels of suppliers. In general, an SUS requires more storage (e.g., especially larger SU assemblies such as an SU bioreactor) than an MUS. The end user must be aware that storage space may be required for the following functions:

- Central site storage
- Local facility storage
- Point-of-use storage
- Storage area for disposal shipping
- Storage area for final disposal off-site

6.12.3 Transportation

An SUS may incur costs associated with the transport of supplies, products, and waste. The specific areas may include:

- Transporting staff and equipment
- Establishing transport routes on-site and within the facility
- Handling equipment dock levelers and jacks

The transportation of biologically contaminated waste on public roads may result in additional costs for an SUS. The physical protection (e.g., trucks capable of cold storage and minimizing vibrations) required during the transportation of frozen product bags may also add costs.

6.12.4 Waste

Waste management costs vary significantly according to the specific requirements of the municipality in which the facility is located. An SUS tends to increase the costs of solid waste disposal while reducing that of liquid waste inactivation and neutralization. It is important for the end user to consider the following cost factors in the evaluation of waste disposal:

- Liquid waste inactivation and neutralization costs
- Solid waste on-site inactivation (chemical or heat inactivation)
- Transport off-site
- Incineration
- Landfill or other disposal method

In addition, the end user must consider other parameters such as extra wrappings, boxes, and storage areas for solid waste.

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6.13 Disposal of an SUS

Despite the recognized benefits in biological and pharmaceutical manufacturing of SU technologies, the end user must still address issues such as residual solid waste from used components and systems. Options for used SU equipment include reuse, material repurposing, or solid waste disposal, including landfill, incineration, and energy recovery. Solid waste is typically measured in tonnage units, so while the implementation of SU manufacturing may appear to introduce a significant increase in solid plastics waste, it is often not perceived as significant by waste management professionals compared to other industrial and municipal plastics waste.

Some SU technologies may be amenable to limited reuse in continuous processing or campaign modes. Examples include nonsterile mixing containers, buffer preparation systems, disposable prepacked preparatory chromatography columns, or membrane chromatography units used in dedicated campaigns.

At the moment, it is not commercially feasible to recycle SU materials because of the potential need to inactivate biohazard waste or separate SU assemblies with multiple components. Furthermore, there is often inadequate volume of total waste for most recycling programs. Recycling is more effective and economically feasible when there is a critical mass of materials that can be recycled. However, the increasing adoption of SU technologies has led to an increasing concern for the amount of clean polymeric/elastomeric material that is either deposited into landfills or incinerated. Most of the SU materials used in the biopharmaceutical industry are either nonhazardous or rendered nonhazardous before being shipped out to either a landfill or an incinerator. Although previous technology was restricted to the recycling of only homopolymers, modern technology now allows for the recycling of mixed plastics. Silicone tubing can be fully converted to base silicone, which can be used to make sealants, gums, and other products. Although some challenges still remain, biopharmaceutical companies are looking for cleaner and greener alternatives, and suppliers are starting to provide alternative solutions ***(68,69).***

Repurposing of mixed-plastic solid waste may require cleaning and sterilization, followed by grinding into particulate form for use as a filler material. Some companies accept clean mixed-plastic wastes for repurposing.

Where SU components or systems cannot be reused, recycled, or repurposed, some form of solid-waste disposal must be explored. Burial (landfilling) or incineration may be either controversial or preferred for a variety of reasons. The waste management professionals of the facility would be familiar with local municipal regulations for the disposal of solid-waste plastics (including wastes from discarded packaging materials, laboratory and cafeteria use, and maintenance). Grind and autoclave systems are widely used to pretreat contaminated medical device and other hospital plastic wastes prior to landfill disposal ***(65).*** This approach is also used by government agencies such as the National Cancer Institute (NCI) and the National Institutes of Health (NIH).

According to the U.S. Environmental Protection Agency (EPA), "incineration is a widely-accepted waste treatment option with many benefits. Combustion reduces the volume of waste that must be disposed in landfills, and can reduce the toxicity of waste" ***(66).*** Incineration is a widely practiced method in many countries and municipalities. In a waste-to-energy process, incineration may result in significant energy recovery. Waste-to-energy technology is employed throughout much of Europe, and it is increasingly being implemented in the Americas. Waste-to-energy is used to capture the trapped energy in municipal and industrial wastes and convert it to heat or electricity. SU waste may be considered for internal waste-to-energy conversion or may be transferred to other municipal or industrial waste-to-energy facilities (e.g., cement manufacturers who are now using mixed plastics instead of coal or used rubber tires to fuel their kilns). Although estimates vary widely, the heat value of mixed-plastics waste is conservatively

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estimated to be about 15,000 to 20,000 BTU/lb (34,890 to 46,520 kJ/kg), which compares favorably to that of coal at 9,000 to 12,000 BTU/lb (20,934 to 27,912 kJ/kg) and to that of gasoline/petrol ***(66,70)***. **Table 6.13-1** shows a comparison of the methods for treating waste from SUSs ***(64)***.

Table 6.13-1 Treating and Discarding Waste from SUSs

Option	Advantages	Disadvantages
Landfill, untreated	Lowest operating cost, no capital cost	Not an option for hazardous waste, perceived as environmentally unfriendly
Landfill, treated	Inexpensive, no capital cost	Perceived as environmentally unfriendly
Landfill, grind and autoclave	Generally accepted as safe, reduced landfill volume	Significant capital cost, extra handling
Recycling	Environmentally appealing	Not practical for mixed materials
Incineration	Generally accepted as safe	May be legally restricted, costly
Incineration with generation of steam or electricity (cogeneration)	Most environmentally benign, some return on investment	May be legally restricted, highest capital cost
Pyrolysis	Produces usable pure diesel fuel; fuel produced is cleaner burning than fuel produced from the refinery	New technology—few options available; subpar efficiency

Despite the generation of solid plastics waste, SU manufacturing is considered environmentally friendly because it reduces energy consumption and eliminates tedious cleaning and validation studies associated with stainless steel systems.

6.14 Sustainability

Green manufacturing and chemistry are fast becoming important business drivers for the sustainability of SUSs. Green chemistry involves the use of chemicals and chemical processes that reduce or eliminate adverse effects on the environment. The use and production of these chemicals may lead to reduced waste products, nontoxic components, and improved efficiency. SU manufacturing has the advantage of being environmentally friendly and energy efficient.

6.15 The Value Added

Unlike an SUS, an MUS requires several preparative procedures (value-added activities) prior to the beginning of each manufacturing batch. In an SUS, these steps are outsourced to the supplier and managed under the supplier's quality control (QC) systems. The cost of these value-added activities is shifted from the company's fixed-cost stream (e.g., labor) to a variable-cost stream (e.g., disposables per batch). This is because in SU manufacturing, high-quality, presterilized, disposable parts are supplied to the end user by the supplier, and this value-added cost and the cost of quality for any deficiencies that the end user encounters must be included in the assessment; otherwise, it would lead to inaccurate results. Capturing the major value-added activities of process design, qualification, operation, and verification across the supply chain ensures that all the required value-added activities are included in the assessment.

During a typical MUS operation, the quality unit verifies the activities performed by operations. Adoption of an SUS eliminates many of the preoperational activities (e.g., assembly, cleaning, and sterilization) from the work performed by operations. However, the quality unit must maintain the same stan-

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dard of quality even though these preoperational activities have been outsourced to the SUS supplier. This quality check may be achieved through a combination of verification and trending activities that allow the supplier's quality unit to determine that the steps were performed successfully. Any deviations or weaknesses in the supplier's quality system or documentation will affect the end user's cost of quality because it is the responsibility of the end user to ascertain that the SUS is fit for purpose.

6.15.1 Value-Added Analysis

It is necessary to assess the value added and how it affects COG and operating cost in order to determine the benefit of an SUS implementation. This can be achieved by mapping both the separate value-added activities and the primary reason for which they were performed. For example, a comparison between an SUS and an MUS should include the entire supply chain—from system design to final process verification activities and product release. **Table 6.15.1-1** depicts a mapping of the separate, value-added activities performed (blue), the primary reason for which they were performed (green), and the group that normally performs the activity for an SUS and an MUS, respectively (yellow). The last column shows the main attributes affected by a change in manufacturing technology.

Table 6.15.1-1 Comparison of Value-Added Activities

Activity	SUS	MUS	Value Added	Value-Added Attribute
Design	Supplier	Engineering	Efficiency	Overall cycle time
Vendor audit	Quality/technology	Quality	Process robustness	Records per cycle
Factory testing	Supplier	Engineering	Process robustness	Records per cycle
Qualification	Technology	Technology	Process robustness	Process capability index
RM release	Quality	Quality	Product safety	Records per cycle
Assembly	Supplier	Operations	Reliability	Records per cycle
Cleaning	Supplier	Operations	Product safety	Records per cycle
Sterilization	Supplier	Operations	Product safety	Records per cycle
Production	Operations	Operations	Product	Process capability index
Investigation	Quality/technology	Quality	Product safety	Investigations per batch
Release tests	Quality	Quality	Product safety	Tests performed per cycle
Disposition	Quality	Quality	Product safety	Records reviewed per cycle
Post-operations	Operations	Operations	Changeover	Records reviewed per cycle
SUS change control	Supplier/quality/technology	Quality	Process robustness	Consistency
Product management	Operations	Operations	Product	Product capacity
Verification	Quality/technology	Technology	Process robustness	Process capability index

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A review of the value-added attributes would show a reduction in the number of preoperational steps required by operations, thus reducing the overall workload. Although there may be a reduction in the number of records produced by manufacturing, the quality unit must still ascertain that these processes were completed satisfactorily; therefore, the type and number of supplier records needed for quality verification may actually increase. For example, the quality unit may periodically audit a vendor to make sure that the vendor's quality systems are acceptable and on par with systems employed for an MUS. This might represent a different type of activity from an MUS, and within this verification step, the associated labor costs and the impact on fixed costs must be considered. As a strategic concern, this shift in production concept represents increased reliance on the supply chain; hence, the outsourcing of related quality costs is likely to increase. Examples of such costs that contribute to the cost of quality are increased auditing, supplier investigation, end-user operational testing of SUS, and quality control activities, as well as increased demand for SU storage in order to avoid interruptions to facility production as a result of unforeseen events in the supply chain.

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7.0 Implementation of a Single-Use System

This section provides a framework for the implementation of an SUS. A successful implementation of an SUS takes into account the unique situation and requirements of the project and then communicates these needs effectively to stakeholders, suppliers, and sponsors. Although good project management skills are essential for any successful project implementation, SUS implementation tends to be enhanced by a stage-gate approach in which the details of the application are evaluated and the results communicated to the broader team. The approach presented in this section allows for important decisions to be taken at appropriate times to facilitate the success of the overall project.

7.1 Implementation Road Map

Figure 7.1-1 shows an implementation road map based on a natural progression of workflows. These workflows use a manufacturing strategy that provides the foundation for the main SUS implementation process described in this section (see also **Figures 3.0-1** and **3.0-2**). **Table 7.1-1** provides a more detailed overview of each of the implementation stages together with key documentary outputs for the entire SUS implementation process.

Stages 3 to **7,** as outlined in the figure below, can also be aligned with ICH Q8 and Q9 (Quality-by-Design and Quality Risk Management) processes/tools: stakeholder management, risk management, and process validation and verification.

The SUS implementation road map consists of a series of gateways (G1 to G8) and stages (S1 to S7). The stages (shown in yellow, blue, and green boxes) describe the natural progression of workflows in which information is gathered and documented. Specific goals (summarized in the drop-down boxes) must be attained before progressing to the next gateway. The process starts with a manufacturing strategy (S1 and S2; described in detail in Section 3.5) and proceeds into the main project-implementation, stage-gateway process via G3.

The implementation process is not necessarily linear: a more in-depth study of elements during S1 and S2 may negate the need for some of these activities at the later stages. For example, if a more elaborate technical assessment and definition of general principles in the SUS scope document are performed in S1 and S2, entry can occur at G4. A parallel approach from G1 to G3 and from G4 to G8 may facilitate or accelerate effective project completion. S3 and S4 may also be pursued in parallel. However, if one or more of these modifications are applied to the road map, care should be exercised that the implementation process does not lose structure and that the comprehensive treatment of all of the elements necessary remains in place.

Sections 4, 5, and **6** of this technical report provide additional information for reviewing the scope and activities of various workflows.

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Figure 7.1-1 SUS Implementation Road Map

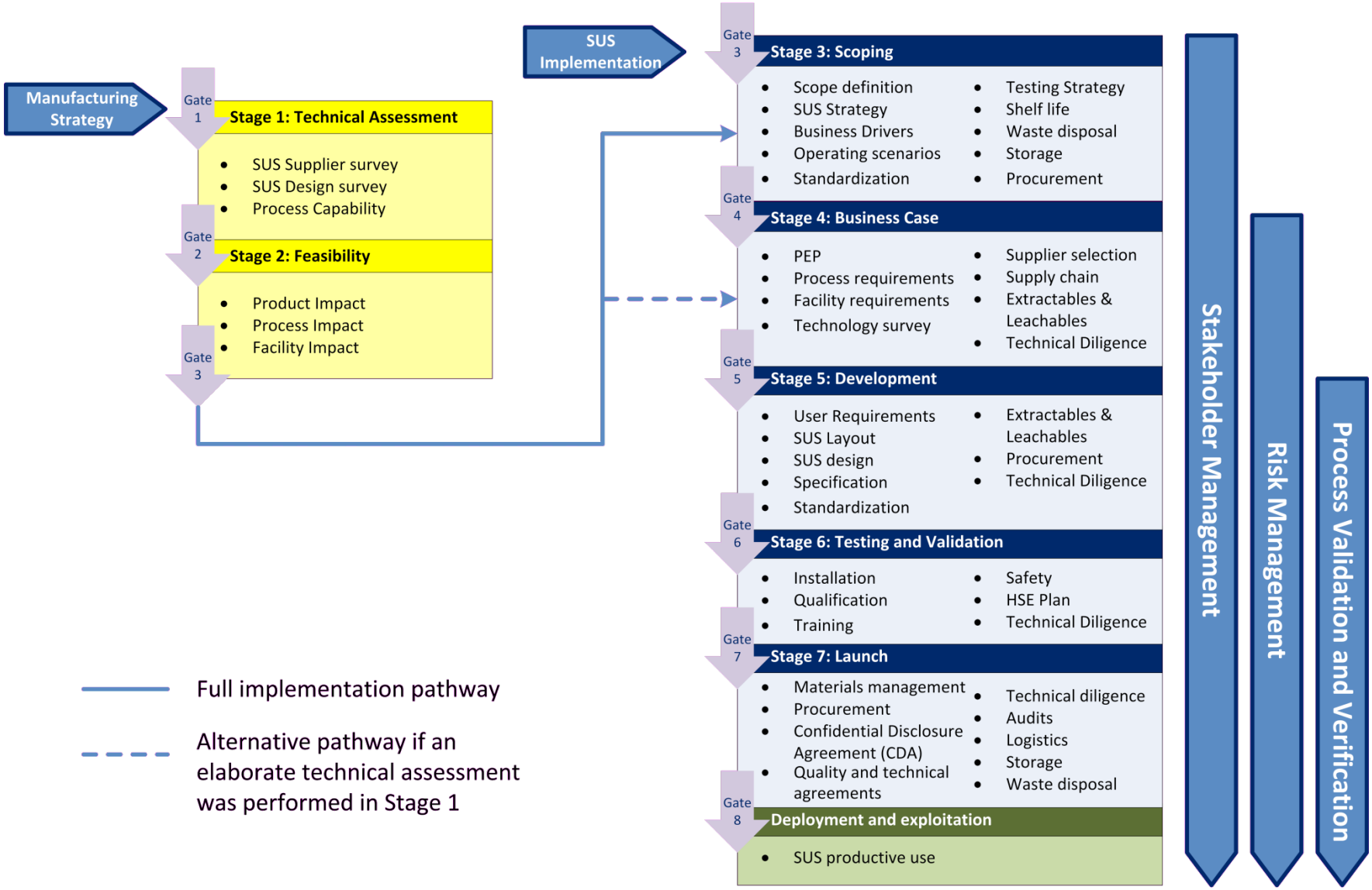

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Table 7.1-1 Focus and Output for Each SUS Implementation Stage

Stage	Section Cross-Reference	Project Implementation Focus	Project Description	Outputs: Document Title(s)	
1	3.5 Implementation of a Single-Use System	Technical survey and alignment with business objectives	Manufacturing Strategy, Part 1. This is a review of SUS suppliers and designs, process capabilities, and compatibility with business objectives.	• Concise review of SUS suppliers • Concise review of SUS designs • Process capability assessment • Business objectives • Confidential disclosure agreement/Nondisclosure agreement (CDA/NDA) with suppliers	
2	3.5 Implementation of a Single-Use System	Impact assessments and implementation approach	Manufacturing Strategy, Part 2. This comprises impact assessments for product, process, and facility.	• Product impact assessment • Process impact assessment • Facility impact assessment	• Implementation strategy • CDA/NDA with suppliers
3	7.3 Scoping	Technical scope definition of the SUS project	The SUS strategy provides general guidance and the policies to be applied for SU project implementations (might be combined with outputs from stages 1 and 2)	• SUS strategy	
4	7.4 Business Case	Articulation and selection of business scenarios that can impact technical choices and preparation of the project plan	Initiation of a project execution plan (PEP) and identification of SUS suppliers The PEP describes the project, its business and technical scenarios and approach, customer deliverables, planning, and quality issues associated with the project	• Project execution plan • Facility-level plan • Implementation-level plan • Operational-level plan • Process mapping • Regulatory assessment tables (RATs) • Good engineering practice (GEP) outputs • Risk management processes • CDA/NDA with suppliers	• Supplier selection and supply chain review • Technology survey • Technical diligence reports • Extractables and leachables data • Validation master plan • PVV documentation • Non-GxP documentation such as local and national government permits
5	7.5 Development	Capture of essential and nonessential user requirements and procurement	A description of the user's requirements for the SUS, including layout, design, and specifications, ready for the procurement process with the raising of a purchase order or equivalent	• PVV documentation • User requirements • Equipment purchase order • Supplier's order confirmation, including the agreed terms and conditions of sale and guarantees	• Import/export permits • Change notifications approved and executed as part of procurement change-control process

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Table 7.1-1 (Continued)

Stage	Section Cross-Reference	Project Implementation Focus	Project Description	Outputs: Document Title(s)	
6	7.6 Testing and Validation	Commissioning and qualification (C&Q), process qualification, and operator training	Development of methods and documentation for equipment SOPs, maintenance, qualification and process validation, impact assessments and risk management processes, HSE processes, and personnel training	• Goods received/Reception and verification checklists • Supplier documentation including operating manual, maintenance, and spare parts lists • Testing and installation reports • Validation master plan • Cleaning validation master plan	• Qualification plans, protocols, and reports • Process validation plans, protocols, and reports • HSE plan • Training plans • Non-GxP documentation such as safety certificates (e.g., CO_2/O_2 detection)
7	7.7 Launch	Materials management	The routine management of suppliers, purchasing, technical diligence, and quality audits, quality and supply agreements, and logistical and waste-handling strategies	• CDA/NDA • SUS supplier qualification programs • Technical diligence reports • Audit reports • Service contract • Quality and supply agreements	• Routine purchase orders • PVV documentation • Non-GxP documentation, including sustainability plans, environmental/GMO assessments, filings, approvals, and • waste disposal contracts

The end user should adjust the outputs from this table to match the desired level of project structure in the SUS implementation process. Strict adherence to this workflow provides a structured and practical approach; however, it may be equally beneficial to operate on an ad hoc basis by selecting specific documents and adding to them as the project lifecycle progresses. The time at which some of the documents are created may be adjusted as required by changes in the circumstances of the end user. A subject matter expert (SME) may implement a simplified SUS project plan whereas a fully established and large company may opt for a complete project implementation. Note that regardless of the approach the end user adopts, there is a minimum level of project organization and documentation necessary to provide structure and clear deliverables without which the implementation process may falter.

7.2 Implementation Themes

As shown in **Figure 7.1-1,** three themes represent a series of concepts and principles that span SUS implementation, and each theme can be applied either as a tool in support or as part of the implementation method.

7.2.1 Stakeholder Management

Stakeholder management is important throughout the SUS implementation process. It helps an organization to achieve strategic objectives by 1) interpreting and influencing both the external and internal environments, and 2) creating positive relationships with stakeholders by appropriately managing their expectations and agreed objectives.

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A structured approach is important and beneficial to every stage of implementation (**Table 7.1-1**). Identifying stakeholders who are involved across multiple functional areas would encourage teamwork and facilitate effective communication. Deficient stakeholder management may lead to adverse conditions such as diverging implementation and missed opportunities for standardization and process streamlining (e.g., the parallel pursuit of several different types of sterile connectors or storage bags, when multiple SUS technologies or suppliers are engaged). It may be helpful to create a "power" grid that classifies stakeholders according to their influence and their interest in the project. **Figure 7.2.1-1** shows an example for preparing user requirements and managing stakeholders.

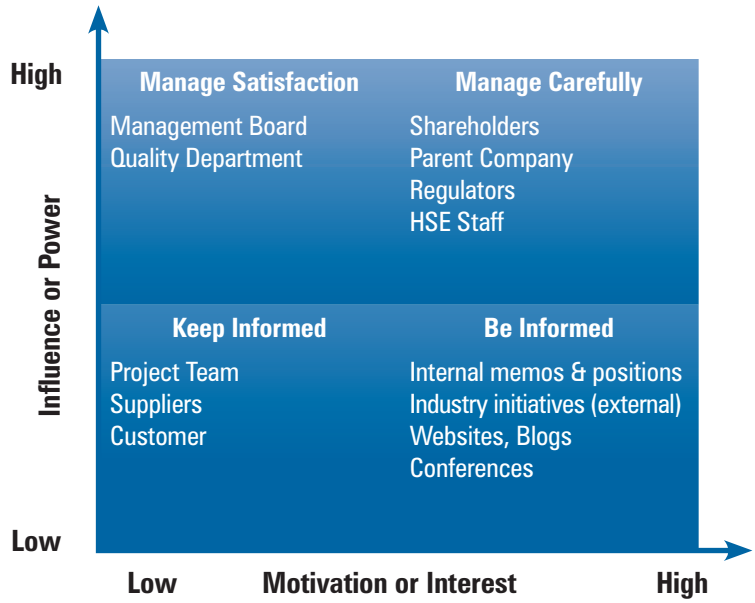

Figure 7.2.1-1 An Example of a Stakeholder Power Grid

The power grid allows the project team to identify, gather input from, and manage internal or external stakeholders. It is also useful for preparing SUS user requirements, and it fosters effective communication among project team members.

Effective stakeholder management may be achieved by 1) evaluating company governance and management processes in order to identify functional areas, and 2) using a responsible, accountable, consulted, and informed (RACI) template to clearly define roles and responsibilities within functional areas and between stakeholders ***(71)***.

7.2.2 Risk Management

As shown in **Figure 7.1-1,** risk management is important throughout the lifecycle of a project (specifically from **Sections 7.4** to **7.7**). Application of a quality risk management (QRM) program during SUS project implementation confers benefits on the manufacturer, patients, quality auditors, and regulators. These benefits come from an understanding of product and process risks, individual assessments, mitigation or acceptance, communication, and review as outlined in Section 3.3.1 ***(4,7,72)***.

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It is critical for an organization to use a QRM program to underpin product quality. An effective QRM program must be organized as a team effort with regular review meetings involving stakeholders. A company charter should provide policies and guidelines for when and how to apply risk management, along with the necessary implementation level and control. The risk assessment and risk mitigation tools used should be commensurate with the objectives of the project and the quality impact of the SUS being addressed.

Risk Assessment Tools

- Information gathering (data mining) and data evaluation
- Ad hoc applications (e.g., using simple tables that delineate some risks associated with predefined implementation scenarios)
- Directional-risk profiles
- Fishbone analysis
- More robust applications such as failure mode and effects analysis (FMEA), in which events are assessed in terms of impact, severity, probability, and detectability

Risk Mitigation Tools

- Identification and ranking of risks, their impact, and acceptance or mitigation (i.e., applying and comparing current and future states)
- At a minor level, mitigation can range from the requirement for further data mining to simple process adjustments
- For more elaborate subjects, quality audits that employ an efficient document management system are appropriate
- A specified risk-review period commensurate with the risk in question

In-depth reviews of common risk-assessment techniques and their applications in risk management are available in specialized literature ***(4,7,72,73)***. A more detailed evaluation of the risks associated with SUS implementation and the necessary qualification and validation studies is reviewed in **Section 5.2.** Some examples of QRM application in SUS implementation are shown in **Tables 7.2.3-1** to **7.2.3-3.**

Once the severity of the risk and the likelihood of its detection have been assessed, the most suitable detection tool can be selected.

Table 7.2.3-1 Guidelines for the Application of Risk Management During SUS Implementation

Severity	Likelihood of Detection	Recommended Tool
Impacts product quality	May or may not be detectable	FMEA
May impact product quality	Detectable after the event	FMEA
May impact product quality	Detectable during the event. There may be latency	Data mining, process mapping, risk profiling, fishbone analysis
Could impact process yield or robustness	Rapid detection	Data mining, process mapping, ad-hoc risk table
No impact	Immediate detection	Continuous verification

Table 7.2.3-2 ranks SUS implementation risks by correlating the complexity and process-specific risks of a potential application. The highest level of risk is shown in red, moderate in orange, and the lowest

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in green. Such a relative risk ranking may be useful in prioritizing the qualifications of SUS suppliers.

Table 7.2.3-2 Directional Risk Profile of SUS Items and Applications

Directional Risk Profile		Complexity of SUS Items		
		Low	Moderate Low	High
Risk of SUS Application	Low	Tubing & Connectors	Sampling Systems: Not a direct impact	Clarification/ Concentration
	Moderate	Manifolds: Externally sourced	Storage using SUS: raw materials, media, supplements, buffers, drug intermediates, product	Drug product formulation
	High	Manifolds: Self-assembled	Sterile connectors	Cell & virus culture

Table 7.2.3-3 shows risk-mitigation tools and the levels of control for current and future states in an SUS. The risk evaluation of future process states would be lower (i.e., a greater level of control over the process or system) if the level of risk were reviewed, assessed, and mitigated on a regular basis.

Table 7.2.3-3 Risk Management and Mitigation in Current and Future SUS Implementation

Process State		Examples of Areas or Subjects Where Risk Mitigation can be Applied	Status
Current	Increased Control/Desired Tendency	Stakeholders	Unknown
		Process status	Not understood or is in initial design phase; no data collection or analysis and no CQAs identified
		Risk assessment	Does not exist
		Corrective actions or control strategies	Do not exist
		Supply or quality agreements or contracts, technical diligence, or audit process	Do not exist
Future		Stakeholders	Identified; stakeholder management on-going
		Process status	Process is understood and CQAs identified; qualification phase and data collection and analysis are in continued verification
		Risk assessment	Valid risk assessments are available
		Corrective actions or control strategies	Corrective actions and/or control strategies exist
		Supply or quality agreements or contracts, technical diligence, or audit process	Agreements and contracts in place; technical diligence is ongoing and quality audit process is in place

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7.2.3 Process Validation and Verification (PVV)

In pharmaceutical manufacturing, process validation (PV) is the collection and evaluation of data from the process design stage through commercial production that establishes scientific evidence that a process is capable of consistently delivering quality product ***(8)***. The U.S FDA and EMA consider PV a requirement in both general and specific terms in Good Manufacturing Practice ***(74,75)***. Verification is a systematic approach for ensuring that manufacturing systems (acting either singly or in combination) are suitable for their intended use, have been properly installed, and are operating correctly. This is an umbrella term that encompasses all types of activities to assuring systems are fit for use, such as qualification, commissioning and qualification, verification, system validation, and others ***(18)***. Verification (or process verification) is an important concept for SUSs and is a primary control strategy for sourcing from suppliers. This remains active throughout the lifecycle of the project.

PVV is a tool that requires an integrated project management approach that involves all pertinent stakeholders (e.g., a cross-company project team) and full support from management. A science- and risk-based approach to PVV tends to generate a thorough understanding of the manufacturing process, which in turn enhances product quality and patient safety ***(8,74)***. ICH guidance documents, such as ICH Q8, Q9, and Q10, provide useful guidelines for development, risk analysis, and product control ***(3–5)***. A draft guideline on process validation for the manufacture of biotechnology-derived active substances and a guideline on process validation for finished products has been published by EMA ***(75,76)***.

SUS process validation and verification is normally performed using a lifecycle approach. The FDA 2011 process validation guidance and PDA Technical Report 60: Process Validation - A Lifecycle Approach provide useful models ***(8,74)***. Within that model, SUS implementation becomes part of process validation stages 1 and 2 (S1 and S2). Productive use of SUS is part of process validation S3. Additional technical guidance is available for this and for a phased approach to quality systems and the application of GMP as well ***(63,77)***.

Continuous quality verification can also be used. The knowledge obtained as the product lifecycle progresses will allow a framework to be established in order to attain mitigation of risks, reduction of process variability, enhancement of process capability, definition of enhancement of process design space, and improvements in product quality. These may lead to other benefits that address both compliance and operational goals (e.g., continuous process improvement) ***(78)***. **Figure 7.2.3-1** illustrates two component elements—validation prerequisites and conformance batches—for process validation S2 that confer greater clarity on SUS implementation.

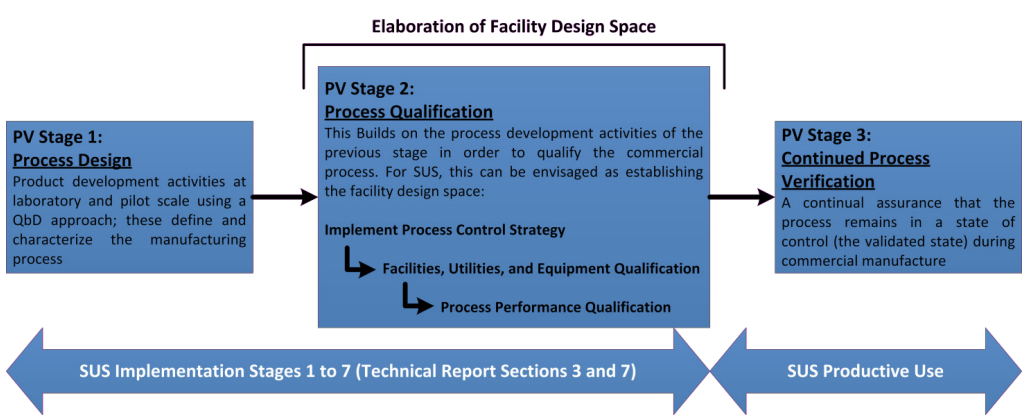

Figure 7.2.3-1 SUS Implementation and the Validation Lifecycle

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7.3 Implementation S3: Scoping

The project scoping process provides a source of information that is complementary to that obtained from a definition of the manufacturing strategy (Section 3). Scoping identifies pertinent information that may affect conception (e.g., potential materials, science, and facility operational requirements) and establishes the principles for conceiving, selecting, and operating the SUS. Once the basic strategy is defined, this information is applied in S4 to define process and facility requirements (**Section 7.4.2).** Technical guides that may assist the SUS scoping process are available from the website of Bioprocess Systems Alliance (BPSA) ***(79).***

7.3.1 SUS Strategy

The output of the scoping process is the SUS strategy document. This is a primary reference document that provides overall guidance and policies for SU applications across the entire project. It may be combined with outputs from S1 and S2 if more elaborate evaluations were performed during those stages. However, if the organization of the road map is changed, care should be exercised that the implementation process does not lose structure and that the comprehensive treatment of all of the elements necessary remains in place. The SUS strategy is normally a short and concise document with a global technical vision outlining how SUS control strategies would integrate with quality systems and international GMPs.

The strategy also highlights key topics and clarifies their roles within the SUS. This is important because propitious decisions made at the beginning of the implementation process can save considerable time, effort, and money in later stages. For instance, it is critical that the selection of SUS equipment and suppliers is not allowed to become an ad hoc process. This issue can be problematic for large companies as well as small to medium enterprises. Lack of control over this decision at an early stage can lead to equipment redundancies and using more suppliers than necessary. Depending on the circumstances, this can be considered to be a form of technology overdosing (i.e., too many suppliers or no coordination across company functional areas). Although having a straightforward relationship with suppliers is essential, dealing with too many suppliers can significantly increase the resources and supplier qualification effort required at later implementation stages.

Equally important, consider the case in which an SUS is supplied as part of a raw material in a ready-to-use format (e.g., a ready-made culture media in which the media container is an SUS). Regardless of whether this was an occasional order or made via a supply agreement, the end user would need to make sure that the supplied components adhere to the same quality standards as all the other SUS components. Such a scenario may represent an SUS-inherited implementation because the end user acquires the product but does not necessarily have a choice about the technology source.

The following paragraphs describe key subjects that should be addressed in the SUS strategy document.

7.3.2 Business Drivers

It is important to review the business drivers for SUS implementation. Sometimes, an SUS may be used for predevelopment prior to a requirement for GMP (e.g., in a small-scale and disposable multi-reactor robotic screening system). Much larger SUSs can be deployed when process development or manufacturing services are required during advanced stages of the product lifecycle. Economic considerations (i.e., the desire to reduce costs) tend to become the most critical business drivers for selecting either an MUS or an SUS. Other factors are quality-improvement drivers such as containment or the need to avoid cross contamination. End users may need to adapt the policies described in this report to their specific situation and review them regularly. This is because it may be necessary to determine how the quality system, GMP, and PVV fit into the overall process.

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7.3.3 Operating Scenarios and Standardization

Developing operating scenarios and standardization offers end users the opportunity to define specific rules, policies, or principles for key issues relevant to the implementation process, for example, the type of multi-purpose operations in use or the selection of production by campaign. More examples are described later in this section. An example of the practical application of standardization can be found in the literature ***(80).***

7.3.4 Process Validation Stage

It is important to define the SUS project within the process validation lifecycle because it affects the subsequent approach to PVV. Predevelopment systems (prior to PVV S1) would not necessarily be subject to the same quality requirements as that for later-stage processes involving GMP. Conversely, the starting materials for some medicinal products, such as those for cell therapy, could be isolated at a very early stage in the product development lifecycle, and their exploitation may require extensive PVV and careful handling to ensure process compatibility and regulatory compliance.

7.3.5 Operating Volumes and Storage Requirements

The end user should define operating and storage volumes for static and mobile requirements. A maximum mobile volume (i.e., the maximum allowable volume for manual transportation within the plant) above 500 L may be difficult to transport and may significantly affect the floor space required in the facility. In addition, such large equipment may be subject to HSE regulations.

7.3.6 Hybrid Systems

The end user should evaluate the requirements for connections that combine SU and MU systems (e.g., CIP or SIP). The principles that ensure operator safety and product integrity must be defined. Any MUS parts that require manual cleaning and autoclaving must be defined and specified.

7.3.7 Connection Principles

The end user should define the requirements and principles of operation that apply to the various connection applications. These principles should include the types of connections allowed or employed, according to process circumstances and regulatory requirements.

7.3.8 Campaigning

Although there are specific situations where an SUS container might be refilled and used in a single campaign (e.g., where a large buffer tank is used to refill a single SUS storage vessel without breaking the sterile barrier), reusing SUS is not recommended. It is preferable to define a position from a risk-based perspective against reusing SU components. Multiple caustic cycles over an SU column may be used to process several pools from a batch or series of harvests, but SU tubing circuits in processing skids may not be able to withstand repeated caustic cycles. Therefore, while the SU columns may be campaigned, the SU tubing circuits may need to be replaced for each cycle.

7.3.9 Future Deployment

Significant productivity gains can be achieved by deploying and exploiting SUS over MUS. For example, there can be significant differences in maintenance costs, batch-to-batch turnaround, and staffing requirements ***(81).*** Standard principles must be reviewed and identified for activities involving the hands-on steps required to successfully assemble and make a system ready for use for a specific SUS application. Including a defined setup and deployment stage, with standardized activities for each part, is a recognized success factor for SUS implementation, and this should be a routine part of the workflow and planning for each SUS application.

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7.3.10 Sourcing

The end user should determine whether the SUS would be sourced from exterior suppliers or vendors (i.e., designed, constructed, and delivered) or self-made (i.e., designed, prepared, and operated on-site). Some applications and business drivers may implicitly require a clear policy, so a risk-based approach should always be used. A self-made SUS would have to meet the following requirements:

- Has specifications
- Has batch process records
- Meets qualification criteria including sterility
- Has a change-control process in place
- Passes stability studies if necessary

It is essential to verify that both internally and externally sourced SUSs meet the same quality standards.

7.3.11 Testing Strategy

An SUS strategy should define the principles and aims of any in-house testing program that would be proposed as part of the implementation process. This may include the collection of supplier testing data and controls. Testing strategies should be discussed in detail with the supplier, and quality agreements should allow for the exchange of test data between a supplier and the end user.

7.3.12 Materials

The strategy should define a position on single- or multiple-film usage. Reducing the number of material specifications reduces confusion and errors. For example, using the same product-contact film may simplify the subsequent evaluation of extractables and leachables. However, it may also limit the range of available technical solutions. Although desirable, standardization within a single-film strategy may not be possible where multiple SUSs from different suppliers are used in an integrated process. Finally, the materials should be biocompatible and the end user should be aware of supplier testing procedures. Harmonizing the various testing methods that suppliers use could make it easier for end users to make comparisons between suppliers (see **Table 5.5.2-1**).

7.3.13 Shelf-Life Policy

The expiration and shelf-life policy needs to be defined. This is particularly important if the end user has a large number of different SUSs (from multiple suppliers or self-made assemblies). Note that some SUS suppliers may have their own definition of shelf life, and the end user should address and resolve these issues during policy definition.

7.3.14 Procurement

The procurement process should be standardized. If the project deals with several suppliers, it may be necessary to make this a uniform process. The end user may need different strategies for purchasing occasional versus routine SUS equipment and components (**Sections 7.4.10, 7.5.3, 7.7.1,** and **7.7.2**).

7.3.15 Storage

The strategy document should define how SUS warehousing and logistical operations would be organized. A dedicated clean and controlled storage area is recommended.

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7.3.16 Waste Treatment

The policy for waste treatment (i.e., how SUS solid waste would be handled and inactivated) should be defined, and the potential impact of carbon taxes must be taken into account. In addition, the end user must explore the feasibility of recycling some of the waste from SUSs.

7.3.17 Non-GxP Issues

The end user should define any non-GxP issues that must be considered as part of the SUS implementation project. These issues are often quite diverse and can be local or country-specific (e.g., permits, tax credits, grants or subsidies, imposed supplier preferences, or safety-related concerns).

7.3.18 Sustainability

A key point is that in general, SUSs are more sustainable and have less adverse environmental effects than traditional MUSs. The SUS strategy should define a policy that includes these principles. Some examples would be fossil fuel depletion and energy use, urban land occupation, pollution, greenhouse gas emissions and ozone depletion, water depletion, and so on ***(82).***

7.4 Implementation S4: Business Case

This stage of implementation concerns the different business and technical scenarios that can influence user requirements and the implementation process. Examples of key business drivers include:

- Whether the project concerns drug substance or drug product
- Whether the project concerns the manufacture of preclinical material, investigational medicinal products (IMPs), or licensed product
- Whether the product is a biological or chemical entity
- Containment requirements

The main documentary output from S4 is the project execution plan (PEP). Important parameters for preparing the PEP (such as facility and process integration, technological survey, supplier selection, and supply chain) are discussed in the paragraphs that follow. Further details on the evaluation of business drivers can be found in **Section 5.**

7.4.1 Project Execution Plan

The purpose of the PEP is to ensure that the goals of the project (i.e., budget, time, and quality) are met. It defines and describes:

- The project parameters
- Execution strategy
- Management
- Compliance

A detailed and structured PEP is critical to a well-organized implementation project. Other project documents such as a validation master plan (VMP) and user requirements may be used in PEP preparation or execution. Typically, the PEP would address the headings shown in **Table 7.4.1-1.** Materials management (**Section 7.7.1**) contains some additional points that may need to be addressed in the PEP. An example of a PEP for an SUS facility can be found in **Section 9, Appendix 2.** Additional examples of checklists, implementation experiences, and points to consider can be found in the literature ***(83–89).***

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Table 7.4.1-1 Typical Contents of the SUS Project Execution Plan

SUS Project Execution Plan		
1. Introduction Purpose Context Document history Languages Document templates	**5. Project Execution Approach** Overview Design Contractors and subcontractors Commissioning and qualification	**9. Health, Safety, and Environment** Context Specific issues HSE plan HSE stakeholders Safety committees
2. Project Overview Introduction Client/user overview History and scope Objectives and drivers Steering committee	**6. Organization** Strategic partnerships Organization and governance Roles and responsibilities Stakeholders Conflict resolution	**10. Implementation** Procurement Delivery Installation Commissioning and qualification Training
3. Contract Documents Deliverables Policies and strategies Impact assessments Technical plans	**7. Quality** Quality strategy Supplier qualifications Quality activities plan Document control	**11. Handover and Operational Support** Phases Planning Materials management Debriefing
4. Scope Engineering scope Equipment and systems Change management	**8. Risk Management:** Approach Risk mitigation strategies Risk mitigation plan	**12. Annex** Abbreviations References Complementary documents

The PEP, sometimes called a project implementation plan (PIP), provides a thorough and detailed description of the project. It complements other project documents such as the VMP, user requirements, and other non-GxP documentation, including environmental safety, waste-handling requirements, and inspections and building permits. Since not all projects involve the same level of complexity, the PEP may need to be modified accordingly. Therefore, this table is for illustrative purposes. The final content of the PEP will depend on the approach to the project, risk-management level, and business drivers such as the nature of the product, lifecycle requirements, engineering policies, and project scale.

7.4.2 Process and Facility Requirements: Basis of Approach

Process and facility requirements are key business drivers, and their consideration forms an important part of the PEP preparation. SUSs can be integrated and deployed within existing processes and manufacturing facilities. However, when a high level of SUS integration is necessary, new facilities may be required because of the radical structural differences between SU and MU systems. For PEP preparation, SUS integration must be considered from the point of view of the engineering service, suppliers, and the end user. The outputs are three relatively simple documents which outline the implementation strategy: facility-, implementation-, and operational-level plans.

7.4.3 Facility-Level Plan

This plan shows how the SUS fits into the facility design space together with other process equipment, and it summarizes different functional situations.

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7.4.4 Implementation-Level Plan

This contains the parameters that affect the supplier during the production, marketing, and supply of SUS for off-the-shelf or customized systems. In addition, it describes the parameters that influence the end user during the creation of self-made or in-house SUSs.

7.4.5 Operational-Level Plan

This describes the parameters for SUS integration (i.e., assembly, connection, and deployment) in readiness for manufacturing. It is essential to review each of these factors thoroughly because they provide the necessary content to complete the PEP and subsequent documents. The operational-level plan contains elements from manufacturing and SUS strategies. The input and output documents for SUS integration are illustrated in **Figure 7.4.5-1.**

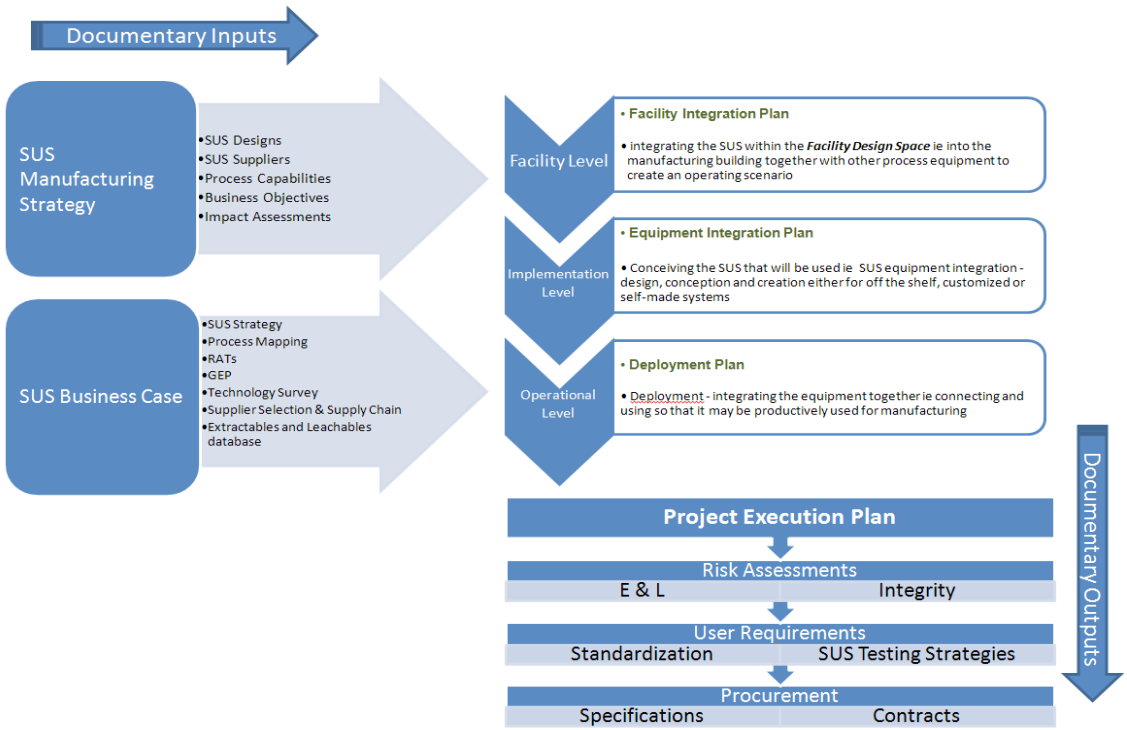

Figure 7.4.5-1 SUS Process and Facility Integration

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7.4.6 Facility-Level Integration Plan

This is normally a concise document that describes the integration of the SUS into the facility design space *(87)*. Robust process knowledge is a prerequisite for defining the various operational situations of the facility. It is critical to adopt a holistic approach because several factors (e.g., compliance, environment, operational requirements, and the compatibility of processes and equipment) may affect the successful operation of the SUS. These factors depend on the nature and properties of the product (e.g., biological or chemical entity) and the requirements for containment. **Figure 7.4.6-1** provides a conceptual representation of how the successive steps in facility-level integration may be achieved. Some of the most relevant parameters to consider include:

- Product type
- Sterile or nonsterile processing
- Product volume
- Lifecycle
- Available surface footprint
- Dependence on utilities
- Flexibility of facility design
- Avoiding cross-contamination
- Level of segregation and dedication
- Environment (e.g., room air grade, containment)
- Personnel interactions
- Process validation (may be necessary when the equipment used in scale-up changes in terms of volume, aspect ratio, or between MUS and SUS)

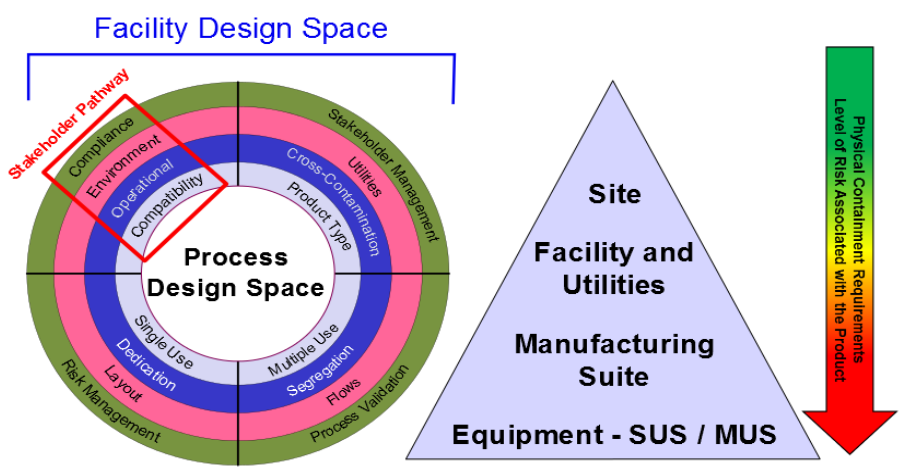

Figure 7.4.6-1 Process and Facility Considerations for SUS Implementation

The chart on the left side of the figure depicts a holistic vision of a facility from the top-level processes that created it (outside rings) to the process design space (inside ring). The key points to address during implementation are illustrated as a stakeholder pathway. The triangle and the down-facing arrow on the right show how the risk from biological or chemical potency may affect the amount of dedication/segregation and containment necessary at all levels within the manufacturing site.

A structured risk-based approach should be used to prepare the facility plan, starting with the manufacturing strategy prepared in S1 and S2, and the SUS implementation strategy prepared in S3.

Process mapping and regulatory assessment tables (RATs) are additional tools for defining facility requirements. Process mapping identifies operating scenarios and potential CQAs, while a RAT enables cross-referencing to key GMP factors and other regulatory requirements where it is appropriate for

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the user to document their policy regarding those regulations (i.e., how they intend to practically address those requirements) ***(87)***. The use of good engineering practice (GEP) also complements this strategy by providing a framework for the application of key concepts (e.g., project engineering, common practices, and operations and maintenance) ***(90)***. Examples include non-GxP compliance, design reviews, change management, equipment maintenance, and lifecycle management. **Table 7.4.6-1** shows an example of a RAT.

Table 7.4.6-1 Example of a Regulatory Assessment Table

ICH Q7A - Basic Requirements for Active Substances Used as Starting Materials: Specific Guidance for APIs Manufactured by Cell Culture/Fermentation		
Section	Excerpt from ICH Q7A	User Policy for Meeting Each Requirement
18.30	"Where aseptic addition of cell substrates, media, buffers, and gases is needed; closed or contained systems should be used where possible. If the inoculation of the initial vessel or subsequent transfers or additions (media, buffers) are performed in open vessels, there should be controls and procedures in place to minimize the risk of contamination."	All aseptic additions will be performed as closed processes within closed systems. SUS must be presterilized; MUS must have received a SIP treatment.
18.31	"Where the quality of the API [active pharmaceutical ingredient] can be affected by microbial contamination, manipulations using open vessels should be performed in a bio-safety cabinet or similarly controlled environment."	A risk-based approach will be used to determine whether a particular manipulation might be affected by microbial contamination. If a process step is considered to be at particular risk, for example, where an open connection should be made, these open steps will always take place in a microbiological safety cabinet under EU-GMP Grade A with background Grade B conditions.
18.34	"Cell culture equipment should be cleaned and sterilized after use. As appropriate, fermentation equipment should be cleaned and sanitized or sterilized."	Process equipment used for upstream production runs will be sourced as an SUS and, as such, will be used only once.
18.38	"Shared (multiproduct) equipment may warrant additional testing after cleaning between product campaigns, as appropriate, to minimize the risk of cross-contamination."	Some equipment used in the production process may be an SUS-MUS hybrid. The CIP procedures for the MUS components (e.g., steel sampling valves) will consist of a decontamination step by SIP or autoclaving followed by a defined and controlled cleaning procedure. These processes will be validated.

This table is an example based on Chapter 18 of ICH Q7A (EU-GMP Part II) ***(2)***. Other regulatory texts can also be exploited in this manner. A RAT is used to cull the sections of a regulatory text that are most applicable to the project and to document the company's position with regard to each chosen paragraph. This helps to identify the key strategies or policies that may be required. A RAT can be a useful supplement as part of a risk-management-based approach to developing situations for facility-operating scenarios.

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7.4.7 Equipment-Level Integration Plan

This brief document describes factors influencing the supplier or end user during the construction, marketing, and supply of the SUS as an off-the-shelf, customized, or self-made system. Some of the parameters for verification are:

- Position in the product lifecycle and making sure that it is compatible with future GMP and quality systems
- Working volumes
- Transport conditions
- Handling or moving the system, and protecting it during this process
- Physical support (e.g., whether the system can support itself or requires a supporting structure)
- Physical parameters
- Flow rates for gases and liquids
- Temperature range (e.g., cold storage to process conditions)
- Heating and cooling requirements
- Agitation and mixing
- Pressure
- Sterilization or chemical sanitation cycles
- Chemical compatibilities (e.g., organic solvents, gas permeability)
- Direct-impact phenomena (e.g., cellular growth kinetics)
- Biocontainment capabilities
- Standardization
- Control systems and their validation
- Autonomy [i.e., does the system require a uninterrupted power supply (UPS) in case of power failure?]

At S4, it is generally assumed that 1) sufficient process knowledge exists to enable a coherent review of the design factors that must be built into the SU, and 2) this would be performed from a risk-based perspective. Some of the most relevant design factors are:

- Construction materials
- Physical properties
- Microbial control methods
- Size and volume
- Physical parameters
- Direct-impact phenomena (e.g., contact time, E&L, biocompatibility)

For many end users, establishing a close partnership with major SUS suppliers allows them to leverage their extensive experience in the process field. Adopting customized SUS or using inexperienced suppliers would increase risk, although this may be mitigated by on-site SUS testing. The onus is on the end user to ascertain the compatibility of process, equipment, and facility design space from a risk-based perspective (**Sections 7.4.10** and **7.7**).

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7.4.8 Operational-Level Integration Plan

This describes the parameters for SUS integration (i.e., assembly, connection, and deployment) in readiness for manufacturing. A risk-based approach should be used because of the structural and functional diversity of SUSs ***(91)***. Some of the important parameters to consider during deployment include:

- **Packaging and container integrity —** It must be determined whether preuse leak testing of SUS is feasible. Manufacturing, shipping, handling, and operational procedures should be qualified to maintain SUS integrity. This can be achieved by implementing control strategies within the end user's organization and between supplier and end user.
- **SUS testing and release —** Testing and release vary with use. The same off-the-shelf system may have different test and release criteria depending on process-, product-, regulatory-, and user-imposed requirements.
- **Reuse of components —** In general, reuse of SUSs is not recommended. The policy for this should have been established in the SUS strategy (**Section 7.2.1.7**). However, there are circumstances in which the reuse of SUS may be acceptable (e.g., use of a container as a mixing vessel or reservoir for buffer concentrates in which there are no broken connections). A risk analysis should be performed to determine if the risks are acceptable.
- **Training —** Appropriate personnel training is essential because the collapse of a bag may lead to stress cracks that could result in a loss of structural and functional integrity. Training topics might include the use of different sterile connection and disconnection devices or techniques for avoiding bag creases or holes during deployment.
- **Shelf life —** A coherent expiration and shelf-life policy should be implemented, especially in situations in which the end user has a large number of different SUSs from multiple suppliers or local, self-made assemblies.
- **Quality assurance —** Ensuring quality during deployment should be addressed at an early stage of implementation. Technical diligence may enhance this goal (**Section 7.4.10** and **7.7.3**).

7.4.9 Technological Survey

The aim of this document is to complement the information gathered as part of the manufacturing strategy (**Sections 3.0, 7.0,** and **7.1**). Depending on when the manufacturing strategy is written, an update to the document may be necessary. However, if a greater level of information is obtained in S1 and entry into the full implementation process takes place at G4, then this survey can be omitted. Adaptations of the implementation road map are addressed in **Section 7.1.** However, if the organization of the road map is changed, care should be exercised that the implementation process does not lose structure and that the comprehensive treatment of all of the elements necessary remains in place.

The components for SUS implementation often come from a range of suppliers, and a SWOT-type assessment (i.e., strengths, weaknesses, opportunities, and threats) may be used to evaluate and compare solutions. The aim is to make sure that the best available technology is selected while also obtaining pertinent information for including any further requirements in the specification process. Sources of information may include:

- Suppliers
- In-house know-how
- Industry networks and associations
- Journals
- Conferences
- Industry guidelines

Suppliers may be considered as stakeholders.

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7.4.10 Supplier Selection and Supply-Chain Review

The aim of this review is to determine a policy for the identification and selection of suppliers in order to complete the PEP and project planning.

The selection of appropriate SUS suppliers can often be an arduous task that can be made less cumbersome with a risk-management approach. For a successful SUS implementation, it is essential to have a detailed and transparent relationship between suppliers, vendors, and the end user. This requires good communication and understanding of stakeholder management in order to garner the expectations of both the suppliers and the end users. Being able to share pertinent information with suppliers at the early stages of the project may speed up the design process. Note that in order to obtain full benefits from this approach, the exchange of uni- or bilateral confidential disclosure agreements (CDA) may be required. The implementation and management of an end user–supplier relationship via technical diligence is described in **Section 7.7.3.** Suppliers provide material data, advice on applying typical designs, and the following documents:

- Process flow diagrams (PFDs)
- Piping and Instrumentation Diagrams (P&IDs)
- Component lists
- Physical layout proposals
- Operation manuals
- Pricing
- Qualification packages, including guidelines on E&L, for example

For both large companies and SMEs, it is critical to prevent the selection of equipment and suppliers from becoming an ad hoc process because that can lead to unnecessary and costly redundancies. In addition, working with too many suppliers may divert limited resources that could otherwise be gainfully employed in other parts of the SUS implementation process. Since the degree of business integration of a particular SUS supplier may depend on the size, location, and SUS assembly concerned, it is important to adopt a risk-mitigation approach to SUS supplier qualification and validation (**Section 7.7.3**).

Starting materials such as resins and films should be treated as critical raw materials as part of the verification process due to the complexity of SUS implementation ***(18)***. It is important to begin the supplier technical diligence (i.e., obtaining a thorough understanding of the SUS product, supplier, and compatibility of the SUS with the end user's process and facility) at this stage. These principles are based on established procedures for critical raw materials as recommended by the Association for Professionals in Infection Control and Epidemiology (APIC) due diligence process, ASTM E2500-07 vendor technical capability as part of the PVV process (**Figure 7.2.7.1-1**), and quality by design (QbD)-inspired raw materials qualification processes ***(18,92–94)***. Full details on how to implement the technical diligence process are described in Section 7.7.3. Note that this task may require the exchange of uni- or bilateral CDAs or nondisclosure agreements (NDA).

The quality of the entire supply chain should be assessed, qualified, and implemented via a structured risk-management process ***(72)***. All parties in the supply chain should make sure that their activities maintain business continuity and support the health of patients. SUS equipment that is purchased occasionally may be done on a contractual basis, but a quality and supply agreement must be used to buy routine SUS equipment (**Sections 7.5.7** and **7.7.12**).

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7.4.11 Extractables and Leachables Database

It may be convenient for the end user to begin collecting E&L guidance and testing data. Note that this can also be implemented in the next stage, Implementation S5: Development. At a minimum, however, the end user should collect extractables data at this stage.

7.5 Implementation S5: Development

During this stage, the SUS user requirements are developed, including layout and design, specifications, and standardization. Complete user requirements should precede the procurement process (e.g., commercial negotiations between strategic project partners or raising a purchase order).

7.5.1 User Requirements for Implementation

User requirements should be prepared at the start of the development process. The principal user and stakeholders must be identified and their requirements obtained. The user requirements are based on product and process knowledge, critical quality attributes, and any relevant regulatory requirements. The principal user should also provide the justifications for the resources required to design, purchase, qualify, and implement any new process, system, or equipment. These justifications can be developed prior to selecting an SUS or supplier, or they may be developed for a particular system. Some or all of the requirements can be based on a QbD approach. In this case, the product and process-control strategy performed earlier in the QbD process will become a prerequisite for defining the SUS requirements.

User requirements and functional specifications may be recorded together in a single document for low-risk, small-scale SUS. However, multiple user requirements may be needed for complex systems or processes. In this case, it would be critical to avoid conflicts across multiple documents. Any necessary conflicts must be identified and documented. The user requirements are created by the end user or on the end user's behalf; however, some supplier documentation may be necessary to supplement this. Supplier documentation is acceptable if the supplier follows GEP ***(90)***, has an appropriate quality system, and is technically capable according to ASTM E2500-07 and *PDA Technical Report No. 60: Process Validation — A Lifecycle Approach* ***(8,18)***. In some cases, the supplier documentation could replace the company document. The user requirements are kept current throughout the system's lifecycle and should be reevaluated after any change in intended use, user needs, or altered materials specifications (e.g., the films used, functional specifications, regulatory assessments or supplier quality audits, technical diligence). An example of a user requirements document is included in **Section 8, Appendix 1.**

7.5.2 Preparing User Requirements

Brainstorming sessions, workshops, interviews, assessments, and ad hoc discussions may be used to gather pertinent information about the needs of the end user. Other useful information may be obtained from suppliers, case histories, and workflow analyses. These requirements may not be fully defined initially but may be developed further during subsequent phases of the project. It is crucial to determine the level of interest of each stakeholder has in the project and ensure that stakeholders receive adequate and regular communications throughout the SUS process lifecycle. User requirements statements must be written to a level of specificity that:

- Describes the SUS function
- Allows end users to verify that the system meets their functional requirements
- Addresses the compatibility of processes and materials (e.g., polymer films)

However, they must be general enough to avoid constraining the system design. If a supplier will provide specific functionality, then the user requirements must be specific enough to determine that functionality.

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Any particular safety concerns and training needs resulting from the requirements should be noted for later implementation. Finally, make sure that a change-control process is in place for this process.

A functional risk-assessment approach may be used for further evaluation of critical requirements for the quality and function of the SUS. For instance, if an integrated SUS is composed of components from multiple sources, it may be necessary to identify the CQA of each component. An example would be a bioreactor system in which the hardware and SUS bags are sourced from different suppliers. Requirements that are not deemed to satisfy CQA may be further evaluated to determine whether they express a regulatory requirement or not. The project team may apply other criteria such as priority, clinical development stage, importance to the business, and other factors that may influence company decisions.

7.5.3 Layout and Design

The layout and design of the SUS can often be achieved using a standard design format; however, in certain cases, process-specific requirements may result in the need for a unique design process (refer to **Section 4**). A structured approach like the one from the S88 batch-automation standard may be used to attain completeness of design and control over interfaces because it focuses on the actual batch operations ***(95)***. This approach examines three key elements:

- Operations
- Equipment (reusable parts)
- Consumable materials (SU parts)

Start the design process with the functional requirements of the SUS in mind. **Figure 7.5.3-1** provides an example of a scope drawing, and the accompanying listing below shows typical considerations for designing a generic SUS within this context.

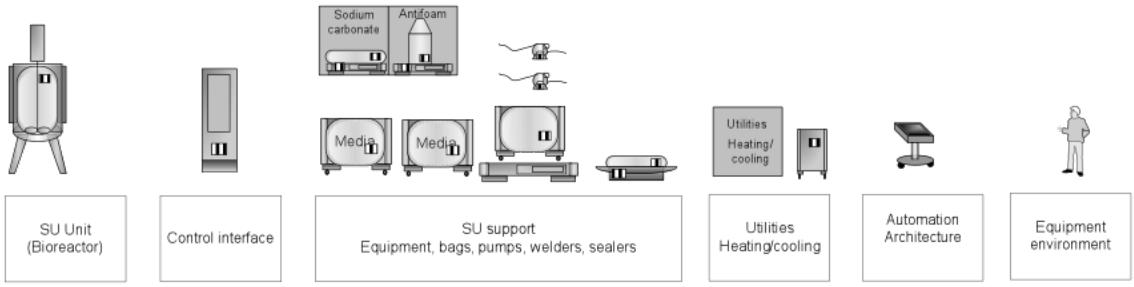

Figure 7.5.3-1 Example of SUS Installation Scope Drawing

This schematic representation of the flow path of an SUS facility shows that design meets both spatial requirements and container staging.

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Typical Considerations:

Operations

- Transportation, delivery, and installation
- Process operations
- Ergonomics
- Sufficient space for operations
- Manufacturing operations, including the following staging processes:
  - Setting up
  - Charging
  - Processing
  - Feeding
  - Discharging
  - Storing
  - Dismantling

Equipment (Reusable Parts)

- Bag support
- Control system
- Heating/cooling device
- Pumps
- Scales
- Tables
- Layout (i.e., the physical arrangement)

Materials (SU Parts)

- Process-containing part (e.g., bag, cartridge, capsule, membrane)
- Volume
- Agitation
- Footprint
- Sampling
- Sensors
- Inlets and outlets
- Films
- Tubing
- Connectors

7.5.4 Specification

The design process produces a series of specifications for the proposed SUS. The user requirements cover both MU and SU parts and can be communicated through the following means:

- Descriptive text
- Calculation
- Diagrams, drawings, and PFDs
- Reference to existing equipment

Implementation of a periodic design review procedure is critical for success. The delivery and installation process must be examined carefully to ensure that the SUS will be delivered in a form that will integrate into the facility.

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7.5.5 Standardization Policy

Although the inherent flexibility at the core of SUS design is an advantage, it could degenerate into a disadvantage by introducing unwelcome variations in the supply chain if left unchecked. Therefore, introducing a standard design policy across the project may reduce the incidence of variations, especially as the scale of operations becomes more complex. Implement a standardization policy as part of the user requirements according to the following guidelines:

- Working volumes may be chosen so that bag ranges cover a series of application ranges.
- Connections and material on bags and tubing may be chosen so that several possibilities for connecting are maintained (e.g., providing weldable tubing and a typical connector together).
- Setting up an SU toolbox consisting of Y pieces, sample manifolds, and lengths of tubing for special or infrequent applications will allow a more specific design element to be attached to a more standardized bag design.
- An active standardization policy may have been defined as part of the SUS strategy (**Section 7.2.1**). If so, it may not be necessary to implement it at this stage.

7.5.6 Extractables and Leachables

In some cases, an E&L database collection may have begun in S4 (**Sections 5.6** and **7.4.11**). If not, E&L data should be compiled now. Although there is no specific regulatory guidance document for E&L in SU systems, the USP, FDA, and EMA have extensive E&L guidance documents for SU technology ***(96-100)***. In addition, a number of cross-industry initiatives are in place to standardize the approach for SUS and achieve industry and regulatory consensus or provide a cross-industry database of current knowledge ***(100–102)***.

7.5.7 Procurement

Although procurement procedures will differ from company to company, it is critical to make sure that the end user requirements and other relevant specifications are correctly used and that the SUS-based equipment adheres to those specifications. It is important to maintain a close partnership between suppliers and end users in order to facilitate the procurement process. Some of the key points to consider are as follows:

- The main procurement process will be outlined and planned in the PEP.
- Key stakeholders must be identified and their roles and responsibilities must be clearly defined. This is extremely important in cases where procurement precedes the implementation of a quality and supply agreement.
- Contractual issues must be clearly defined. The end user's legal team should have reviewed and, if necessary, negotiated any modification(s) to the supplier's general terms and conditions of sale to ensure that the user's business, financial, and operational expectations are met (e.g., compliance with Incoterms 2011) ***(103)***.
- Good project planning and management principles must be carefully followed so that the project remains on track throughout its life.
- A change-notification process should be in place between the equipment supplier and the end user in order to monitor any design changes that take place at this point.
- Appropriate verifications must be conducted and the acceptance criteria met on a timely basis in order to avoid unnecessary delays.

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- A technical diligence and quality audit process should be implemented. By now, the supplier and SUS should have been evaluated by a site visit. Any design changes should be assessed for their compliance, the supplier's ability to adhere to the modifications, and the end user's process compatibility (**Section 7.7.1**).
- The delivery and installation process should be agreed to with the supplier. Please note that not all suppliers include these facilities in their basic offers; hence, they can take time to define properly and add significant costs to the total investment (**Sections 7.6.1** and **7.6.2**).
- It may be appropriate at this stage to initiate quality and supply agreements for routine SUS procurement (**Section 7.7**).
- Business contingency planning and crisis management should be reviewed for potential problems or the need to make necessary changes to project planning, construction, and delivery or receiving because these can have profound impact on the end user's startup and C&Q processes.

7.6 Implementation S6: Testing and Validation

During this stage, methods and documentation must be developed for equipment SOPs, maintenance, qualification and process validation, HSE processes, and personnel training. Some documents (e.g., the validation master plan) may require updates at this stage to maintain compliance.

7.6.1 Delivery

The PEP must cover the delivery process, including responsibilities of the supplier and the end user. Transport logistics are sometimes complex, and the procedure by which the equipment will be safely shipped (transported) and unloaded from its delivery vehicle must be considered. Some suppliers do not take unloading into account, and the end user may need to take particular precautions here or ensure that the supplier contractually takes responsibility. There may be specific non-GxP issues to resolve such as documentation (delivery notes), inspection for inventory or damage, and insurance or the removal and appropriate disposal of crates and other packaging materials ***(103)***.

7.6.2 Installation

The PEP will also cover all aspects of the startup process, including post-delivery storage, handling prior to installation unpacking, and the responsibilities of the supplier and the end user in a more general manner. Transferring the equipment to its place of use can be problematic if equipment dimensions or facility access has not been reviewed in advance. Preparing the area, particularly in cases where the SUS equipment will be used in manufacturing operations, may require specific training. The overall strategy and approach will depend on the nature of the SUS and the level of risk associated with this process (**Section 7.4.1**). In some cases, non-GxP documentation may be generated or supplied and work permits (e.g., fire safety) secured during the installation processes (e.g., removals).

7.6.3 Qualification

Qualification of an SUS should include lifecycle testing that reproduces the most impactful steps of the system's lifecycle (e.g., sterilization, transport, freezing/thawing) followed by a verification of the key functionalities and performance attributes (e.g., integrity testing). The lifecycle testing must be designed to cover the intended working space. Please refer to **Sections 5** and **7.7** for a discussion of the SUS qualification process and supplier qualification, respectively.

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7.6.4 Training

Training is a GMP requirement and an integral part of SUS technical implementation. The goal of any training program should be to provide specific knowledge and understanding that will support successful operations. Although, the specific training needs are normally described in the user requirements, other training requirements (e.g., for project management and risk assessment) may be critical for the successful implementation of the project. Training programs for SUS should include concepts and principles of operation that might be unique to SUS and unfamiliar to MUS operators.

7.6.5 Training Workflow

Guidelines for a training workflow are as follows:

- Examine each part of the entire SUS PEP individually and identify the functional areas concerned.
- Identify the training needs in each functional area and assign stakeholders.
- Assemble training requirements in a training matrix (example in Section 10, Appendix 3).
- Determine the scope of each training module. Some subjects may require more than one level of presentation and may require periodic retraining.
- Make sure that a trainee assessment or evaluation process is included at the end of the training module; this is a GMP requirement.
- Take the resources as well as the subject and complexity of the training requirement into account when planning training procedures.
- Some training modules may be delivered by competent team members or operators who are familiar with SUS implementation. Other training modules may be delivered by suppliers.
- Add completed training modules to the individual training records of each stakeholder.

7.6.6 An Example of SUS Training

Factors that may contribute to the leakage of bags are:

- Poor handling during deployment
- Mechanical stress on films or welds
- Accidental puncture or stressing of the SUS during handling and setup

Incidences of bag leakage may be reduced through the following interventions ***(104,105)***:

- Changing storage bin and bag design, if appropriate
- Optimizing the method of deployment (i.e., installation)
- Providing operational training
- Providing bag-handling awareness training
- Implementing a control strategy (e.g., remedial procedures by the supplier)
- Monitoring location and type of bag damage for trending

The in-house training program should be based on operational parameters and operator feedback. Operators should be educated and engaged about:

- Working areas and surfaces
- Bag installation and filling

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- Prior root causes for failure
- Leakage history and trends
- Root-cause analyses

Additional details on training are provided in **Section 6.**

7.6.7 Safety

Health and safety issues involving the control of both recognized and unidentified hazards in order to achieve an acceptable level of risk in the workplace are the responsibility of all stakeholders. Ensuring safety in the workplace as required by local laws can be best addressed by preparing an HSE plan.

7.6.8 Preparation of a Health, Safety, and Environment Plan

The HSE plan addresses health, safety, and environmental concerns throughout the SUS project. It should be prepared at an early stage and incorporated into the user requirements. Published examples of an HSE plan (although not related to SU technology) can be found in the literature ***(106,107)***. The HSE plan lists stakeholders and those responsible for performing specific HSE activities. These activities may range from mapping and analyzing the impact of HSE to developing and implementing process improvements. A workflow for developing an HSE plan may consist of the following steps:

- **Screen —** Quick clarification of any HSE aspects related to the project
- **Define —** Overall evaluation of significant HSE concerns during the project lifecycle
- **Plan —** Final HSE plan with stakeholders, activities, responsibilities, and deadlines

The screening and defining activities involve the complete evaluation of the project from an HSE point of view in order to determine HSE concerns that require immediate action. Some of the HSE concerns for a typical SUS project are presented here.

Health

- Ergonomics
- Manual operations

Safety

- Flammables
- Equipment malfunction
- Improper operation of equipment

Environment

- Emissions
- Waste
- Energy
- Water
- Chemicals
- Fossil fuels and raw materials
- Consumption of raw materials
- Wrappings and other support materials
- Transport

7.6.9 Health and Safety Issues Related to SUS

SUSs often consist of plastic components that have low resistance to physical stress, pressure, and temperature extremes. A typical setup includes a range of support equipment (e.g., pumps, agitators, scales) that requires manual transportation, installation, and dismantling. This generates a unique set of HSE concerns, as presented in the following lists.

Safety

- Tubing ruptures at eye level
- Bag ruptures at eye level
- Spills and slippage
- Emergency stops for moving parts

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- Automatic restarts after power cuts
- Tubing on floor that may result in tripping
- Loose parts (e.g., media containers, peristaltic pumps) that can fall
- Mobile containers over feet or hands, mobile containers squeezed in doorways
- Mobile containers' finish, corners, and handles
- Connections failing for compressed air and other gases (e.g., CO_2 and nitrogen)
- Retention of infectious material
- Failure in connecting to CIP/SIP equipment
- Position of emergency stop during filling operations may be out of sight from dispensing vessel

Health

- Ergonomics for installation and removal of SUS components (e.g., installing bags, filters)
- Ergonomics for handling bags that contain liquid
- Stress of communicating during filling operations that may be out of sight from dispensing vessel
- Lifting heavy or cumbersome objects
- Insufficient work space and surfaces

7.7 Implementation S7: Launch

The PEP leads to an initial procurement of equipment for the SUS project in addition to setting out a secure supply chain for continuous operation of the SUS. Initial supplier evaluations are part of the PEP preparation in S4 (**Section 7.4.10**). In the following paragraphs, these first steps are consolidated with a detailed evaluation of the appropriate tools for the routine management of suppliers.

7.7.1 Management of Materials

Since continuous sourcing of supplies is critical to keeping the SUS in a productive state, it is important to maintain an efficient routine supplier management (e.g., technical diligence, quality audits, quality and supply agreements, and purchasing, logistical, and waste-handling strategies) to create a robust support for SUS manufacturing operations.

7.7.2 Routine or Operational Procurement

Routine procurement for SUS (e.g., single-use bioreactors or chromatography systems) requires a supplier qualification program for each of the SUS suppliers included in the PEP. Whenever the risk and complexity of the component in question is considered significant, the qualification program should be implemented using a risk-management approach. Some of the factors to consider are as follows:

- The end user may have to adopt a different approach to the procurement of integrated SUS equipment versus routine SUS components (**Sections 5.2, 5.4, 5.5, 6.12.1, 6.14, 7.5.7,** and **7.6.2**).
- A backup policy of sourcing the same component from at least two different vendors should be adopted, and the suitability for the backup to be interchangeable with the original must be defined.
- The end user must evaluate and audit the supplier's manufacturing capabilities and supply chain to determine the supplier's ability to increase output on demand.
- The relationship between the end user and the supplier should be as transparent as possible in order to enhance the unimpeded operation of the SUS.
- The quality systems of the supplier should be examined as part of a quality audit process, and quality and supply agreements should be implemented (**Section 7.7.4**).

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- The supplier should be provided with a detailed description of the manufacturing process of the SUS, film quality, biocompatibility, and testing (e.g., bioburden monitoring of the SUS), particulates, extractables and leachables, and testing of leakages ***(108)***.
- A contract (supply or manufacturing agreement) describing the commercial conditions that relate to the supply chain, guarantee of delivery, and transport quality should be implemented ***(109)***.
- Both the supplier and the end user must institute robust change-control processes in order to prevent ad hoc modifications to the SUS.
- Time lines from the design to arrival of both MUS and SUS parts should be monitored.
- Evaluation models (preferably at the end user's site) that use mock-ups of preliminary SUS designs (available for training and simulations) should be tested. Testing is an important risk-mitigating procedure. This process may have been started as part of the SUS strategy (**Section 7.2.3**).
- The end user should adhere to green and sustainable manufacturing policies throughout the entire supply chain and evaluate waste disposal for carbon footprint reduction.

7.7.3 Technical Diligence

A QbD approach is the best way to qualify suppliers for SUSs. It is based on ICH principles (ICH Q8) and it embraces the complementary concepts of quality-risk management and pharmaceutical quality-system management (as detailed in ICH Q9 and Q10), process validation and verification, and stakeholder management (**Sections 7.2.1, 7.2.2** and **7.2.3**). These approaches lay the foundation for quality-driven processes built on a science- and risk-based approach and understanding. This is the case not just for the process for which the SUS will be used but also for the SU technology that will be deployed, thus supporting patient safety and product availability ***(3–5,18,77,106,110-114)***. The QbD approach allows for the identification and management of critical sources of variability in which control strategies can be used to maintain a state of control and facilitate the application of continuous improvement throughout the SUS product lifecycle ***(114)***.

Technical diligence can be started at S4 (**Section 7.4**). Based on established principles, it involves a carefully documented visit to the supplier's site ***(18,92–94)***. The outcome should be an appropriate and measured understanding of all the technical issues pertaining to the SUS manufacturing process, readiness for installation, and operation at the end user's facility. Technical diligence should be considered an important step in the supplier qualification process and a complement to the quality audit process. The principal differences between technical diligence and a quality audit are summarized in **Table 7.7.3-1.**

Table 7.7.3-1 Principle Differences Between a Quality Audit and a Technical Diligence Assessment

Description	Quality Audit	Technical Diligence
Purpose	Compliance, quality systems	Extensive technical understanding of design, manufacture, fitness for purpose, and control strategy
Personnel involved	Quality team(s)	Scientists, engineers, SMEs, and the supplier's technical and manufacturing staff
Scope	Focused, supplier's quality systems	Open-ended scope will vary depending on project, supply-chain transparency, subsuppliers back to source

Some of the important factors to consider when performing technical diligence are reviewed in **Table 7.7.3-2.**

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Table 7.7.3-2 Some Pertinent Factors for a Technical Diligence Assessment Checklist

Supplier Information	Supplier Technical Capability	End-User Product Suitability and Compatibility
CDA implemented	Design: engineering drawings and specifications	Fit with the end user's process
Company and stakeholders	Risk to the end user	Fit with the end user's facility and warehousing
Level of supplier integration	Resins, films, and other components	Prototype testing procedure
Sourcing and supply chain	Manufacturing process overview	Qualification
Contingency planning	Irradiation or other form of sterilization	Deployment method
Plant tour	SUS shelf life	Training and HSE

Determine the supplier's degree of integration:

- **Vertical:** The supplier manufactures practically everything on-site, from the film to the finished and packaged SUS
- **Horizontal:** These activities are spread over more than one site
- **Mixed model:** A combination of vertical and horizontal integration

The mixed model has become increasingly common. Resin is nearly always sourced as a bulk from larger industrial companies, and the SUS supplier company may be specialized in the production of SUS for pharmaceutical operations and have facilities on more than one continent. In some cases, it may require more than one technical diligence visit to the supplier's sites to address all the relevant technical concerns (e.g., the film's blow-extrusion may be performed at a different site than that used for SUS assembly and testing). The end user should pay careful attention to supply chain issues in this context. Depending on sourcing from another continent or from multiple continents could easily be disrupted by many factors beyond the end user's control and should be assessed from a risk-based approach. A quality audit may be executed either following or in parallel with the technical diligence process (e.g., an audit questionnaire can be part of the technical diligence process for phase I IMPs). A practical example of the technical diligence process can be found in Brown's presentation on *Implementation and Technical Diligence (115)*. **Table 7.7.3-3** shows an interpretation of the SUS implementation process for a drug substance using a lifecycle approach.

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Table 7.7.3-3 An Interpretation of the SUS Implementation Process

Clinical Status of Product Manufactured	SUS Implementation Stage According to TR Section 7.1, Figure 7.1-1	FDA Process Validation Guide Stage	Supplier Qualification Activity	Presence of a Quality Agreement	Risk and Complexity
Preclinical material	Technical assessment (1), detailed investigation (2), and scoping (3)	PV S1	Collect supplier and SUS product information	Optional	High
IMPs for toxicology and phase I	Business case (4), development (5), and testing and validation (6)	PV S1–2	Perform technical diligence	Yes	High
IMPs for phases II and III PPQ conformance batches	Launch (7)	PV S2	Perform technical diligence and an audit	Yes	Moderate
Licensed product	SUS exploitation stage	PV S3	Continue supplier verification	Yes	Low

7.7.4 Quality and Technical Agreements

Quality and technical agreements clarify the functional aspects of a business relationship by delineating the responsibilities of quality, regulatory, and production personnel within each company in order to enhance communication and prevent ambiguity as referred to in Section 6 ***(109,110–114)***. A quality agreement should not be confused with a business, supply, or manufacturing agreement. Several templates for APIs are available for chemical and biological products, and the end user may adapt them to an SUS from the Bulk Pharmaceuticals Task Force (Society of Chemical Manufacturers and Affiliates or SOCMA) or the International Pharmaceutical Excipients Council (IPEC) ***(116,117)***.

7.7.5 Logistics and Storage

The storage and warehouse facilities required for the efficient operation of the SUS should be defined in the PEP. In general, the amount of pallet space required may be calculated as a function of consumption and the dimensions of the shipping container. A risk-based approach may be used to allocate storage space for critical consumables with limited or no alternative suppliers, while standard consumables can be sourced as and when they are needed. Value-stream mapping (a lean manufacturing technique used to analyze the flow of materials and information required to bring a product or service to the user) can be used to identify improvements in lead time or to identify inefficient operations. For example, a value-stream map can show where excess materials are accumulating. Some of the factors to consider are:

- High-risk consumables
- Standard consumables
- Storage policy
- Pallet space
- Frequency of delivery to consumers
- Environmental storage conditions monitoring and control (a GMP requirement)

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7.7.6 Waste

The SUS strategy and PEP address waste management as an important part of SUS implementation because the amount of solid waste increases significantly for an SUS compared to an MUS. Some of the topics the PEP covers include:

- Handling and elimination of process fluids before disposal ***(118)***
- Collection and storage of waste in the facility and avoiding cross contamination
- Inactivation of waste before it leaves the facility or site
- Dual-stream handling of waste where necessary (i.e., split infectious and noninfectious waste)
- Feasibility of recycling SUS components, including the infrastructure for collection and disassembly ***(119)***
- Collection and storage of waste outside of the SUS facility
- Transport routes
- Frequency of pickup
- Training
- Safety
- The impact of local, national or regional regulation

Section 5 also addresses the issue of waste for SUS.

An example of when regional or national regulations would be important would include an SUS facility that must eliminate waste such as biologics (i.e., naturally occurring pathogens, including transmissible agents, or GMOs) or highly active, potent, or sensitizing compounds. The use of landfills for disposing of SUS waste is legal in the United States but is not permitted in the European Union, where waste must be disinfected and incinerated prior to disposal.

Other end-of-lifecycle processes (e.g., the retirement or decommissioning process, as per GEP) can also be addressed at this stage. In addition, consider non-GxP issues (e.g., refurbishing and reusing an SUS, sale of an SUS, the effect of applicable national tax credits). The end user must be aware of international hazard guidance for biologics and chemical entities ***(120–122).*** A BPSA guide to waste disposal for SUS is available for reference ***(65).***

7.7.7 Post-launch Review

Regular performance review audits (**Figure 7.1-1**) between stakeholders after launching the SUS may provide valuable feedback for the implementation of risk-based change-control procedures in the current SUS as well as new SUS projects.

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8.0 Appendix I: Overall User Requirement Specification Example

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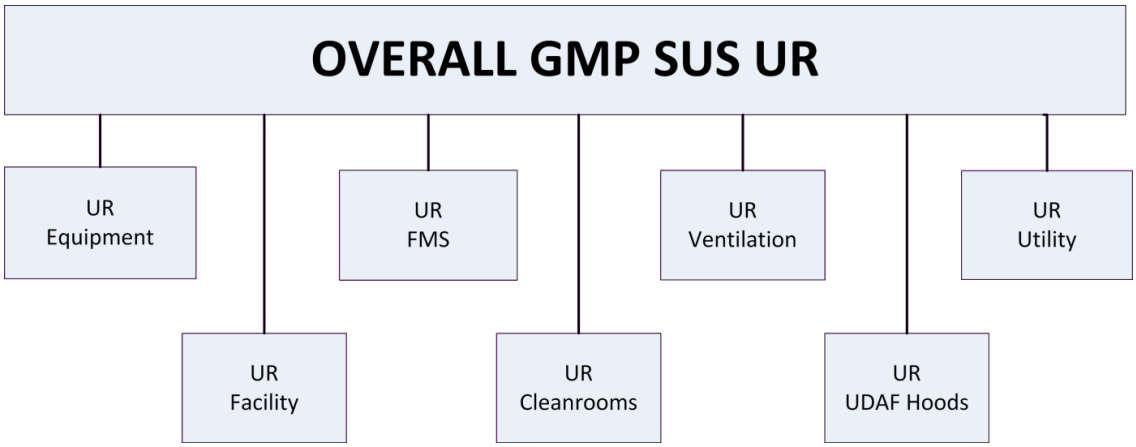
8.1 Description of Document

This appendix contains an example of user requirements (UR) written for an SUS facility. This material is referenced in **Section 7.5.1,** User Requirements, and should be used as an example for how this type of document is structured. The complexity and size of the document will change according to the type of SUS and complexity of the project. These types of documents should be considered as living (i.e., they can be updated as appropriate should the project or UR change). The document can be prepared by the end user or a sponsor of the end user such as an engineering company or supplier.

This document describes the UR concerning the new production suite, GMP, SUS (shown in figure below). This is an overall UR specification and detailed UR will be written. The requirements stated in this document may not be qualified but will form the basis for design.



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8.1.1 Abbreviations

API Active Pharmaceutical Ingredient
ASME American Society of Mechanical Engineers
BMS Building Management System
BSL Biosafety Level
CFR Code of Federal Regulations
CIP Clean-in-Place
CNC Controlled but Not Classified
CPE Cytopathic Effect
EMEA European Agency for the Evaluation of Medicinal Products
EN European Norm
FDA United States Food and Drug Administration
FMS Facility Monitoring System
GAMP Good Automated Manufacturing Practice
GEP Good Engineering Practice
IP Ingress Protection
ISO International Organization for Standardization
ISPE International Society of Pharmaceutical Engineering
HEPA High-Efficiency Particulate Air
HMI Human-Machine Interface
HVAC Heating, Ventilating, and Air-Conditioning
MAA Marketing Authorization Application
MCB Master Cell Bank
MVSS Master Viral Seed Stock
NAM National Agency for Medicines
PC Personal Computer
PEP Project Execution Plan
Ph. Eur. European Pharmacopoeia
PLC Programmable Logic Control
PMS Particle Monitoring System
QA Quality Assurance
QAP Quality Activity Plan
QC Quality Control
QMS Quality Management System
SCADA Supervisory Control and Data Application
SIP Sterilize-in-Place
SOP Standard Operating Procedure
SPA Special Protocol Assessment
UDAF Unidirectional Airflow
UR User Requirements
USP United States Pharmacopeia
VHP Vaporized Hydrogen Peroxide
WFI Water for Injection
WHO World Health Organization

8.2 Introduction

8.2.1 Description of Facility

Company X needs to equip a new facility GMP, SUS for the manufacture of injectable products for viral particle gene therapy. This document outlines the user requirements of the facility. In this project and UR document, abbreviation T4 stands for the main building site and GMP, SUS for the cleanroom area. The new production suite should be designed for biosafety level 2 (BSL2) and comply with EU GMP and FDA cGMP.

The products of this facility shall be based on modified live viral particle technologies (gene therapy). The manufacturing process shall cover production of biological bulk and final filled products for sterile, as parenteral applications. The initial adenovirus production portfolio covers in the first instance two bulk and two final products, which are known as Product 1 and Product 2.

Product 1 is a novel gene-based product that is given as a suicide therapy for tumors. Product 2 is a novel product being developed to deliver a growth factor for use in surgery.

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8.2.2 Environment

As a design basis for the HVAC in Company X, an upper limit of +30°C at 50% RH (63 kJ/kg) in summertime and lower limit of –37°C in wintertime for the outside climate will be used.

8.2.3 Capacity

The target capacities to fulfill the expected need of the market for the facility are summarized in the following list. Total number of vials will be about 130,000.

The process capabilities are limited by the drug substance manufacture. The drug substance manufacture will need 24-hour, 7-day-week control. However, the filling operation will be run on an 8-hour, 5-day-week basis.

8.2.4 Process

8.2.4.1 Production Process

The planned process for large-scale manufacturing of Product 1 and Product 2 vectors for the commercial scale process is shown below.

Thawing and expansion of master cell banks (API clean process)

- T-flask, roller bottles, and rocker bags

Infection of expanded cells with working virus seed stock (WVSS) (API clean process)

- Infection of the cell suspension with WVSS
- Full CPE monitored with analytical and process tools

Cell lysis and clarification of crude viral lysate (cCVL) (API clean process)

- Benzonase® and surfactant treatments to lyse the cells
- Depth filtration to remove cell debris

Benzonase® is a registered trademark of EMD Millipore

Buffer conditioning by crossflow ultrafiltration (API clean process)

- Diafiltration to reduce the amount of host cell– and growth medium–originated impurities
- Buffer conditioning suitable for chromatography

Chromatography (API clean process)

- Capture and polishing of recombinant adenoviruses

Crossflow ultrafiltration (API clean process)

- Concentration
- Diafiltration to remove the chromatography elution components (salts)
- Change to final formulation buffer standardization of concentration
- Freezing and holding in –70°C

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Fill and finish (aseptic process)

- Thawing final sterilization filtration prior to filling
- Washing and sterilization of vials
- Filling, stoppering, and crimping

Inspection, packaging, and storage

- Checking of container seals and visual inspection
- Labeling of primary containers and packaging
- Freezing and storage of the drug product

8.2.4.2 Assumptions for planning

- Disposable equipment is used during processing of the product.
- The production suite should be equipped with its own WFI facility and solution manufacture.
- Glass vials for sterile filling are brought into the facility ready to be washed and depyrogenated. Stoppers and aluminum crimps for filling are bought from suppliers presterilized.

8.2.4.3 Storage in GMP, SUS

There shall be storage areas for incoming raw materials and packaging material within T4. The GMP, SUS area shall provide storage of a batch material between process steps in the production. (Storage outside the process area is described in Section 4.9.)

8.2.4.4 Inactivation

All floor and process drains, such as product-containing waste waters, elution buffers, and so on, from the manufacturing areas shall be led to the collection tanks in the inactivation system. The connection to the collection tank shall be a closed system. Inactivation shall be carried out chemically or thermally in the balance tanks. After inactivation and neutralization, the drain liquid is transferred to the common drainage for transportation to the municipal wastewater treatment plant.

Floor and process drains from other areas shall be led directly to the common drainage or, when appropriate, to existing T4 balancing tank.

8.2.4.5 Waste Handling

Used equipment from the aseptic area and reusable equipment from the API areas are placed and sealed into plastic bags and decontaminated in a decontamination area (BSL2). The risk assessment will show the need for decontamination in the autoclave.

8.2.4.6 CIP

No CIP station is necessary due to the use of disposables. Elution solutions are used in order to clean columns and filter systems. A washing machine is used for washing filling-machine parts and filter holders.

8.2.4.7 Washing and Sterilization

Sterilization of reusable equipment shall be performed in an autoclave. Stoppers and aluminum crimps are bought from suppliers presterilized. Vials are to be washed and depyrogenized locally on entering the filling area.

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8.2.4.8 SIP

No sterilization-in-place (SIP) is necessary due to the use of disposables and the supply of media, buffer, and elution solutions.

8.2.5 Material Flows

The risk of cross-contamination shall be minimized in the layout of the plant area and in the manufacturing areas. Special attention shall be put on straight flows so that possibility for mix-ups between products from different process phases or different products is minimized. Line clearance shall be performed between production runs of different products. The facility shall be designed for sequential batches of cells to be expanded in cell culture at the same time. Attention shall also be put on keeping the BSL2 requirements in the layout.

8.2.5.1 Traffic to and from T4

The incoming raw materials shall be transported to the goods-receiving area of T4. In the same way, the final products are removed for transportation from the same T4 building. Rejected material is transported to incineration.

The following shall be delivered to or brought out of T4:

Incoming

- Raw material for the process
- Utensils, disposables, and components for production
- Clothing for cleanrooms
- Packaging materials
- Material for cleaning of cleanrooms

Outgoing

- Final products in cardboard boxes
- Clothing from cleanrooms
- Production waste
- Additional waste

Material flows for the production shall be separated from the other material flows.

8.2.5.2 Storage areas in GMP, SUS and T4

The main function of the storage areas in T4 is to receive and store the raw material needed in production and daily work and, when approved, to distribute it for use. Released final products shall be stored and dispatched to the customer in pallets or smaller packages. The storage areas also work as storage for hazardous and medical material and collection points for waste. The storage areas shall be designed for storage of at least two months' needs in production and consumption. The storage for finished products shall be dimensioned for six months' production.

There shall be storage areas for the following materials and products:

- Incoming raw material, packaging material, and pallets (existing)
- Raw material in cold storages (MCB in liquid nitrogen and MVSS in freezer)
- Intermediates during process (cold storages)
- Quarantined material and packaging material (existing)
- Approved bulk material and packaging material (existing)
- Rejected raw material and packaging material (existing)
- Finished products (freezers)
- Quarantined finished products (existing freezers)
- Approved finished products (existing freezers)
- Rejected finished products

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- Reference samples (existing)
- Spare parts (existing)
- Samples for stability studies (existing)
- Lining materials and used boxes (existing)

8.2.5.3 Material Airlocks

Transport of material or products between unclassified and classified rooms and rooms with different classifications shall be done through extracted airlocks or pass-throughs. The airlocks shall be of the one-way type because personnel and the material flow shall be unidirectional. The material or product shall be unpacked and sprayed with ethanol inside the airlock when transporting to a room with class C classification. Filling components passed into the class B area shall be washed and depyrogenated, or packages of presterilized components shall be subject to a vaporized hydrogen peroxide cycle.

8.2.6 Personnel Flows

The risk of cross-contamination shall also be minimized in the personnel flows. The main office building area has existing entrance, lunchroom, offices, and other social rooms.

There shall be emergency exits from the T4 and GMP, SUS buildings. If possible, the use of emergency exits should not compromise contained operation.

Personnel lockers and changing rooms for factory suit shall also be situated in T4. There shall be authorization control at the entrance to the manufacturing area, allowing only authorized personnel to enter. The working areas should be visible by inspectors and QA personnel for observation of the activities.

8.2.6.1 Gowning

Personnel working in the production areas and in laboratories shall wear suitable protective clothes for classified rooms and working in a BSL2 containment area. Personnel working in office areas, such as the central facility control room, may wear common clothes.

8.2.6.2 Personnel Airlocks

The first main objective with airlocks is to maintain a positive pressure in the classified area and, by that, ensure a proper separation of the individual rooms or areas when only cleanliness grades are to be considered. The second main objective with airlocks is to prevent contaminants from BSL2 active rooms from spreading to nonactive cleaner areas. The personnel entering or leaving the cleanrooms shall go through airlocks where gowning is performed to suit the classified area gowns. The airlocks can be of the one-way type. There should be unidirectional personnel flow in the active areas.

8.2.6.3 Estimated Number of Personnel

About 8–11: cell culture rooms, 2–3 persons; downstream, 3–4 persons; filling, 3–4.

8.2.7 Training

In order to assure that the personnel have updated information of their working area, all personnel shall be trained in GMP procedures and in using SOPs and instructions concerning them.

Sufficient training for the personnel shall be provided by the supplier of every process equipment and utility and automation system. The education level shall be maintained with additional training during the whole lifecycle of the plant.

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8.3 Regulatory Requirements

The production facilities shall be designed to meet applicable requirements in the US CFR, EU GMP, PIC/S, and applicable EU directives and Finnish legislation. All components and systems shall be designed and commissioned according to GEP, GAMP, and ASME standards and shall have a CE mark and a declaration of conformity.

The substances used and produced at the site shall meet the effective versions of monographs in the Ph. Eur., USP, or registered specifications. The compressed air and gases shall meet the demands in ISO 8573-1.

The cleanrooms shall be designed, constructed, and commissioned according to ISO 14644 and 14698, to meet cGMPs above.

8.4 User Requirements

8.4.1 Functional Area Requirements

The following areas shall be available in the manufacturing suite:

1. Material receipt and product dispatch
2. Storage for incoming materials (room temp and cold)
3. Quarantine area
4. QC laboratory control area
5. Released and rejected material storages
6. Weighing room (optional)
7. Master cell bank storage
8. Master virus seed stock storage
9. Airlocks for personnel and materials going between areas of different classification
10. Personnel locker room/changing room
11. Column packing room
12. Cell culture room, clean
13. Cell culture room, infected (upstream)/harvest
14. Purification room (downstream)
15. Filter integrity testing
16. Filling
17. Visual inspection, labeling, and packaging
18. Washing area
19. Decontamination
20. Clean area
21. Intermediate storage
22. Media/buffer preparation area
23. Storage for the final drug product (-70˚C)
24. Area for production utilities
25. Area for waste inactivation
26. Technical space for building HVAC and cleanroom HVAC
27. Area for cleanroom clothes
28. Area for maintenance equipment
29. Storage for cleanroom cleaning equipment
30. Control room/IT (CNC—external to change room routine)

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8.4.2 Utilities and Services

Type	Purpose	Area	Required?
Compressed air	Instrument air	All	Yes
CO_2	Rocker system	Fermentation	Yes
O_2	Rocker system	Fermentation	Yes
Electrical	Facility power	All	Yes
Process air	Rocker system	Fermentation	Yes
Plant steam	Heating of autoclaves, WFI	Decontamination	Yes
Clean steam	Sterilization	Sterilization	Yes
WFI	Washing machine, final rinse	Washing	Yes
Tap water	Washing, PW production	Changing rooms, utility production, process equipment	Yes
Hot water	Washing	Changing rooms	Yes
Chilled water	Cooling	HVAC	Yes
Drainage	Common wastewater	Changing rooms, utility production, process equipment	Yes
Waste active	Process waste solutions	Manufacturing areas	Yes
Waste inactive	Process waste (including solids)	Manufacturing areas	Yes
Liquid nitrogen	Cooling down and freezing of solutions	Storage before filling	Yes
Vacuum			No
Heating	Temperature control	Radiators (already installed), HVAC	Yes, existing municipal heating system
Solvents (EEx)			No
VHP			Has to be considered

8.4.3 Cleanroom and HVAC

8.4.3.1 Room Classification

The rooms are classified according to the needs of the process and BSL2 containment requirements. The cleanrooms are designed according to ISO 14644-1 but validated, operated, and monitored according to EU GMP and PIC/S. The protection classes of electrical equipment in the cleanrooms shall be IP54 and in other areas according to room activity. **Table 8.4.3.1-1** gives classification examples.

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Table 8.4.3.1-1 Sample Room Classifications

Activity	Class (ISO 14644-1)	Grade (EU GMP)
UDAF-hood and aseptic filling	Core area 5	A
Aseptic filling room	7	B
Aseptic formulation rooms, chromatography and filter stations, cell growth, buffer preparation	8	C
Sampling and weighing of raw material, clean corridors, and washing rooms	8 (at rest)	D
Visitors corridors, clean utility production, utility production, packaging hall, and all other controlled production areas	Unclassified	Unclassified
HVAC, utility production, social areas	Unclassified	Unclassified

ISO classes are, unless separately mentioned, defined as during operation.

Airlocks shall be clean-room classified according to cleaner classification when at rest.

8.4.3.2 Ventilation Equipment

The ventilation system is divided into separate systems, according to use (one for cleanrooms and one for technical basement). Space will be reserved for future expansion of ventilation systems, which will serve cleanrooms on the first floor.

The cleanroom ventilation systems consist of fresh-air-handling units to obtain the air changes specified for each room or classified area. The ventilation system shall be designed as a "full fresh air" system in order to ensure the BSL2 and prevent cross-contamination in the production suite. The cleanrooms and technical spaces with high heat generation shall be equipped with separate circulation air-handling units to maintain room temperature within acceptable limits. Fan filter units shall be used in nonactive (BSL2) cleanrooms. There shall be a redundancy in the HVAC for the primary fan motors at a minimum. Due to the fact that operators will be in the cleanroom only eight hours a day, there shall be a possibility to have a lower airflow at night.

8.4.3.3 Air Filtration

The air for basement ventilation is filtered in three steps: G4, F7, and F9 filter, according to EN 779, within the air-handling unit.

Cleanroom air is prefiltered in four steps: G4, F7, F9, and H12, according to EN 779 and EN 1822, within the air-handling units. The terminal filter is minimum H13.

Exhaust air from the area where live virus is processed is filtered in two steps: F7 and H14 filter. Exhaust air from the basement is F7 filtered.

8.4.4 Control and Automation System

8.4.4.1 Process Control System

The process equipment is of the stand-alone type, which means that every unit has its own control system. The system shall consist of a local PLC/PC and an HMI and it shall be possible to handle at least three user levels. Every user shall have an individualized user name and password to restrict the access and maintain the audit trail. The ability of a control system to communicate with a SCADA

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system shall be stated in the UR. There shall be electronic records from the process but no electronic batch records.

All utility systems and appropriate equipment shall have separate control systems, which control production, distribution, and quality of the utility. It shall be possible to connect critical parameters to a SCADA system.

There should be an overall data collection and viewing system for the entire production plant.

8.4.4.2 Facility Monitoring System

The cleanrooms shall have a facility monitoring system (FMS) consisting of a PLC and an HMI. The FMS shall monitor the critical parameters for room conditions [e.g. temperature, pressure, viable and nonviable particles, microbial presence, and moisture (some areas)] in every classified room. The critical refrigeration devices also have to be connected to the FMS. The whole GMP, SUS suite is controlled by the building monitoring system (BMS). Cleanrooms in grades A and B will be equipped with a continuous online particle monitoring system. The operational C grade areas are also equipped with online particle monitoring (e.g., manifold system).

8.4.5 Health, Safety, and Environmental Requirements

8.4.5.1 General

An HSE risk analysis shall be performed for the plant. Special attention shall be put on taking care of the environment. Contribution to the sustainable development shall be made by using natural resources rationally and by preventing health and environmental damage caused by waste. The generated waste shall primarily be reused, secondarily used as energy, and only as a final solution be transported to the dump. Special attention shall also be put on biosafety containment.

8.4.5.2 Air

Air from the production areas shall be filtered before leaving the building.

8.4.5.3 Drain

After inactivation, the drain liquid is processed further and neutralized in the municipal effluent plant together with all other drain systems.

8.4.5.4 Material

Waste material (e.g., biowaste, cardboard, plastics, paper, glass, and other waste) shall be sorted and reused if possible. If not, the material shall be transported to a dump or burned.

Hazardous and dangerous waste shall be gathered separately and moved to a storage area for hazardous waste. The waste shall be transported to incineration. Hazardous waste can be, for example, medical waste from production and equipment or returned medicines.

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8.4.5.5 Emergency Systems

Type	Purpose	Area	Required
Fire alarm	Personnel safety	All	Yes
Intruder alarm system	Security / Biosecurity	Outer walls	Yes
Sprinkler	Fire control		No
Cell phones	Personnel safety	All	Yes
Oxygen sensors	Personnel safety	In liquid N_2, CO_2 presence, excess O_2	Yes
UPS (uninterrupted power supply)	• Business contingency • Maintenance of bio-containment • Personnel safety	UDAF-hood, HVAC FMS including PMS, control systems	Yes
Diesel generator	• Business contingency • Maintenance of bio-containment • Personnel safety	UDAF-hood, HVAC FMS including PMS, refrigerators, and freezers	Yes
Waste active failure alarm	• Business contingency • Maintenance of bio-containment • Personnel safety • Pollution control	Basement	Yes
Leak and other appropriate detectors	• To avoid damage to equipment • HSE issues	Basement	Yes

8.4.5.6 Noise

A risk assessment regarding noise levels shall be performed. Both internal and external noise shall be analyzed.

8.4.6 Maintenance and Service

A maintenance program for the whole suite shall be developed before start of production. The program shall include the facility and technical equipment within, the ventilation system and equipment, the production equipment, and all laboratory equipment. The maintenance system shall produce, in advance, weekly reports of equipment that needs service.

All equipment in the area shall be assigned an item number, which shall be fed to the maintenance system. The maintenance personnel are responsible for service of all equipment. Two times a year (summer and winter) there shall be production stops for maintenance of the production equipment.

Spare parts for process equipment for two years of continuous production shall be stored.

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8.5 Documentation Requirements

8.5.1 General

All documents shall be marked with a unique name and identity according to the QAP. Documentation shall be in English; user manuals and instructions shall be in English. Documents and backup copies of electronic files shall be archived in a fireproof archive.

8.5.1.1 Version Handling

All GMP documentation and software shall be version handled according to internal procedures after the first approved version.

8.5.2 Risk Assessment

Every system, equipment, and room shall be evaluated in a GMP risk assessment in order to classify the demands for the commissioning and qualification work.

Depending on impact criticality on the product, the systems, equipment, and rooms shall be evaluated to be:

- Direct-impact systems
- Indirect-impact systems
- No-impact systems

Risk assessments for evaluation of safety of personnel and economical risks shall be carried out for every system, piece of equipment, and room.

8.5.3 SOPs

Existing SOPs are used according to Company X's internal QMS.

8.6 Validation and Testing

Validation and testing are performed according to VMP and internal SOPs. Qualification philosophy during the construction, commissioning, and qualification (IQ/OQ) project is according to QAP.

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9.0 Appendix II: Project Execution Plan Example

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9.1 Introduction

9.1.1 Purpose

This appendix contains an example project execution plan (PEP). This material is referenced in **Section 7.4.1,** Project Execution Plan, and should be used as an example for how this type of document is structured. The complexity and size of the document will change according to the type of SUS and complexity of the project. These types of documents should be considered as living (i.e., they can be updated as appropriate should the project change). The document can be prepared by the end user or by a sponsor of the end user such as an engineering company or supplier.

The purpose of this PEP is to define and describe the project execution strategy as well as how the project will be managed and how quality control is applied in order to ensure that the project meets its goals regarding cost, time, and quality.

9.1.2 Context of this Document

This document provides a road map for the work to be completed. The PEP is a "fit-for-purpose" and living document that defines the supervising engineering contractor's (SEC's) scope of services and responsibilities for the project.

This plan describes and defines the project scope, the execution strategy, quality control systems, and management plan.

9.1.3 Revision of the PEP

All information included in this PEP is current as of the latest issue date of each document referred to. If changes are made that significantly affect the execution strategy, project management, and/or quality control activities as defined herein, a revision shall be made with rationale for the change explained in the new revision. The approval of the new revisions follows the same procedure as the approval of the original PEP. If there is a change to an appendix or a referred document to this PEP, only the document will be revised.

9.2 Project Overview

9.2.1 General

Company X wishes to design and equip a new production suite, GMPX, for manufacture of injectable products for viral particle gene therapy.

The products manufactured in the new production suite shall be based on modified live viral particle technologies (gene therapies). The manufacturing process shall cover production of biological bulk and final filled products for sterile, liquid drug applications. The viral developments are based on adenovirus. The production portfolio covers in the first place two bulk and two final products, which are known as Product 1 and Product 2.

9.2.2 Client

The client for the GMPX project is Company X.

9.2.3 Project History

A basic design was earlier developed for a traditional non-SUS approach.

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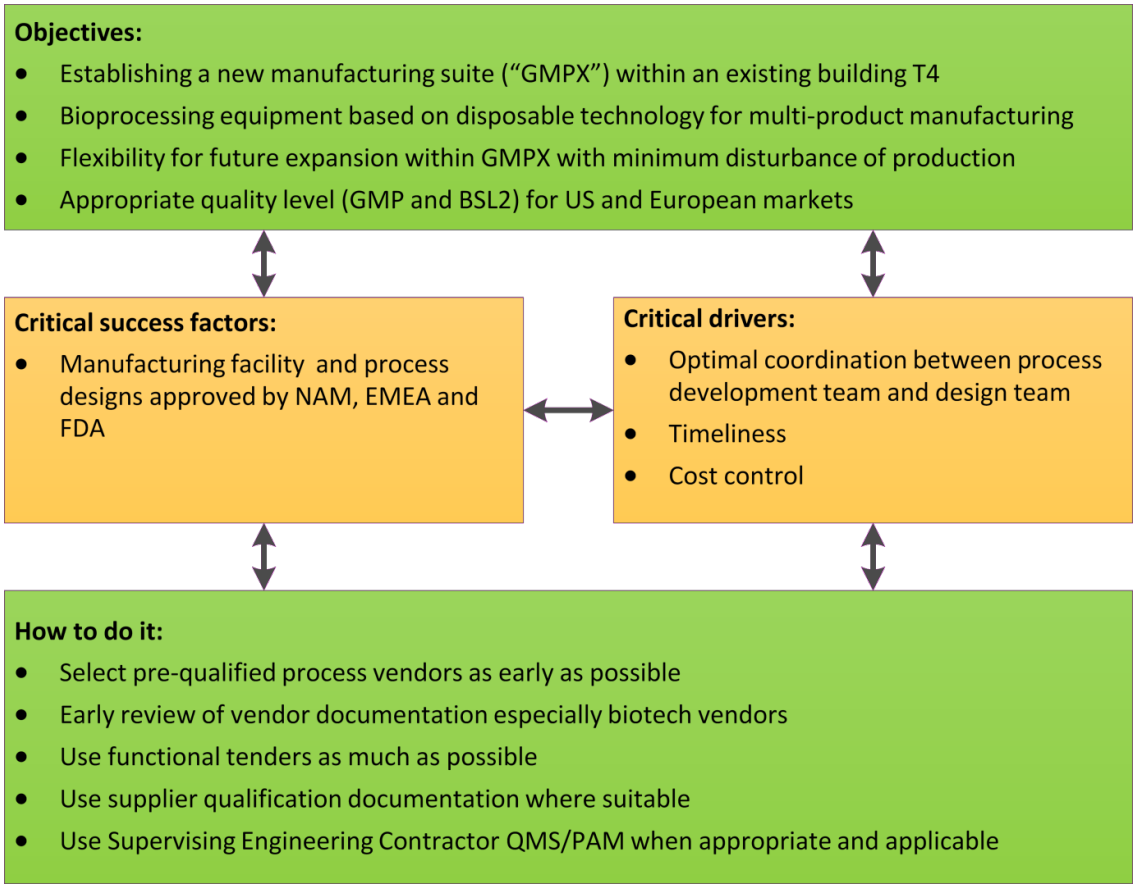
The building for GMPX is already constructed alongside building T4, and all basic utility installations are available there.

9.2.4 Project Scope

The overall purpose of the project is:

- To design and equip a new manufacturing suite, to be integrated into the T4 complex, for manufacture of injectable products for viral particle gene therapy with disposable technology
- To ffinalize OQ for the manufacturing suite within 18 months from start of basic design within agreed project budget

9.2.5 Project Objectives, Success Criteria, and Focus Areas



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9.2.6 Steering Committee

The SEC and Company X will implement a common steering committee for the project. The purpose and responsibilities of the steering committee are to:

- Assure that the project goals are met
- Create a common picture of questions, problems, and limitations within each organization
- Create a common picture of roles and responsibilities between parties
- Create a common picture of how project risks should be handled
- Approve project execution plan, schedules, economy, and planning
- Supervise project progress
- Approve monthly project reporting

9.3 Contract Documents

Basic Design:

- Contract document
- Project scope
- Project organization
- Reference group

Detailed Design and Execution:

- Contract document
- Project proposal
- Project contents (basic design report)
- Project UR
- Project methodology
- Process module diagram
- Layouts for floors 0 and 2 with BSL and GMP classification
- List of equipment and materials
- Impact assessment GMP
- Permits to be submitted
- Main project time schedule
- Definition of milestones
- List of man-hours and travel time
- Procurement strategy
- Project organization
- Manuals and training documents
- Definitions of GMP and GEP

9.3.1 Customer X Deliverables

Customer X is responsible for providing:

- Drawings of existing building
- Information on available utilities and other information in existing building
- Process know-how and product requirements and specifications
- User Participation
- Reviewer and approver of documents

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9.4 Scope

9.4.1 Objectives of the Scope

The GMPX project activities such as value engineering and thorough schedule control of all design, construction, commissioning and qualification (C&Q), and changes will have a high importance throughout the project.

9.4.2 Scope Supervision

The scope and progress will be supervised by the steering committee. Changes in the scope will be handled by change orders as described in **Section 4.6.**

9.4.3 Scope of Engineering

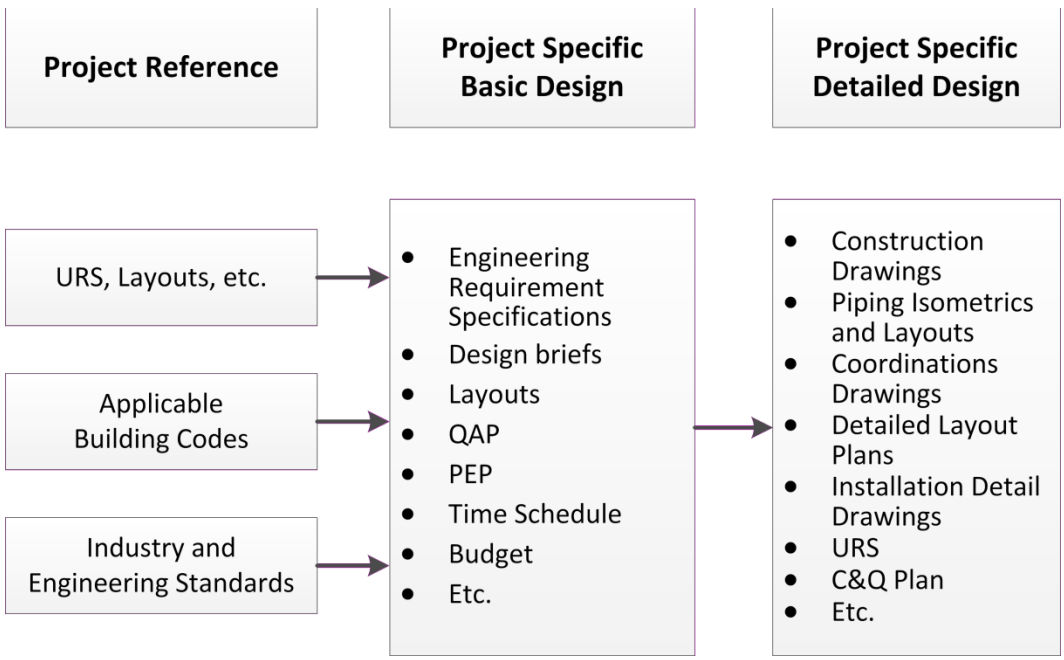
All engineering required to supply and build the process building is within the scope of the SEC. The scope comprises design development of the technical documents prepared during basic design to a detailed design status (i.e., the status required for manufacturing, installation, commissioning, operation, and maintenance of the facility), with all the installed equipment.

The basic design documents have been reviewed and approved by Company X and constitute the project baseline, setting the specific requirements for the project. Any changes from this baseline are to be taken up through the change management procedure.

Reference documents (including client inputs) used during basic design become backup documents. In case of contradiction between requirements specified in the basic design documents and requirements specified in any of the reference documents, then the latter shall take precedence, unless such requirement is superseded by a subsequent change management.

During detailed design the information set by the basic design package is used to develop construction documentation.

Above principles are illustrated in the schematic that follows, but the lists are not exhaustive.



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9.4.3.1 Design Steps Overview

The SEC will provide a complete design of the new manufacturing suite in the following steps:

- Front-end definition
- Basic design
- Detailed design

9.4.3.2 Basic Design

The SEC will prepare:

- Project proposal with executive summary project contents
- Project plan
- Biosafety level impact assessment

9.4.3.3 Detailed Design

The SEC will provide engineering with:

- Approved and signed piping and instrumentation diagrams
- Approved building layout
- Approved main documents such as UR, PEP, and QAP

9.4.4 Scope of Equipment and Systems

9.4.4.1 General

The scope includes all architectural considerations for cleanroom and mechanical spaces (with exception of the outer building walls) and finishes, building installations for HVAC, utility piping, safety, process piping, process equipment, electrical, and automation.

9.4.4.2 Spare Parts

The SEC will provide spare parts required for commissioning, testing, and qualification.

The SEC will also provide critical spare parts for two years from installation date.

9.4.4.3 Consumables

The SEC will supply the consumables, such as filters, gaskets, lubricant, lamps and lights, fluorescents, and bags, that are required until end of commissioning. Subsequent consumables shall be provided by Company X.

9.4.4.4 Disposables

Disposable bags for cell expansion will be supplied by the SEC based on the UR for the bags in a limited amount only if necessary for IQ and OQ purposes.

9.4.4.5 Utilities

The running cost for utilities during construction and commissioning and qualification (C&Q) will be accounted directly to Company X. All temporary utilities needed only during construction and C&Q will be supported by the project.

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9.4.5 Change Management

Management of project change is essential to successful project execution. The project will implement a change management process that will provide for timely identification of deviations or changes in project scope.

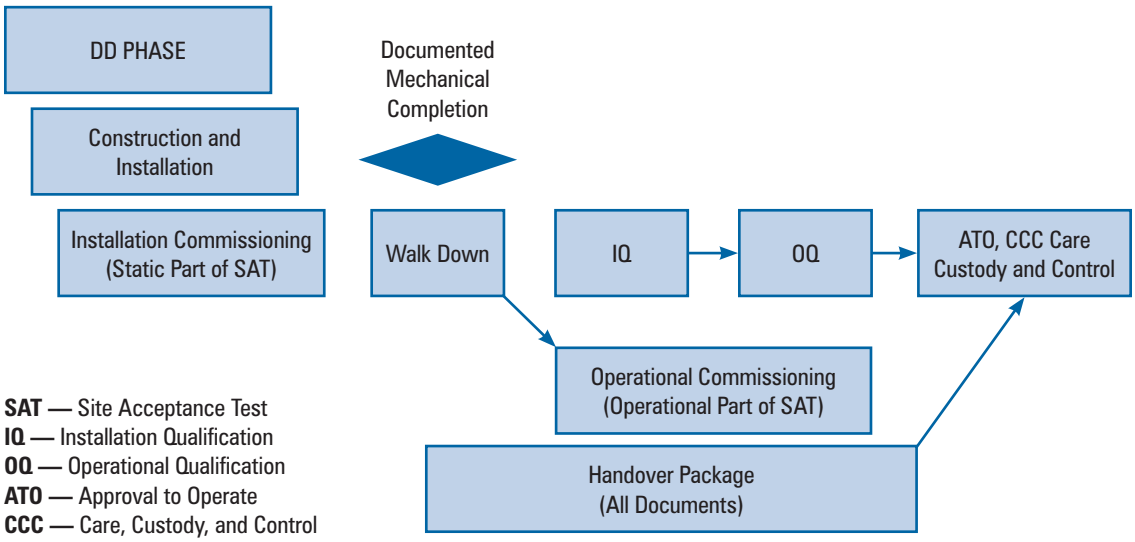
All change orders shall be recorded in a change-order log.

See QAP for change control of documents and C&Q activities.

9.5 Project Execution Approach

9.5.1 General

As shown in the diagram below, many project activities will overlap. Detailed design, construction and installation, and C&Q will overlap. This figure is schematic and does not apply to the project time frame.



9.5.2 Detailed Design (DD)

9.5.2.1 General

The SEC's operation department appoints and enrolls projects' design resources as are required by a project's scope and schedule. The request comes from the project manager or project leads by the project manager's approval.

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9.5.2.2 Design Disciplines

The SEC splits engineering into the following disciplines for this project:

Architectural

- Cleanroom

Process technology

- Upstream (cell culture and virus infection)
- Downstream (product purification)
- Fill and finish
- Packaging (inspection and packaging)
- Labeling
- Support (dishwasher, autoclave)

Utilities

- Black piping
- HVAC
- Electrical
- Clean utilities

Automation

QA is involved during the DD phase to review and watch over GMP requirements as an independent part of the project. See QAP, Doc. No. GMPX-QAP-07-001.

9.5.2.3 Engineering Organization

The project engineering manager leads and directs all engineering activities. The engineering manager is primarily responsible for providing backup to the leads in matters of coordination between the disciplines and/or for each discipline. There is a project design lead appointed for each discipline.

9.5.2.4 Design Execution

Detailed design and construction will be overlapping and partially concurrent activities.

During detailed design, detailed construction and installation drawings, specifications, schematics, test sheets, and other documents will be generated at a pace and content as is required by construction and installation.

The release of construction and installation documents is preceded by a series of reviews and approvals, internal to the SEC as well as together with the client's representatives.

See the QAP and C&Q plan for more details about the review and approval process.

9.5.2.5 Design Documents

Governing design documents will be sent to Company X for review and approval. Such documents are listed in the document progress table.

Detailed design documents that are required for construction and installation only will not be issued to Company X for review and approval unless such issuing is required to obtain building permits.

9.5.2.6 Permits and Authority Approvals

- Building permit: QVIM June 2007
- Authority review and approval before any construction activity starts
- Occupancy permits (if applicable); need to be applied for by Company X

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9.5.2.7 As-Built Drawings

Approved documents that need to be as-built have to be identified in a later stage.

These documents will be redlined at the site during construction and C&Q.

Redlined documents can be used as as-built documents for C&Q and handover of the systems. The work will be a joint effort of the SEC, vendors, and, if applicable, Company X.

These documents will thereafter be reissued (CAD) as revisions as-built by the SEC.

9.5.3 Supervising Engineering Contractor's Subcontracting and Procurement

9.5.3.1 General

The SEC will subcontract and procure material and equipment.

Subcontractors for design, support material, installations, and transports will be appointed by the SEC without prior information to Company X.

9.5.3.2 Subcontract Documents

Such subcontract documents that constitute technical specifications will be presented to Company X insofar as such documents relate to GMP equipment, systems, or components.

9.5.3.3 Equipment Log

Status of purchasing, design, fabrication, testing, and delivery of major equipment will be reported at project meetings.

9.5.3.4 Expediting

9.5.3.4.1 Equipment

During the complete delivery phase from contract to completed handover, the SEC supervises and checks the vendors' work and progress. This is made according to the contract with the specific vendor. A delivery of equipment or system is expedited in the following steps:

- Follow up milestone achievements according to agreed schedule continuously during design, construction, and testing at the vendor's workshop. This is performed by visiting the vendor and/or following up progress reports from the vendor. If deemed to be required, such follow-up will encompass the vendor's major subcontractors also.
- Review and approve the vendors' design documents.
- Perform and report FAT.
- Perform receiving check of equipment.
- Supervise and check installation and start-up.
- Supervise and attend qualification work, and review and approve qualification documentation (if applicable).
- Check and approve final project documentation.

The monitoring, expediting, and reporting of progress of work is executed by the respective design leads.

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9.5.3.5 Material

Purchasing and supply of material other than equipment will be monitored, expedited by the project's purchaser.

9.5.3.6 Subcontractors

All subcontractors for construction and installation activities will be contracted by the SEC on behalf of Company X.

9.5.3.7 Procurement

All purchase orders during detailed design and orders of equipment will be signed by the SEC's project manager and finally approved by Company X and placed in the name of Company X.

This will be described in more details in the procurement strategy and the purchase plan.

9.5.4 Commissioning and Qualification

Execution of C&Q is described in the commissioning and qualification plan GMPX-C&Q-07-001 and in the QAP Doc. no. GMPX-QAP-07-001. The execution can be summarized as follows.

9.5.4.1 Factory Acceptance Test (FAT) and Site Acceptance Test (SAT) for Systems Assessed as Indirect Impact and Direct Impact

Suppliers and contractors of indirect- and direct-impact systems must, if applicable according to SQP during FAT/SAT, demonstrate to the SEC that all requirements, including functions' "as-built" documentation, are fulfilled and available and that the deliverables operate as required. In situations where the supplier or contractor is not in a position to perform FAT/SAT as expected, the SEC, in cooperation with the supplier or contractor, must perform additional tests, possibly as commissioning.

FAT/SAT shall be prepared in a manner that ensures that all C&Q requirements are tested and documented and fulfill the acceptance criteria at the completion of SAT.

When the machine or product has been assembled at the site and adequately connected and run-in tested by the supplier or contractor, the SAT is carried out. This takes place in a way similar to the FAT, according to SEC-approved SAT protocols. For indirect-impact and direct-impact systems, SEC QA should review protocols and test plans in the planning of SAT to ensure that approval of the SAT report is obtainable. The SAT is finished with the SEC and SEC QA approval of the SAT report.

9.5.4.2 Commissioning Indirect-Impact Systems and Direct-Impact Systems

For systems assessed as direct or indirect impact, fulfillment of C-requirements shall be demonstrated through commissioning. The work package owner (WPO) has the responsibility to ensure that commissioning activities are well planned and adequately executed, documented, and reported. The Q-coordinator is responsible for overall coordination and follow-up.

Vendor SAT is part of the commissioning.

9.5.4.3 No-Impact Systems

For systems assessed as having no impact, fulfillment of requirements must be demonstrated and documented according to GEP based on the tradition in the discipline.

The WPO is responsible for planning and executing the commissioning activities for no-impact systems and to ensure that the activities are documented in a way that allows the SEC to illustrate that all requirements are met.

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9.5.4.4 Qualification

For systems assessed as direct impact, fulfillment of Q-requirements in relation to installations and operations shall be demonstrated through installation qualification and operational qualification (IQ/OQ). Qualification and especially prerequisites before the qualification can start shall be described in the protocols. IQ and OQ must be carried out according to the approved protocols.

9.6 Organization

For details of the Supervising Engineering Contractor Organization (SEC), refer to the SEC organogram in (4286) S9.13 References.

9.6.1 Supervising Engineering Contractor's Roles and Responsibilities

An overview of the SEC's roles and responsibilities is presented in the following table.

Role	Description
Project Manager and Execution Manager	Has the overall responsibility for the project
Project QA	Is responsible for setting up and maintaining quality assurance and quality control procedures and for aligning the supervising engineering contractor's quality routines with Company X and regulatory requirements
Procurement	Is responsible for setting up an overall procurement strategy and making sure that the strategy is implemented
Project Document Controller	Is responsible for setting up and implementing document control procedures within the project
Engineering Manager	Is responsible for ensuring that the overall design of the facility is done in a cost-effective sequence and for physical coordination (e.g., clash control procedures) and also to ensure that partners, suppliers, and the client are incorporated in the design process in an effective way
Construction Manager	Has the overall responsibility to establish and maintain the necessary organization and facilities for the construction taking part at Kuopio, Finland
C&Q Manager	Has the overall responsibility to plan and coordinate testing of the facility
Project leads (Engineering and installation leads)	Are responsible for time and quality in their subparts of the project

9.6.2 Company X Organization

For details of the Company X organogram, refer to the Company X organogram in (4286) S9.13 References. Roles and responsibilities between the SEC and Company X are defined in **Section 2.6** Steering Committee.

9.7 Quality

9.7.1 Project Quality Strategy

A validation master plan (VMP) is produced by Company X for the project. This VMP will be used to provide the basis for all validation and quality activities during the lifecycle of the project.

The SEC is responsible for carrying out the commissioning and qualification activities (IQ and OQ) unless otherwise described in the VMP. The subsequent process qualification, cleaning validation, and process validation are carried out by Company X and are not included in this project.

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The VMP is the overall document for validation in the project and will be master and basis for qualification of equipment facilities and utilities. A separate VMP will be written for process validation.

The qualifications sequence of CP, IQ, OQ, and PQ activities are described in details in the QAP.

9.7.1.5 Partner/Vendor/Supplier Qualifications

Partners and suppliers describe their "normal" test in the tender documents (for instance, via examples of test plans). If possible, the tests are performed on the GMP level when relevant.

- Tests must address all requirements.
- Scope of the test activity is determined in the contract (the partners and suppliers draft QAP).
- After this, the SEC prepares a draft of IQ and OQ protocol in which test plans from the supplier may be entered, if relevant.
- The suppliers test planning may be adjusted to achieve maximum synergy.
- The SEC approves supplier's final QAP.

Time schedule and economy of the project includes a completed IQ and OQ with the SEC as the final step. The validation involves the active participation of the end user and Company X-QA. The scope of this is described in the basic design phase.

9.7.2 Quality Activity Plan (QAP)

The project's QAP manages all quality activities and quality documentation to be prepared. The QAP is made in compliance with the requirements in the VMP.

9.7.3 Document Control

Document control is handled in accordance with the QAP.

9.8 Risk Management

There will be a number of risk analyses performed during the project. Their main purpose is to identify risk early and create activities to reduce the risk.

The risk analyses are:

- BD Project Risk Analysis 1
- BD BSL and GMP Risk Analysis with Focus on the Layout
- DD Project Risk Analysis 2
- DD HSE Risk Analysis on the Execution Phase

9.8.1 Documentation

- **Language and templates—** Project language is UK English with the exception of documents used for Finnish craftsmen. The SEC templates are used when adequate.
- **Organizing project data—** Data storage must be planned for and organized at the start of the project in order to properly archive the information during project closeout. Proper setup at the start of the project will help make it possible to utilize data efficiently, minimize rework, and ensure complete electronic records retention.

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- **Project documentation —** In basic design the planned documents are described in the object breakdown structure.
  The quality managed documents are described in the QAP.
  The document delivery of the suppliers and partners are described in the tender documents or in the contract of the project.
- **Technical tools and standards —** For each skill or profession (construction, architecture, bulk process, utility, etc.) it is necessary to determine and develop tools and standards to ensure continuous engineering of the project parts. The lead engineers manage these tools and standards. Specification tools and standards are being prepared in basic design.

9.9 Health, Safety, and Environment

An initial HSE evaluation has been started. The first version of this plan will be issued and approved before any construction activity starts. This plan will be updated during the construction with new items and activities.

9.10 Commissioning and Qualification

When the SEC finds it beneficial, an all-day meeting will be called in order to approve documents and/or carry out reviews. This will be scheduled no less than one week in advance and with duration from one day to several days.

When the SEC finds it beneficial, an all-day commissioning day will be called in order to correct errors and carry through cleanroom documents. This will be arranged no less than one week ahead and with duration from one day to several days.

The SEC is responsible for carrying out or supervising and reporting all qualification activities up to and including OQ.

Company X will participate in the test activities.

Company X provides QA resources during the OQ test period.

Company X is responsible for the calibration and maintenance after OQ.

9.11 Handover and Operation Support

The project is handed over to the customer in three steps: 1) ready to run, 2) handover, and 3) final handover. Final handover is the last part of the project.

The project is handed over to the customer with the participation of the SEC project manager, relevant SEC and/or contract project engineers, and representatives from the customer of his or her own choosing.

If the customer wishes to put the area into use before the handover has taken place, the SEC project manager shall ensure that a physical examination of the area takes place together with the customer or the user and that an agreement on operation and maintenance responsibilities is made.

Prior to final handover, deviation lists will be resolved by a cleanup team.

Warranties from third-party vendors, suppliers, and subcontractors are handed over to Company X.

PQ and PV activities are not part of this project. The SEC recommends the client to plan on a PQ support team. SEC will supply resources for support in PQ/PV phase if Company X so wishes. The SEC resources will work under Company X supervision and management during PQ/PV.

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Company X has full responsibility for running and maintaining equipment after handover. This must be carried out without invoking any delays to the project.

Documentation will be handed over in binders and/or electronic copy as specified by Company X at final handover, and direct to Company X document control. Where the SEC has produced or obtained from a supplier an electronic copy, this will be provided with the final documentation.

Optimizing program and parameters is not within the scope of this project. Education and training are to be defined. (See C&Q plan.)

9.12 Abbreviations

BD — Basic Design
BSL — Biosafety Level
BMS — Building Management System
Company X — Hypothetical Company
C&Q — Commissioning and Qualification
CR — Cleanroom
DD — Detailed Design
FMS — Facility Management System
FAT — Factory Acceptance Test
GDP — Good Documentation Practice
GEP — Good Engineering Practice
GMP — Good Manufacturing Practice
HSE — Health, Safety, and Environment
IQ — Installation Qualification
MC — Mechanical Completion
OQ — Operational Qualification
QA — Quality Assurance
QAP — Quality Activity Plan
QMS — Quality Management System
SAT — Site Acceptance Test
SEC — Supervising Engineering Contractor
UR — User Requirement
VMP — Validation Master Plan

9.13 References

The following reference documents should be read in association with this project execution plan:

- Level A Time Schedule
- Project Quality Activity Plan (QAP)
- Project Validation Master Plan (VMP)
- Object Breakdown Structure
- Procurement Strategy
- Purchase Plan
- Travel and Costs Principles

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10.0 Appendix III: Training Requirements Example

SUS IMPLEMENTATION TRAINING PLAN																										ORGANIZATION AND PLANNING	
Department	Name 1	Name 2	Name 3	Name 4	Name 5	Name 6	Name 7	Name 8	Name 9	Name 10	Name 11	Name 12	Name 13	Name 14	Name 15	Name 16	Name 17	Name 18	Name 19	Name 20	Training Module	INT/ EXT	Duration or Frequency	Training Organization: Trainer's Name	Provisional Date	Realization Date
Warehousing	x	x	x	x	x	x															Bioreactor Bags: reception and storage	INT	2 hours	John	Q1 2015	
Production Support		x					x	x	x	x	x	x									Bioreactor Bags: preparation and integration	EXT	4 hours	Brian	Q1 2015	
Production Support							x	x	x	x	x	x									Calibration and Qualification	EXT	4 hours	Dave	Q2/Q3 2015	
Production													x	x	x	x	x	x	x		SUS Bioreactor Utilization 1	EXT	4 hours	Susan	Q1 2015	2015 January 09
Production														x	x	x	x	x	x		SUS Bioreactor Utilization 2	EXT	4 hours	Alan	Q3 2015	
Production Support							x	x	x	x	x	x		x	x						SUS Disposal	INT	2 hours	Lucy	Q2/Q3 2015	
Production Support							x	x	x	x	x	x	x	x	x						SUS Integrity and Leak Testing	EXT	4 hours	James	Q12015	

Approvals	Date/Signature
Manufacturing	
Quality	
Qualified Person	
Human Resources	
Finance	

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11.0 References

1. Quality Guideline Q6A: *Specifications: Test Procedures and Acceptance Criteria for New Drug Substance and New Drug Product: Chemical Substances*; International Conference on Harmonisation: 1999. www.ich.org (accessed Feb 4, 2014).
2. Quality Guideline Q7: *Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients*; International Conference on Harmonisation: 2000. www.ich.org (accessed Feb 4, 2014).
3. Quality Guideline Q8 (R2): *Pharmaceutical Development: Technical Requirements for Registration of Pharmaceuticals for Human Use;* International Conference on Harmonisation: 2009. www.ich.org (accessed Feb 4, 2014).
4. Quality Guideline Q9: *Quality Risk Management;* International Conference on Harmonisation: 2005. www.ich.org (accessed Feb 4, 2014).
5. Quality Guideline Q10: *Pharmaceutical Quality Systems*; International Conference on Harmonisation: 2008. www.ich.org (accessed Feb 4, 2014).
6. Quality Guideline Q11: *Development and Manufacture of Drug Substances (Chemical Entities and Biotechnological/Biological Entities);* International Conference on Harmonisation: 2012. www.ich.org (accessed Feb 4, 2014).
7. *Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations*; Parenteral Drug Association: 2012. www.pda.org/bookstore (accessed Jan15, 2013).
8. *Technical Report No. 60: Process Validation – A Lifecycle Approach*; Parenteral Drug Association: 2013. www.pda.org/bookstore (accessed Jan 15, 2013).
9. ISO 14644-1:1999. Cleanrooms and associated controlled environments — Part 1: Classification of Air Cleanliness; International Organization for Standardization: 1999. www.iso.org (accessed Feb 4, 2014).
10. General Chapter <788> Particulate Matter in Injections. USP 29/NF 24; U.S. Pharmacopeia: 2007. www.usp.org.
11. General Chapter <85> Bacterial Endotoxins Test. USP 29/NF 24; U.S. Pharmacopeia: 2011. www.usp.org.
12. European Pharmacopoeia 2.6.14 *Bacterial Endotoxins Test*; European Medicines Agency: 2010. www.edqm.eu/ (accessed August 11, 2014).
13. WHO Monograph 3.4 *Test For Bacterial Endotoxins,* World Health Organization: 2012. www.who.int.
14. Japan Pharmacopoeia 4.01 *Bacterial Endotoxins Test*; JP Sixteenth Edition. The Ministry of Health, Labour and Welfare Ministerial Notification No.65. 2011. www.pmda.go.jp/english/service/regulation.html (accessed August 11, 2014).
15. Pharmaceutical Validation website: pharmaceuticalvalidation.blogspot.com/2007/08/validation-of-rapid-sterile-transfer.html (accessed August 18, 2014).
16. International Safe Transit Association Publications Page. www.ista.org/pages/resources/publications.php (accessed February 27, 2014).
17. ASTM D4169-09. Standard Practice for Performance Testing of Shipping Containers and Systems; American Society for Testing and Materials: 2009. www.astm.org (accessed August 18, 2014).
18. ASTM E2500-07. Standard Guide for Specification, Design, and Verification of Pharmaceutical and Biopharmaceutical Manufacturing Systems and Equipment; American Society for Testing and Materials: 2007. www.astm.org (accessed Jan 15, 2013).
19. ISO 10993-1:2009. Biological evaluation of medical devices — Part 1: Evaluation and Testing within a Risk Management Process. International Organization for Standardization: 2009. www.iso.org (accessed Feb 19, 2014).
20. ISO 9001:2008. Quality management systems – Requirements. International Organization for Standardization: 2008. www.iso.org (accessed Feb 19, 2014).
21. U.S. Code of Federal Regulations Title 21Part 820: Quality System Regulations. www.ecfr.gpoaccess.gov (accessed Feb 19, 2014).
22. ISO 13485:2003. Medical devices — Quality management systems — Requirements for Regulatory Purposes. International Organization for Standardization: 2003. www.iso.org (accessed Feb 19, 2014).
23. U.S. Code of Federal Regulations Title 21 Part 210. Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs. www.ecfr.gpoaccess.gov (accessed Feb 19, 2014).
24. General Chapter <87> Biological Reactivity Tests, In Vitro. USP 31/NF 26; U.S. Pharmacopeia: 2008. www.usp.org.
25. General Chapter <88> Biological Reactivity Tests, In Vivo. USP 31/NF 26; U.S. Pharmacopeia: 2008. www.usp.org.
26. *U.S. FDA Public Health Notification: Safety Assessment of Di(2-ethylhexyl)phthalate (DEHP) Released from PVC Medical Devices*; U.S. Food and Drug Administration: 2012. www.fda.gov/MedicalDevices/Safety/AlertsandNotices/PublicHealthNotifications/ucm062182.htm. (accessed February 20, 2014).
27. Cafmeyer, N.; Wolfson, B. The Possible Leaching of Diethyl Phthalate into Levothyroxine Sodium Tablets. *Am. J. Hosp. Pharm.* **1991**, *48, 735-739*.

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28. Ostenberg, R. Potential Toxicity of Extractables and Leachables in Drug Product. *Am. Pharm. Rev.*. **2005**, *8, 64-67*.

29. Gayathri N., et al. Changes in Some Hormones by Low Doses of Di(2-ethyl hexyl) phthalate (DEHP), a Commonly Used Plasticizer in PVC Blood Storage Bags and Medical Tubing. *Indian J. Med. Res*. **2004**, *119, 139-144*.

30. Wood, J., et al. Strategy for Selecting Disposable Bags for Cell Culture Media Applications Based on a Root-Cause Investigation, *Biotech. Pro.*. **2013**, *29, 1535-1549*.

31. Wang, Q. Selection of Analytical Techniques for Pharmaceutical Leachables Studies. *Am. Pharm. Rev.* **2005**, *8(6), 38-44*.

32. General Chapter <661> Containers - Plastics. USP 29/NF 24; U.S. Pharmacopeia: 2004. www.usp.org.

33. General Chapter <381> Elastomeric Closures for Injections. USP 32/NF 27; U.S. Pharmacopeia: 2009. www.usp.org.

34. ASTM D5774-09. Standard Test Methods for Rubber – Chemical Analysis of Extractables; American Society for Testing and Materials: 2009. www.astm.org (accessed August 18, 2014).

35. BioPhorum Operations Group Home Page. www.biophorum.com/ (accessed Feb 27, 2014).

36. Extractables and Leachables Subcommittee of the Bio-Process Alliance. Recommendations for Extractables and Leachables Testing. *BioPro Intl* [Online] December 2007, p 36-49. www.bpsalliance.org/assets/files/BPSA-Extractables-101-BPI-Suppl-May-2008.pdf. (accessed Jan 27 2014).

37. 37. Quality Guideline Q3C(R5): *Impurities: Guideline for residual solvents*; International Conference on Harmonisation: 2011. www.ich.org (accessed Feb 4, 2014).

38. PQRI Leachables and Extractables Working Group. Safety Thresholds and Best Practices for Extractables and Leachables in Orally Inhaled and Nasal Drug Products. Product Quality Research Institute: 2006. www.pqri.org/pdfs/LE_Recommendations_to_FDA_09-29-06.pdf (accessed February 19, 2014).

39. Hammond, M., et al. Identification of a Leachable Compound Detrimental to Cell Growth in Single-Use Bio-Process Containers. *PDA J Pharm Sci Technol*. **2013**, *67(2), 123-34*.

40. Ding, W. Determination of Extractables and Leachables from Single-Use Systems. *Chem. Eng. Tech*. **2013**, *85(1-2), 186–196*.

41. EMEA/CHMP/QWP/251344/2006: Guideline On The Limit of Genotoxic Impurities; European Commission: 2006. www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500002903.pdf (accessed February 12, 2014)

42. Meyer, B.; Vargas, D. Impact of Tubing Material on the Failure of Product-Specific Bubble Points of Sterilizing-Grade Filters. PDA J Pharm Sci and Tech. 2006, 60, 248 - 253.

43. See also www.pharmamanufacturing.com/blogs/onpharma/at-bpi-some-straight-talk-about-single-use-bioprocess-equipment/

44. Markovic, I. Challenges associated with extractable and/or leachable substances in therapeutic biologic protein products. *Am. Pharm. Rev*. **2006**, *9(6), 20-27*.

45. Markovic, I. Evaluation of safety and quality impact of extractable and leachable substances in therapeutic biologic protein products: A risk-based perspective. *Exp. Op. Drug Safe*.. **2007**, *6(5), 487-491*.

46. Markovic, I. Risk management strategies for safety qualification of extractable and leachable substances in therapeutic biologic protein products. *Am. Pharm. Rev*. **2009**, *12(4), 96-101*.

47. U.S. Code of Federal Regulations Title 21Part 189.5: Prohibited Cattle Materials. www.ecfr.gpoaccess.gov (accessed Feb 19, 2014).

48. EMA 410/01 rev 3: *Note for Minimising the Risk of Transmitting Animal Spongiform Encephalopathy Agents via Human and Veterinary Medicinal Products;* European Medicines Agency, Official Journal of the European Union: 2011. www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500003700.pdf (accessed Feb 22, 2014)

49. ASTM F1980 – 07. Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices; American Society for Testing and Materials: 2007. www.astm.org (accessed Feb 27, 2014).

50. General Chapter <790> Visible Particulates in Injections. USP 35/NF 30; U.S. Pharmacopeia: 2012. www.usp.org.

51. ISO 11137-2:2012. Sterilization of health care products — Radiation — Part 2: Establishing the Sterilization Dose Standard. International Organization for Standardization: 2012. www.iso.org (accessed Feb19, 2014).

52. AAMI TIR 33:2005. Sterilization of health care products – Radiation – Substantiation of a Selected Sterilization Dose – Method VDmax. Association for the Advancement of Medical Instrumentation: 2005. my.aami.org/ (accessed Feb 19, 2014).

53. Kapp, T. et al. Roadmap to Implementation of Single-Use Systems. *BioProcess Intl*. 2010, April, 10-19. www.bpsalliance.org/assets/files/BPSA-Economics-White-Paper-2010.pdf (accessed Feb27, 2014).

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54. *Technical Report No. 1 (Revised 2007): Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control*; Parenteral Drug Association: 2007. www.pda.org/bookstore.

55. *Technical Report No. 61: Steam In Place*; Parenteral Drug Association: 2013. www.pda.org/bookstore.

56. Pethe, V. et al. Integrity Testing of Flexible Containers. *Biopharm. Intl.* **2011**, *24(11), 42-49*.

57. Beh, Wooiliam and Wong, Russell, "Case Study: Improving the Robustness of a 2000 L Bag, Disposables for Biopharm Production (IBC), December 12-13, 2005, Reston, Virginia. (Later IBC Single Use Conference, data from Bayer)

58. Lakhdar, K., et al. Multiobjective Long-Term Planning of Biopharmaceutical Manufacturing Facilities. *Biotechnol. Prog.* **2007**, *23 (6), 1383-93*.

59. Barnoon, B. In *Cost Considerations: Stainless Steel vs. Bag Assemblies*, Proceedings of the IBC Single-Use Applications for Biopharmaceutical Manufacturing Conference, Boston, MA, June 9-10, 2014.

60. Lim J., et al. Economic Benefits of Single-Use Membrane Chromatography in Polishing: A Cost of Goods Model. *BioProcess Intl.* [Online] February 2007, p 48-56. www.bioprocessintl.com/wp-content/uploads/bpi-content/070205ar06_77096a.pdf (accessed Feb 20, 2014).

61. Sinclair, A. How to Evaluate the Cost Impact of Using Disposables in Biomanufacturing. *Biopharm Intl. 2008, 21 (6)*. www.biopharminternational.com/biopharm/Disposables+Advisor/How-to-Evaluate-the-Cost-Impact-of-Using-Disposabl/ArticleStandard/Article/detail/522179 (accessed August 18, 2014).

62. LoMonaco, J.; Rumsey, T. The Economics of Single-Use Systems – a Media Prep ROI Analysis, *Biopro. Intl.* **2006,** October, pp 64-68. www.bioprocessintl.com/wp-content/uploads/bpi-content/06049ar08_76409a.pdf (accessed August 18, 2014).

63. Sinclair, A.; Monge, M. Quantitative Economic Evaluation of Single-Use Disposables in Bioprocessing. *Pharm.Eng.* **2002**, *22(3), 20-34*.

64. Sinclair, A. et al. The Environmental Impact of Disposable Technology. *Biopharm Intl.* [Online] November 2008. www.biopharminternational.com/biopharm/article/articleDetail.jsp?id=566014&pageID=1&sk=&date=. (accessed Feb 20, 2014)

65. Disposals Subcommittee of the Bio-Process Systems Alliance. Guide to Disposal of Single-Use Bioprocess Systems. *BioPro. Intl.* [Online] November 2007, p 22-28. www.bpsalliance.org/assets/files/BPSA-Disposal-Article-BPI-1107.pdf (accessed February 10, 2014).

66. U.S. Environmental Protection Agency. Wastes — Hazardous Waste — Treatment & Disposal: Combustion. 22 Feb. 2006. 12 June 2007 www.epa.gov/osw/hazard/tsd/td/combustion.htm (accessed July 8, 2014).

67. Haughney, H.; Hutchinson, J. Single –Use Systems Reduce Production Timelines. *Gen. Eng. News.* **2004**, *24(8)*.

68. Rawlings, B.; Pora, H. Environmental Impact of Single-Use and Reusable Bioprocess Systems *BioPro. Intl.*. **2009**, *7*, pp 18-25. www.pall.com/pdfs/Biopharmaceuticals/Allegro_Article_6_-_BioProcess_International_-_February_09.pdf (accessed August 18, 2014).

69. American Chemistry Council. Plastics Education Campaigns and Resources Page. http://plastics.americanchemistry.com/Education-Resources/ (accessed Feb 19, 2014).

70. Abdullah, G. Making the Most of a Powerful Nuisance. *Penn State Agri. Mag.* **2003**, *17-19*.

71. Bright Hub Project Management Constructing a RACI Matrix Page. www.brighthubpm.com/templates-forms/53552-constructing-a-raci-matrix/ (accessed Feb 27, 2014).

72. Mire-Sluis A, et al. Practical Applications of Quality Risk Management; *Biopro. Intl.* **2010**, *8(3), 20-32*.

73. *A Guide to Supply Chain Risk Management for the Pharmaceutical and Medical Device Industries and their Suppliers*; The Chartered Quality Institute: London, 2010. www.pqg.org/pharma/wp-content/uploads/2011/09/PQG-Guide-to-Supply-Chain-Risk-Management-V-1-0-2010.pdf (accessed Feb 27, 2014).

74. *Guidance for Industry, Process Validation: General Principles and Practices*; U.S. Food and Drug Administration, Revision 1: January 2011 www.fda.gov/downloads/Drugs/Guidances/UCM070336.pdf (accessed Feb 20, 2014).

75. EMA/CHMP/BWP/187338/2014, *Guideline on process validation for the manufacture of biotechnology-derived active substances and data to be provided in the regulatory submission - Draft*. 2014. www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2014/04/WC500165805.pdf (accessed August 14, 2014).

76. EMA/CHMP/CVMP/QWP/70278/2012-Rev1, *EMA Draft Guidance on Process Validation; European Commission Guideline on process validation for finished products information and data to be provided in regulatory submissions:* 2014. www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2014/02/WC500162136.pdf (accessed August 18, 2014).

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77. *Technical Report No. 56: Application of Phase-Appropriate Quality Systems and cGMP to the Development of Therapeutic Protein Drug Substance*; Parenteral Drug Association: 2012. www.pda.org/bookstore (accessed Jan 15, 2013).

78. ASTM E2537-08: Standard Guide for Application of Continuous Quality Verification to Pharmaceutical and Biopharmaceutical Manufacturing; American Society for Testing and Materials: 2008. www.astm.org (accessed Jan 15, 2013).

79. BioPhorum Operations Group Home Page. www.biophorum.com/ (accessed Feb 27, 2014).

80. Kelleher C. in *Standardization in Single-Use Technologies – Need for Consistency in Design*, Proceedings of the: Pharma IQ Disposable Solutions for Manufacturing Conference, Brussels, Belgium. Feb 19-20, 2013.

81. Blanc, G. et al. Defining the Implementation of Single-Use Technology: A Comparative Evaluation of a Single-Use Bioreactor with a Stainless-Steel Bioreactor; *Pharm. Biopro.*. **2013**, *1(4), 1-9*.

82. Royce, J. in *An Environmental Life Cycle Assessment Comparison of Single-Use and Conventional Process Technology for the Production of Monoclonal Antibodies*, Proceedings of the PDA Europe Workshop on Single-Use-Systems for Pharmaceutical Applications, Uppsala, Sweden, Nov 29-30, 2011.

83. Pora, H.; Rawlings, B. A User's Checklist for Introducing Single-Use Components Into Process Systems; *Bioprocess Intl.* **2009**, 7(4), *9-16*.

84. Olbrich, C.; Burkhard, A. in *Selection and Implementation of Single-Use Systems (SUS) for Clinical Trial Manufacture*, Proceedings of the PDA Workshop on Single-Use Systems for Pharmaceutical Applications, Uppsala, Sweden, Nov 29-30 2011.

85. Schreiber, B. in *Hybrid SU Facilities–Optimizing Existing SS Facilities & Applying SU to Create Value*, Proceedings of IBC's 10th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference, Durham, NC, June 3-4 2013.

86. Restelli M. in *Single-use Technology for a Bacterial Vaccine Production,* Proceedings of the PDA Europe Workshop on Single-Use Systems for Pharmaceutical Applications, Milan, Italy, Jan 15-16 2013.

87. O'Connor, K. in *Challenges of Implementing Single-Use Systems into Existing Commercial Manufacturing Facilities*, Proceedings of IBC's 10th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference, Durham, NC, June 3-42013.

88. Shen, K. et al. Implementing Single-Use Technology in Tangential Flow Filtration Systems in Clinical Manufacturing. *BioPharm Intl.* [Online] November 2, 2010. www.biopharminternational.com/biopharm/Disposables/Implementing-Single-Use-Technology-in-Tangential-F/ArticleStandard/Article/detail/695156 (accessed Feb 27, 2014).

89. Brown, S. A Risk-Based Approach to Single-Use Systems Selection and Implementation for IMP Manufacture ; Disposables for Biopharmaceutical Manufacturing, 2010, October 27; *Informa Bioproduction Conference* 2010, Barcelona, Spain

90. ISPE Good Practice Guide: Good Engineering Practice; International Society for Pharmaceutical Engineering, 2008.

91. Neeleman, R. GMO Containment Risks Evaluation of Single-Use Bioreactors, Dutch agency for implementing GMOs. *Xendo Bioprocessing*. [Online], 2010. www.ggo-vergunningverlening.nl/dsresource?type=pdf&objectid=rivmp:193530&versionid=&subobjectname (accessed Feb 27, 2014).

92. APIC Supplier Qualification and Management Guideline; Chapter 2, Due Diligence; Appendix 3, Due Diligence checklist: Active Pharmaceutical Ingredients Committee; 2009. apic.cefic.org/pub/GuidelineSupplierQualification_200912_final.pdf. (accessed July 7, 2014).

93. Rathore, A. and Low, D. Managing Raw Materials in the QbD Paradigm, Part 1: Understanding Risks. *BioPharm Intl.*, [Online] Nov 1, 2010. www.biopharminternational.com/biopharm/Quality/Managing-Raw-Materials-in-the-QbD-Paradigm-Part-1-/ArticleStandard/Article/detail/695163 (accessed January 27 2014).

94. Rathore, A. and Low D. Managing Raw Materials in the QbD Paradigm, Part 2: Risk Assessment and Communication. *BioPharm Intl*, [Online], Dec 1, 2010. www.biopharminternational.com/biopharm/Quality/Managing-Raw-Materials-in-the-QbD-Paradigm-Part-2-/ArticleStandard/Article/detail/698526 (accessed January 27, 2014).

95. ANSI/ISA-88, Batch control standard parts 1-4 and 5; International Society of Automation: 1995. (accessed Feb 15, 2014).

96. Littlefield, B., In *Determination of Extractables/Leachables From Integrated Single-Use Systems, Workshop III, Single-Use Technologies in Manufacturing Process*, Proceedings of the PDA Biopharmaceutical Development and Manufacturing Conference, Munich, Germany, June 16-17,2009.

97. Gupta, V. In *Risk-Based Assessment of Extractables and Leachables in Single-Use Systems, Disposable and Single-Use Biopharmaceutical Manufacturing Systems*, Proceedings of the PharmaED Conference, Philadelphia, PA, Sept 27-28, 2010.

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98. Smith, E. In *Extractables and Leachables for Disposables and Process Components: Disposable and Single-Use Biopharmaceutical Manufacturing Systems*, Proceedings of the PharmaED Conference, Philadelphia, USA, Sep 27-28, 2010.

99. Morren K. In *Extractables & Leachables*, Proceedings of the Challenges of Single-Use Technology Biopractis Workshop, Nantes, France, 21 April, 2013.

100. Ball D. In *Establishing Threshold Levels of Extractables & Leachables Parenteral and Ophthalmic Drug Products – Application to Single-Use Manufacture Discussion of Proposed PQRI Qualification Plan*, Proceedings of the Pharma IQ Disposable Solutions for Manufacturing Conference Brussels, Belgium. Feb 19-20, 2013.

101. Wong KM. In *Extractable Protocol and Vendor Expectation Document Standardization Efforts from BPROG Working Group*, Proceedings of IBC's 10th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference, Durham, NC, June 3-4, 2013.

102. The Extractables and Leachables Safety Information Exchange (ELSIE) Home Page: www.elsiedata.org/ (accessed Jan 27 2014).

103. Incoterms® 2010 (8th version). International Commercial Terms. International Chamber of Commerce (ICC). January 2011: www.iccwbo.org/products-and-services/trade-facilitation/incoterms-2010/ (accessed July 7, 2014).

104. Bei W. In *Operational Considerations for Single-Use Systems*, Proceedings of IBC's 5th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference. San Diego, CA, , June 2 2008.

105. Wong R. In *Preventing Product Loss – Beyond Assembly Design*, Proceedings of IBC's 10th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference, Durham, NC, June 3-4 2013.

106. Example HSE Plan Page. www.csemag.com/single-article/how-to-make-a-health-safety-and-environmental-plan/712141194725dbdcfded6357f17e08f7.html. (accessed Jan 27 2014).

107. Example HSE Plan Page. idagroup.com.tr/pdf/ida_project_specific_action_plan_hse.pdf. (accessed Jan 27 2014).

108. Handel, T. Identifying and Selecting Outsourcing Providers, what to look for in a contract service provider and what to expect once you've found one. *Contract Pharma* [Online], August 22, 2005. www.contractpharma.com/issues/2000-04/view_features/identifying-and-selecting-outsourcing-providers. (accessed Jan 27 2014).

109. Blasini, R. Quality Agreements Between Pharmaceutical/Biopharmaceutical Companies and Their Contractors; *Biopharm Intl*,. [Online], April 1, 2005, biopharminternational.findpharma.com/biopharm/article/articleDetail.jsp?id=157266. (accessed Jan 27 2014).

110. 110. O'Korn, D. Negotiating a Manufacturing Agreement – What do you need to know? *Contract Pharma*, March 2, 2007. www.contractpharma.com/issues/2007-03/view_features/negotiating-a-manufacturing-agreement. (accessed Jan 27 2014).

111. Schniepp, S. Insider Solutions: The Quality Agreement; *Pharm. Tech.* **2010**, *34(8), 69-72*.

112. Pluta, PL. "QbD and Process Validation – Complementary Lifecycle Approaches, in Quality By Design," Chapter 6 of Putting Theory Into Practice, S. Schmitt, Ed. (PDA/DHI, 2011).

113. Winkle, HN. In *Implementing Quality by Design* (refer to the table on page 10), Proceedings of the PDA/FDA Joint Regulatory Conference, Washington, DC, Sep 23-24 2007.

114. Blomberg, M. Julien, C. Continuous Quality Verification as a Control Strategy for Biocontainers, *Pharmpro*, [Online], Jan 18, 2012. www.pharmpro.com/articles/2012/01/biopharmaceutical-processing-Continuous-Quality-Verification-as-a-Control-Strategy-for-Biocontainers. (accessed Jan 27 2014).

115. Brown S. In *Implementation and Technical Diligence.* Proceedings of IBC's 10th Annual Single-Use Applications for Biopharmaceutical Manufacturing Conference, Durham, NC, 3-4 June 2013.

116. Example of a Quality Agreement template (SOCMA). www.socma.com/governmentRelations/?subSec=26&articleID=1831#Quality-Agreement-Template. (accessed Jan 27, 2014)

117. Example of a Quality Agreement template (IPEC Europe). www.ipec-europe.org/page.asp?pid=59. (accessed Jan 27 2014).

118. BS EN 12740:1999. — Biotechnology — Laboratories for research, development and analysis - Guidance for handling, inactivating and testing of waste; British Standards Institution: 1999. http://shop.bsigroup.com/ProductDetail/?pid=000000000030003193 (accessed July 7, 2014).

119. Markarian, J. Waste Management of Disposable Biopharmaceutical-Manufacturing Components. *Pharm Tech* [Online], August 15, 2012.

120. Laboratory Biosafety Manual 3rd Edition; World Health Organization, Geneva, 2004.

121. Biosafety in Microbiological and Biomedical Laboratories, 5th edition; U.S. Government Printing Office, Washington, DC, 2007.

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122. QAS/0.8.256 Rev 1: *Good Manufacturing Practices for Pharmaceutical Products Containing Hazardous Substances, Draft for Comment*; World Health Organization: 2009. www.who.int/medicines/services/expertcommittees/pharmprep/CleanRev-290709-GMPHazardous_QAS08_256.pdf (accessed Jan 27, 2014).

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